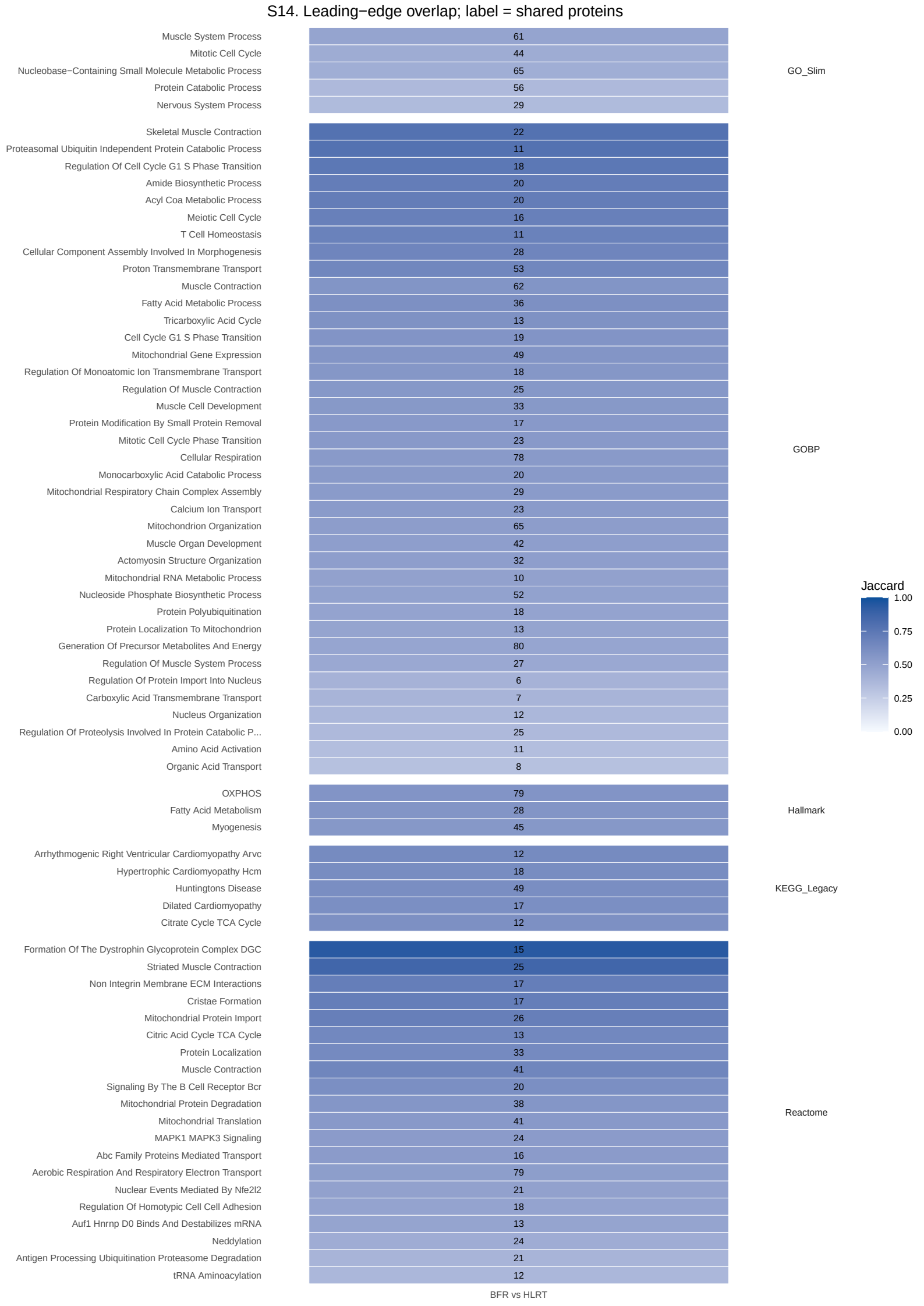
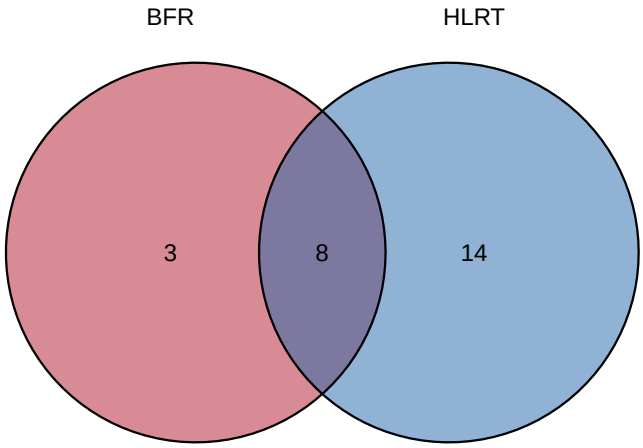

S14. Leading−edge overlap; label = shared proteins
Muscle System Process
Mitotic Cell Cycle
Nucleobase−Containing Small Molecule Metabolic Process
Protein Catabolic Process
Nervous System Process
GO_Slim
Skeletal Muscle Contraction
Proteasomal Ubiquitin Independent Protein Catabolic Process
Regulation Of Cell Cycle G1 S Phase Transition
Amide Biosynthetic Process
Acyl Coa Metabolic Process
Meiotic Cell Cycle
T Cell Homeostasis
Cellular Component Assembly Involved In Morphogenesis
Proton Transmembrane Transport
Muscle Contraction
Fatty Acid Metabolic Process
Tricarboxylic Acid Cycle
Cell Cycle G1 S Phase Transition
Mitochondrial Gene Expression
Regulation Of Monoatomic Ion Transmembrane Transport
Regulation Of Muscle Contraction
Muscle Cell Development
Protein Modification By Small Protein Removal
Mitotic Cell Cycle Phase Transition
Cellular Respiration
Monocarboxylic Acid Catabolic Process
Mitochondrial Respiratory Chain Complex Assembly
Calcium Ion Transport
Mitochondrion Organization
Muscle Organ Development
Actomyosin Structure Organization
Mitochondrial RNA Metabolic Process
Nucleoside Phosphate Biosynthetic Process
Protein Polyubiquitination
Protein Localization To Mitochondrion
Generation Of Precursor Metabolites And Energy
Regulation Of Muscle System Process
Regulation Of Protein Import Into Nucleus
Carboxylic Acid Transmembrane Transport
Nucleus Organization
Regulation Of Proteolysis Involved In Protein Catabolic P...
Amino Acid Activation
Organic Acid Transport
GOBP
OXPHOS
Fatty Acid Metabolism
Myogenesis
Hallmark
Arrhythmogenic Right Ventricular Cardiomyopathy Arvc
Hypertrophic Cardiomyopathy Hcm
Huntingtons Disease
Dilated Cardiomyopathy
Citrate Cycle TCA Cycle
KEGG_Legacy
Formation Of The Dystrophin Glycoprotein Complex DGC
Striated Muscle Contraction
Non Integrin Membrane ECM Interactions
Cristae Formation
Mitochondrial Protein Import
Citric Acid Cycle TCA Cycle
Protein Localization
Muscle Contraction
Signaling By The B Cell Receptor Bcr
Mitochondrial Protein Degradation
Mitochondrial Translation
MAPK1 MAPK3 Signaling
Abc Family Proteins Mediated Transport
Aerobic Respiration And Respiratory Electron Transport
Nuclear Events Mediated By Nfe2l2
Regulation Of Homotypic Cell Cell Adhesion
Auf1 Hnrnp D0 Binds And Destabilizes mRNA
Neddylation
Antigen Processing Ubiquitination Proteasome Degradation
tRNA Aminoacylation
Reactome
Jaccard
1.00
0.75
0.50
0.25
0.00
BFR vs HLRT

S15. Leading edges by leg

Organic Acid Transport

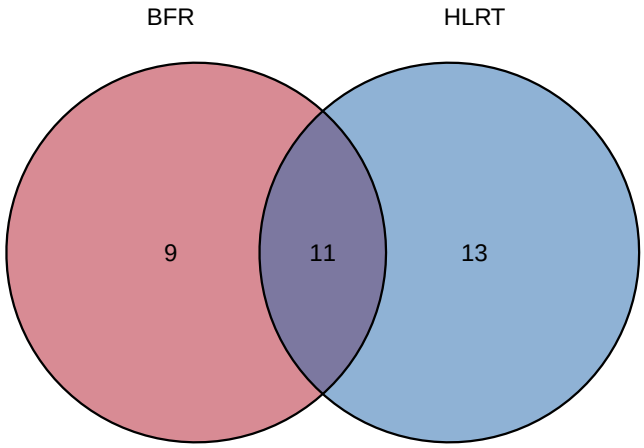
BFR: NES 1.74, padj 0.013
HLRT: NES 1.72, padj 0.024
shared 8, expected 5.1



Shared: ACSL1, AKT2, GOT2, MPC1, RBP1, SFXN1, SLC16A3, SLC25A12

Amino Acid Activation

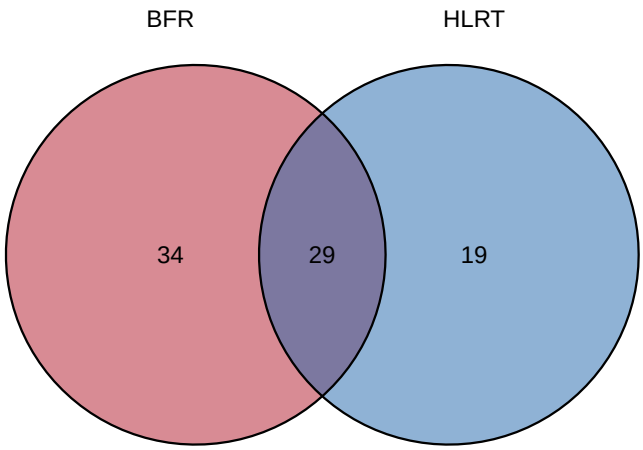
BFR: NES 2.31, padj 2.6e−05
HLRT: NES 2.21, padj 0.00025
shared 11, expected 12.3



Shared: DARS2, HARS1, IARS1, IARS2, LARS1, LARS2, NARS1, QARS1, SARS1, SARS2, TARS1

Nervous System Process

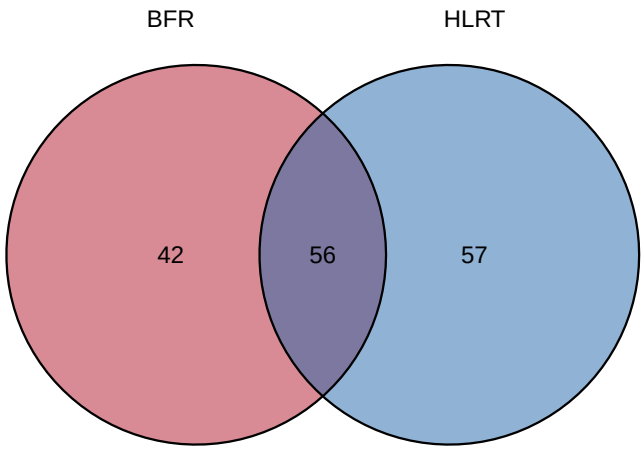
BFR: NES −1.45, padj 0.033
HLRT: NES −1.44, padj 0.048
shared 29, expected 17.9



Shared: ACSS2, ALDH7A1, ATP2A1, ATP2B2, CASQ1, CDKN1B, CRYM, DAG1, DMD, EEA1, EML2, GPX1, GSK3B, GSTM2, JSRP1, LAMB2, MECP2, MYH1, MYH14, MYH7, OAT, PPP3CB, PRKN, SCRN3, STAC3, SYNM, TNNC2, TNNI2, TNNT3

Protein Catabolic Process

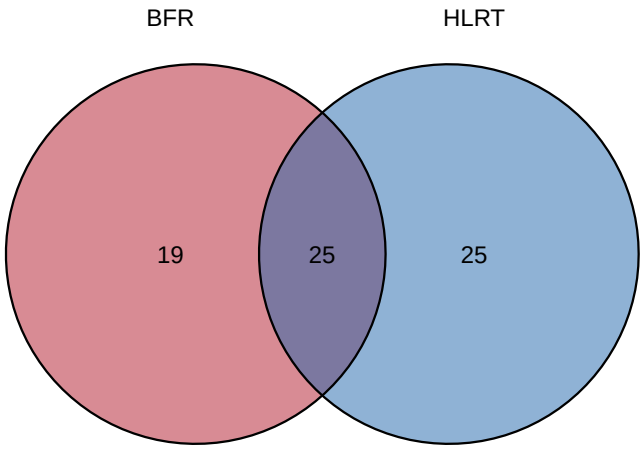
BFR: NES −1.67, padj 0.00013
HLRT: NES −1.39, padj 0.028
shared 56, expected 35.5



Shared: ALAD, ARIH1, ARIH2, ASB11, BAG3, CDK5RAP3, CDKN1B, CHMP1B, CHMP3, CUL5, DNAJB2, ECPAS, ERLIN2, FBXO42, GABARAPL2, GPC1, GPX1, GRN, HSPA8, HUWE1, LAMP2, NPLOC4, NRDC, NSFL1C, PARK7, PIN1, PLAA, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD10, PSMD4, RAB12, RAD23A, RDX, RNF170, SAYSD1, SKP1, SQSTM1, UBE2B, UBQLN1, UCHL3, VPS28, WDR77

Regulation Of Proteolysis Involved In Protein Catabolic Process

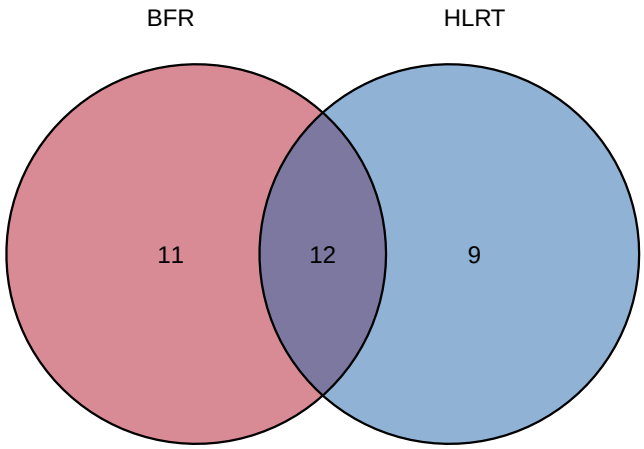
BFR: NES −2.05, padj 4.6e−05
HLRT: NES −1.69, padj 0.0052
shared 25, expected 19.8



Shared: ALAD, CDK5RAP3, DNAJB2, GABARAPL2, GPX1, PARK7, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD10, PSMD4, RAD23A, UBQLN1

Nucleus Organization

BFR: NES −1.73, padj 0.02
HLRT: NES −1.68, padj 0.031
shared 12, expected 9.7

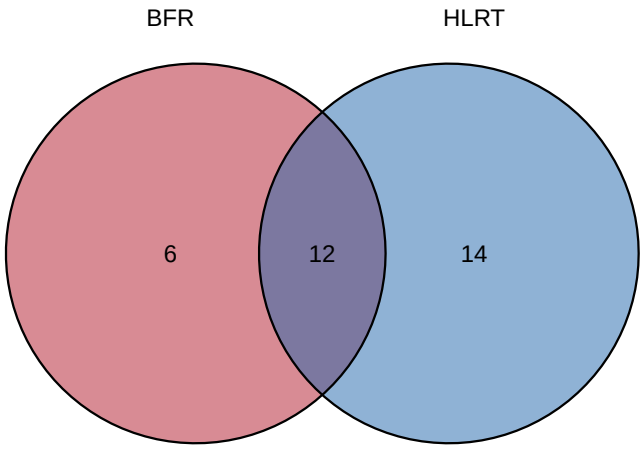


Shared: AGFG1, BANF1, CHMP1B, CHMP3, EMD, FXR1, LMNB2, NSFL1C, RTN4, SUN1, SUN2, TPR

S15. Leading edges by leg

tRNA Aminoacylation

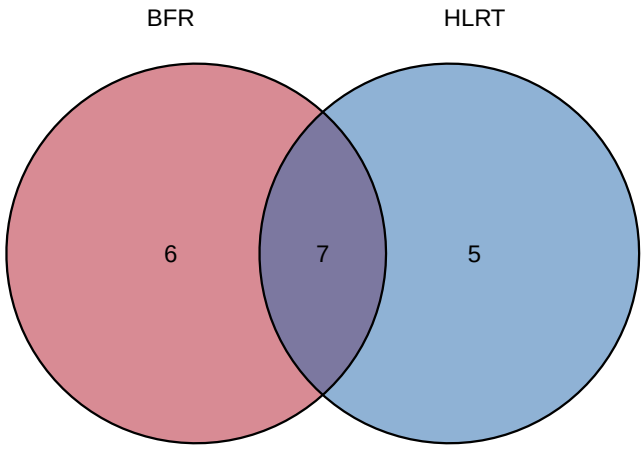
BFR: NES 2.21, padj 7.8e−05
HLRT: NES 2.31, padj 6.8e−05
shared 12, expected 13.0



Shared: DARS2, HARS1, IARS1, IARS2, LARS1, LARS2, NARS1, PPA2, QARS1, SARS1, SARS2, TARS1

Carboxylic Acid Transmembrane Transport

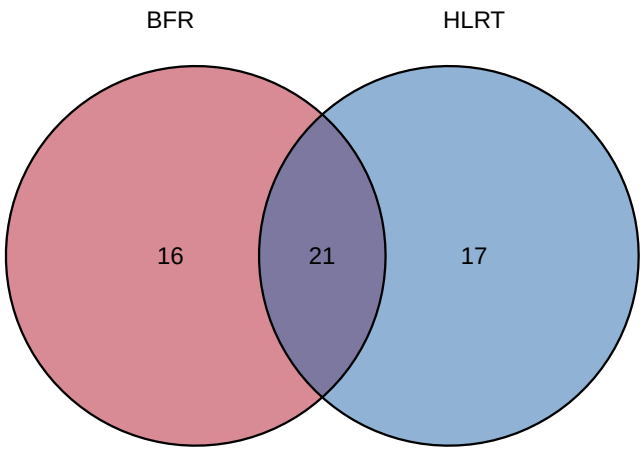
BFR: NES 1.87, padj 0.0083
HLRT: NES 1.70, padj 0.037
shared 7, expected 5.8



Shared: ACSL1, AKT2, MPC1, SFXN1, SLC16A1, SLC16A3, SLC25A12

Antigen Processing Ubiquitination Proteasome Degradation

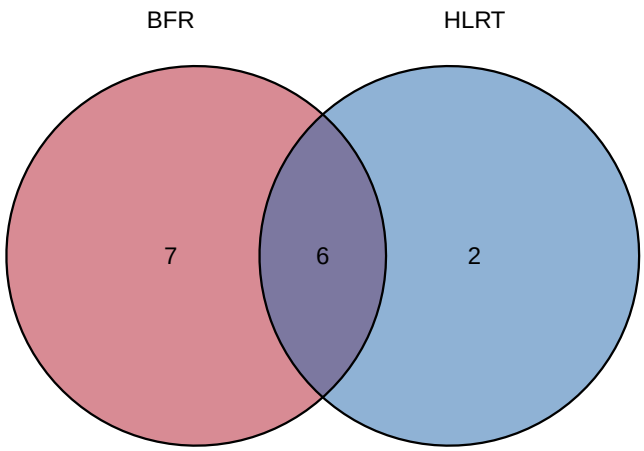
BFR: NES −2.11, padj 2.7e−05
HLRT: NES −2.04, padj 5.4e−05
shared 21, expected 15.0



Shared: ARIH2, ASB11, BLMH, CUL5, HUWE1, NPEPPS, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB6, PSMC2, PSMC5, SKP1, UBE2V2

Regulation Of Protein Import Into Nucleus

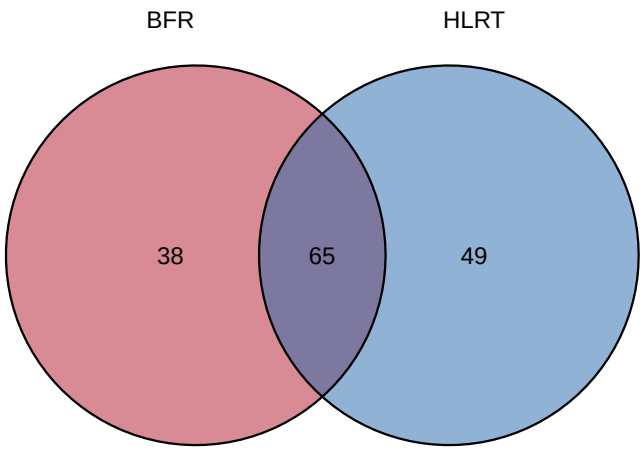
BFR: NES −1.90, padj 0.0079
HLRT: NES −2.08, padj 0.0023
shared 6, expected 5.2



Shared: BAG3, CHP1, PIK3R1, RAN, TPR, YWHAE

Nucleobase−Containing Small Molecule Metabolic Process

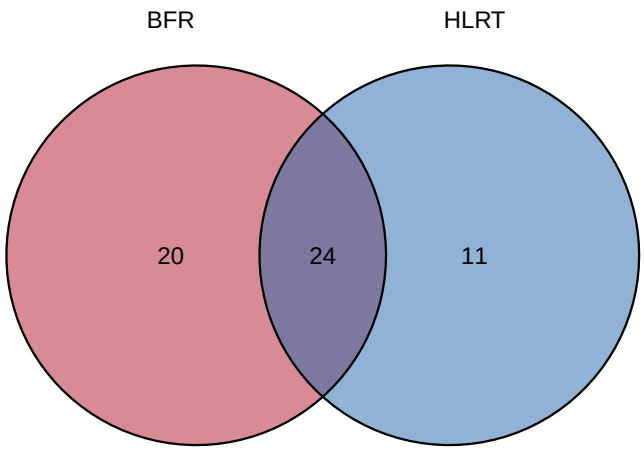
BFR: NES 1.79, padj 4.7e−05
HLRT: NES 1.53, padj 0.0019
shared 65, expected 36.5



Shared: ACOT9, ACSF3, ACSL1, ACSS1, ADSS1, AK2, AK3, ATP5F1A, ATP5F1B, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, BCKDK, CBR4, DCTPP1, DLAT, DLST, FHIT, GFPT1, GOT2, HK1, HSD17B12, MCCC2, MDH2, ME2, MMUT, MTCH2, NADK2, NAMPT, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB9, NDUFS1, NDUFS3, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNMT, NNT, NUDT19, OGDH, OPA1, PCCA, PDHA1, PDHB, PDHX, PDK2, PDK4, PDP1, PDPR, PFKFB1, PRKAA1, SDHA, SLC25A12, STOML2, SUCLA2, SUCLG1, SUCLG2, TSPO, UGDH

Neddylation

BFR: NES −2.19, padj 1.7e−05
HLRT: NES −2.05, padj 6.8e−05
shared 24, expected 18.8

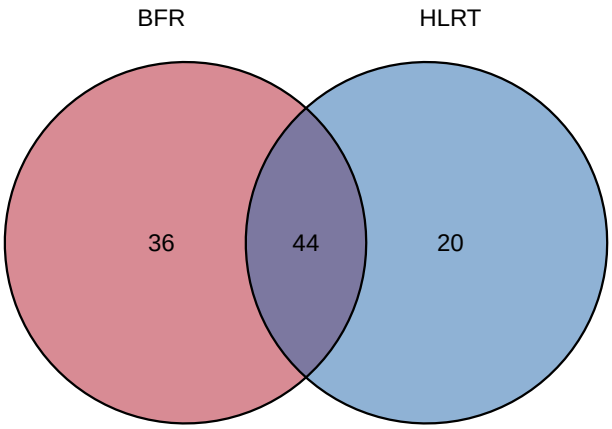


Shared: ASB11, COPS4, COPS7A, COPS7B, CUL1, CUL5, NPLOC4, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7, SKP1, SQSTM1

S15. Leading edges by leg

Mitotic Cell Cycle

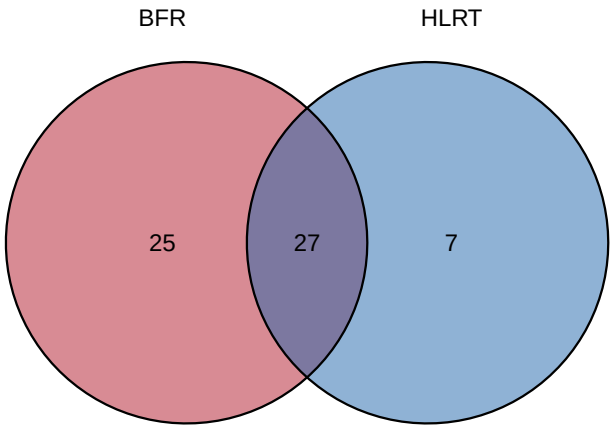
BFR: NES −1.83, padj 7.8e−05
HLRT: NES −1.46, padj 0.045
shared 44, expected 26.4



Shared: BANF1, BCL2, CAMK2A, CDK5RAP3, CDKN1B, CHMP3, CLIP1, CUL1, CUL5, DGKZ, EIF4G1, EML1, HSPA2, MAP3K20, MYH14, NEK7, NSFL1C, PPP2CA, PRMT5, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PTPA, RAN, RDX, RPL24, RPS6KB1, SKP1, STAT5B, SUN2, TACC2, TOM1L2, TPR, TUBB4A, YWHAE

Regulation Of Muscle System Process

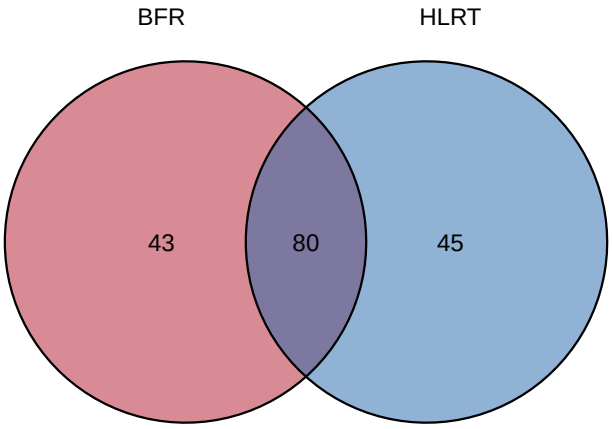
BFR: NES −2.15, padj 2.1e−05
HLRT: NES −2.22, padj 3e−06
shared 27, expected 20.1



Shared: ATP2A1, ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, DAG1, DMD, ENO1, GSTM2, GSTO1, JPH1, MAP2K1, MLIP, MYH7, MYL2, MYL3, MYLK2, PRKAG3, RYR2, TMEM38B, TNNC1, TNNC2, TNNT1, TNNT3, TPM1

Generation Of Precursor Metabolites And Energy

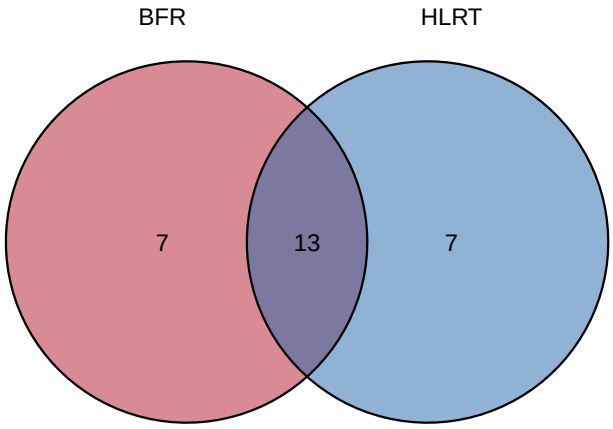
BFR: NES 2.64, padj 5e−19
HLRT: NES 2.46, padj 8.7e−16
shared 80, expected 58.2



Shared: ACO2, AKT2, ATP5F1A, ATP5F1B, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, BCKDK, CHCHD2, CISD1, COQ10A, COX5B, COX6B1, COX7A1, COX7A2, COX7B, CS, CYC1, CYCS, DLAT, DLST, ETFA, ETFB, FH, GAA, GBE1, GFPT1, GHITM, GYS1, HCCS, HK1, HMGCL, IDH3A, IDH3B, IDH3G, LYRM7, MDH2, MTCH2, NDUFA4, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB9, NDUFS1, NDUFS3, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHX, PDK2, PDK4, PDP1, PDPR, PFKFB1, PHKA1, PHKB, PHKG1, PNPT1, PRKAA1, SDHA, SHMT2, SIRT3, SLC25A4, SOD2, STOML2, SUCLA2, SUCLG1, SUCLG2, TRAP1, UQCRC1, UQCRC2, UQCRH

Protein Localization To Mitochondrion

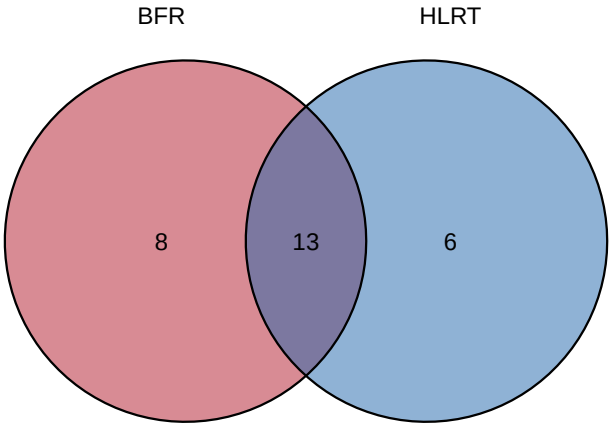
BFR: NES 2.07, padj 0.00035
HLRT: NES 1.91, padj 0.0049
shared 13, expected 8.9



Shared: AIFM1, HK1, HSPD1, MTCH2, MTX1, OXA1L, TIMM13, TIMM21, TIMM44, TIMM50, TIMM8A, TOMM40L, TSPO

Auf1 Hnrnp D0 Binds And Destabilizes mRNA

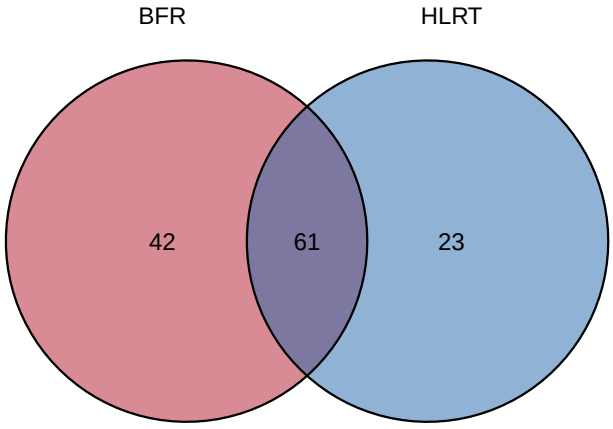
BFR: NES −2.38, padj 3.9e−06
HLRT: NES −1.93, padj 0.0032
shared 13, expected 10.8



Shared: EIF4G1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB6, PSMC2, PSMC5

Muscle System Process

BFR: NES −1.90, padj 1.5e−05
HLRT: NES −1.84, padj 3.4e−05
shared 61, expected 39.7

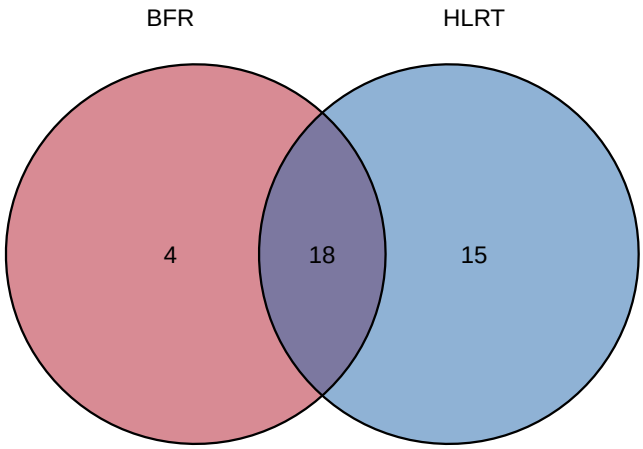


Shared: ABCC9, ACTA1, ATP2A1, ATP2A2, CA3, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, CD38, CERT1, DAG1, DMD, DTNA, EMD, ENO1, GSTM2, GSTO1, HRC, JPH1, JSRP1, LMOD1, MAP2K1, MLIP, MYH1, MYH13, MYH14, MYH4, MYH7, MYL1, MYL2, MYL3, MYL4, MYL5, MYL6B, MYLK2, MYOM2, MYOZ2, PDLIM5, PRKAG3, PVALB, RPL24, RYR1, RYR2, SCN4A, SGCA, SGCD, SNTA1, SSPN, STAC3, SYNM, TCAP, TMEM38B, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TRDN

S15. Leading edges by leg

Protein Polyubiquitination

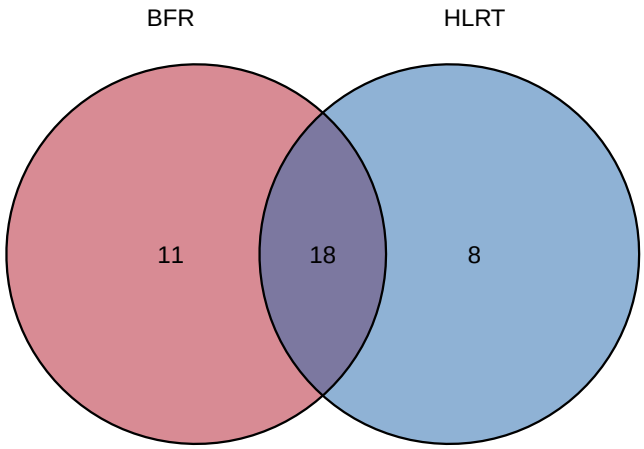
BFR: NES −1.91, padj 0.0013
HLRT: NES −1.96, padj 0.0014
shared 18, expected 13.0



Shared: ARIH2, ASB11, BCL2, BLMH, CUL5, HUWE1, MKRN2, NPEPPS, PLAA, PRKN, RNF126, RNF170, SKP1, UBE2B, UBE2K, UBE2R2, UBE2V2, WDR77

Regulation Of Homotypic Cell Cell Adhesion

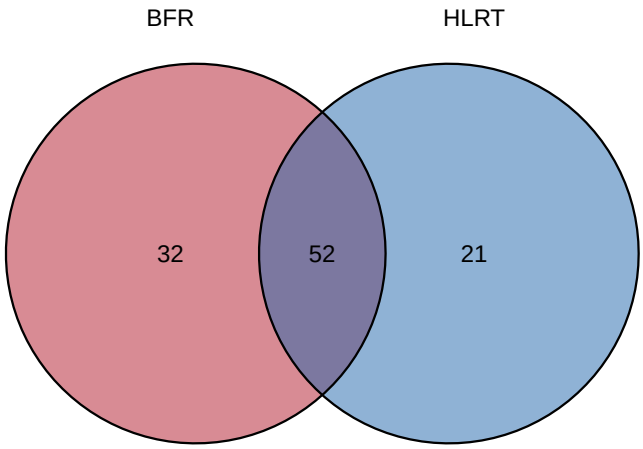
BFR: NES −2.12, padj 4.9e−05
HLRT: NES −1.60, padj 0.037
shared 18, expected 10.9



Shared: AGO2, CTNNA1, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, RBBP7, RPN2

Nucleoside Phosphate Biosynthetic Process

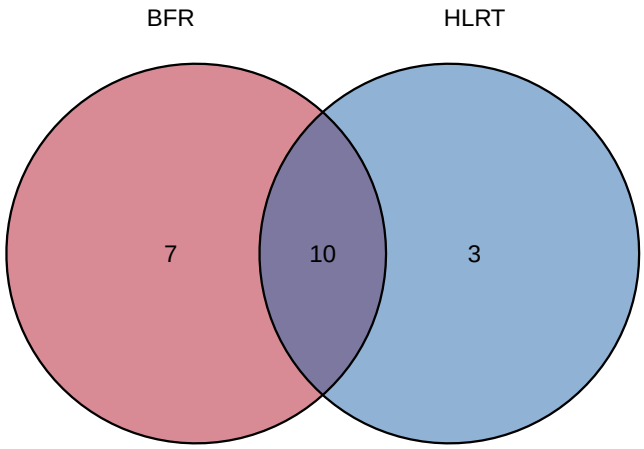
BFR: NES 2.16, padj 2.3e−07
HLRT: NES 1.86, padj 9.9e−05
shared 52, expected 35.2



Shared: ACSF3, ACSL1, ACSS1, ADSS1, AK2, AK3, APRT, ATP5F1A, ATP5F1B, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, BCKDK, CBR4, DLAT, DLST, GFPT1, HK1, HSD17B12, MMUT, NADK2, NAMPT, NDUFA10, NDUFA9, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, OGDH, PCCA, PDHA1, PDHX, PDK2, PDK4, PDP1, PDPR, PPCS, SDHA, STOML2, UGDH

Mitochondrial RNA Metabolic Process

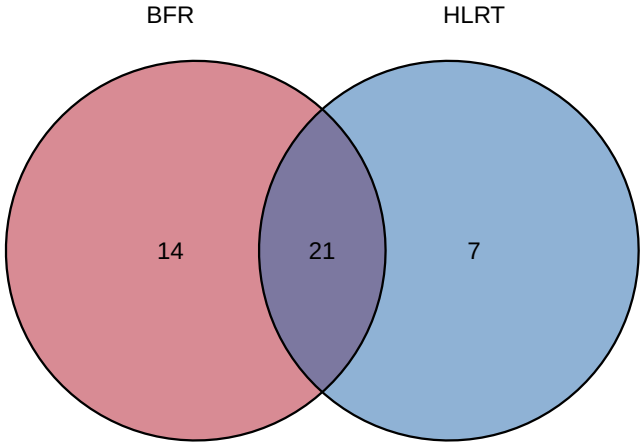
BFR: NES 2.32, padj 8.1e−05
HLRT: NES 2.03, padj 0.0029
shared 10, expected 10.0



Shared: DARS2, GRSF1, HSD17B10, LRPPRC, MRPL12, PNPT1, PRKAA1, SARS2, SLIRP, SUPV3L1

Nuclear Events Mediated By Nfe2l2

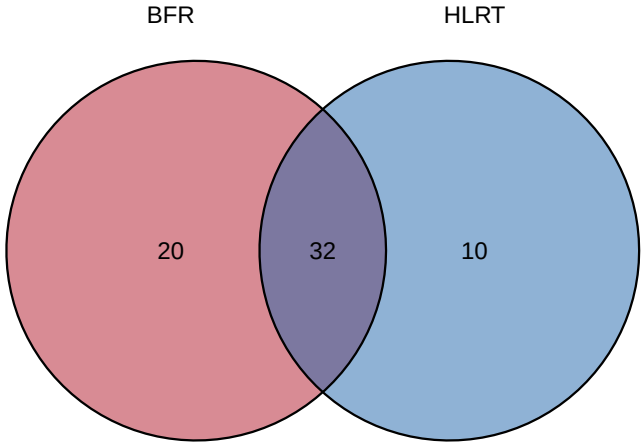
BFR: NES −2.31, padj 4.3e−06
HLRT: NES −2.13, padj 0.00012
shared 21, expected 17.5



Shared: BCL2, CUL1, GSK3B, ME1, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1, SQSTM1, TALDO1

Actomyosin Structure Organization

BFR: NES −1.77, padj 0.0021
HLRT: NES −1.72, padj 0.005
shared 32, expected 23.2

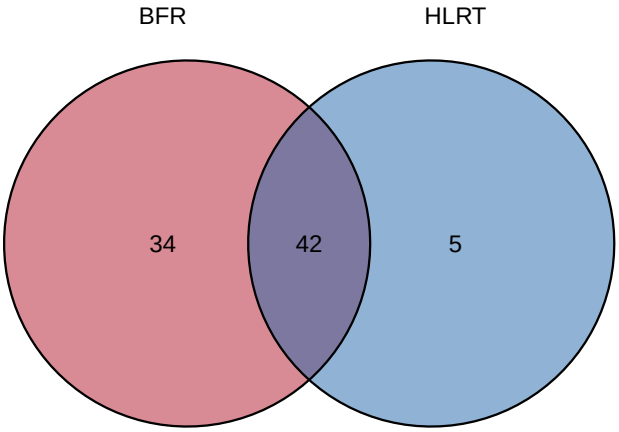


Shared: ACTA1, CASQ1, CD47, CFL2, LDB3, LIMCH1, LMOD1, MYBPC1, MYBPC2, MYH14, MYL2, MYLK3, MYOM2, MYOM3, MYOZ2, MYPN, NEB, PDLIM1, PFN2, PGM5, PIK3R1, PLEC, PXN, RAC1, SYNPO, SYNPO2L, TCAP, TMOD4, TNNT1, TNNT3, TPM1, TTN

S15. Leading edges by leg

Muscle Organ Development

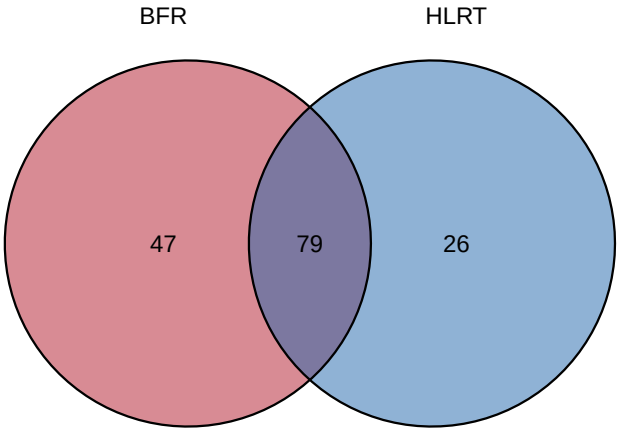
BFR: NES −2.24, padj 1.4e−06
HLRT: NES −2.10, padj 1e−05
shared 42, expected 30.8



Shared: ACTA1, BCL2, CACNA1S, CASQ1, CFL2, CTF1, DAG1, DMD, EMD, FHL3, FXR1, GPC1, GPX1, HDGFL2, ITGA7, JPH1, LAMA2, MAPK12, MAPK14, MYH14, MYH15, MYH7, MYL2, MYL3, MYL6B, MYLK2, NEB, PLEC, PPP3CB, RYR1, RYR2, SGCA, SGCB, SGCD, SGCG, SMTNL1, STAC3, SVIL, TCAP, TNNC1, TPM1, TTN

Aerobic Respiration And Respiratory Electron Transport

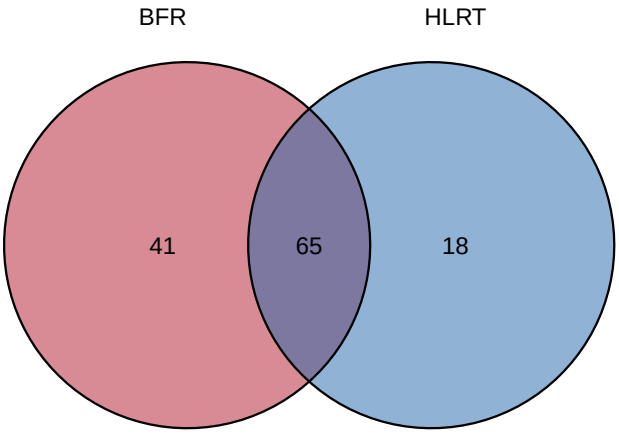
BFR: NES 3.23, padj 2.2e−30
HLRT: NES 2.79, padj 4.1e−20
shared 79, expected 70.7



Shared: ACAD9, ACO2, ATP5F1A, ATP5F1B, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, COQ10A, COX17, COX20, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CS, CYC1, DLAT, DLST, ETFA, ETFB, FAHD1, FH, GOT2, HCCS, HSPA9, IDH3A, IDH3B, IDH3G, LETM1, LYRM7, MDH2, ME2, MPC1, NDUFA10, NDUFA4, NDUFA9, NDUFAF2, NDUFAF3, NDUFAF4, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHX, PDK2, PDK4, PDP1, PDPR, SDHA, SIRT3, SLC25A12, SLC25A4, SMIM20, SUCLA2, SUCLG1, SUCLG2, TACO1, TIMM21, TRAP1, UQCRC1, UQCRC2, UQCRH

Mitochondrion Organization

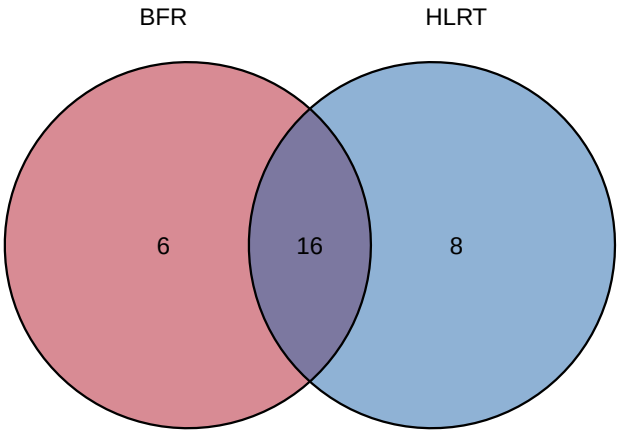
BFR: NES 2.61, padj 4.5e−16
HLRT: NES 2.11, padj 6e−08
shared 65, expected 44.0



Shared: ACAD9, AFG3L2, AIFM1, ATP5F1D, ATP5IF1, ATPAF1, CCDC93, CHCHD10, CHCHD2, CHCHD3, CLPP, COA4, COA6, COA7, COX17, COX20, COX7A1, COX7A2L, DAP3, ENDOG, GHITM, HAX1, HMGCL, HSD17B10, HSPA9, HSPD1, HTRA2, LETM1, LONP1, LYRM7, MICOS13, MTCH2, MTX1, NDUFAF2, NDUFAF3, NDUFAF4, NDUFC2, NDUFS1, NDUFS3, NDUFS6, NDUFS8, OPA1, OPA3, OXA1L, PHB2, PNPT1, POLDIP2, PPIF, PRDX3, SIRT5, SLC25A4, SLIRP, SMIM20, SOD2, SSBP1, STOML2, SUPV3L1, TACO1, TIMM13, TIMM21, TIMM50, TIMM8A, TMEM11, TRIAP1, VDAC2

Abc Family Proteins Mediated Transport

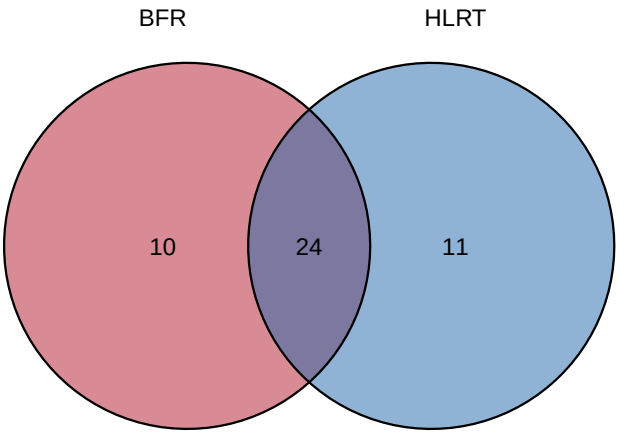
BFR: NES −1.94, padj 0.0027
HLRT: NES −2.14, padj 0.00045
shared 16, expected 11.0



Shared: EIF2S2, ERLIN2, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

MAPK1 MAPK3 Signaling

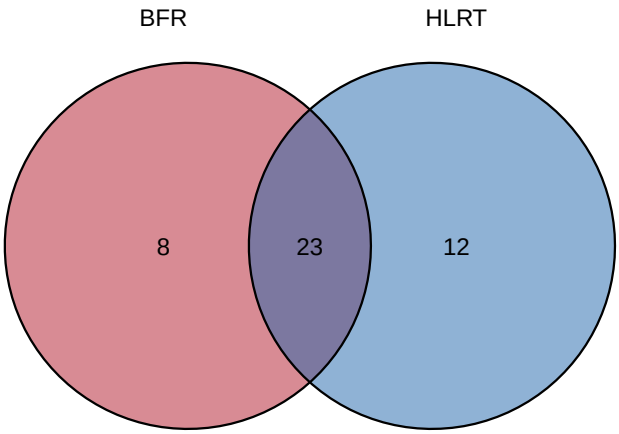
BFR: NES −2.12, padj 2.6e−05
HLRT: NES −1.54, padj 0.047
shared 24, expected 14.3



Shared: CAMK2A, FNTA, GRB2, MAP2K1, MAPK12, PEBP1, PIK3R1, PPP2CA, PPP2R5A, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SPTB

Calcium Ion Transport

BFR: NES −1.97, padj 0.00014
HLRT: NES −1.71, padj 0.0044
shared 23, expected 11.0

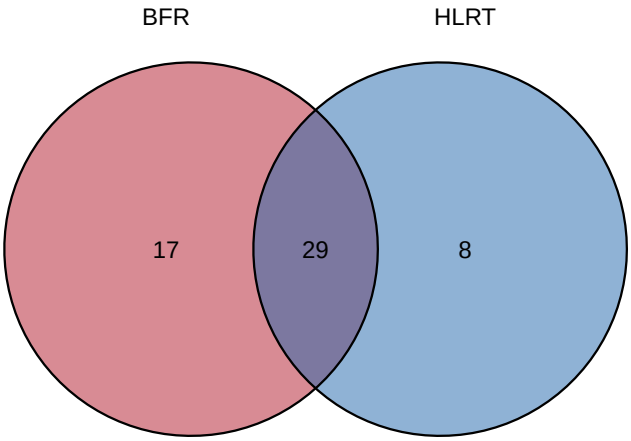


Shared: ATP2A1, ATP2A2, ATP2B2, ATP2C1, BCL2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CAMK2A, CASQ1, DHRS7C, DMD, GSTM2, HRC, JPH1, PPP3CB, PPP3R1, RYR1, STIM1, TMCO1, TPT1, UBQLN1

S15. Leading edges by leg

Mitochondrial Respiratory Chain Complex Assembly

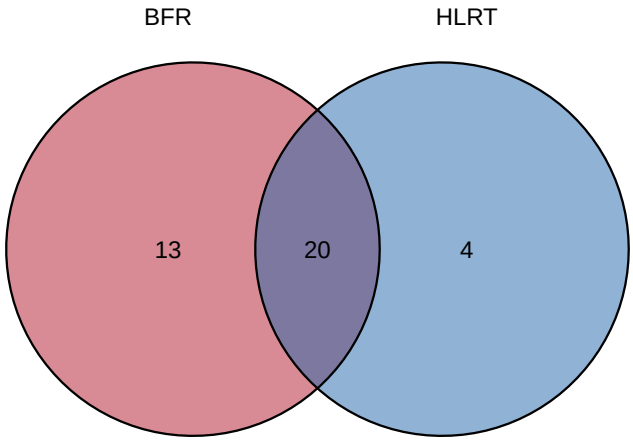
BFR: NES 2.86, padj 6.2e−13
HLRT: NES 2.34, padj 3.2e−06
shared 29, expected 25.8



Shared: ACAD9, AIFM1, ATP5F1D, ATPAF1, COA4, COA6, COA7, COX17, COX20, COX7A1, COX7A2L, LYRM7, MT−ND2, NDUFAF2, NDUFAF3, NDUFAF4, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, OXA1L, SDHAF2, SMIM20, TACO1, TIMM21, TMEM70, UQCRFS1

Monocarboxylic Acid Catabolic Process

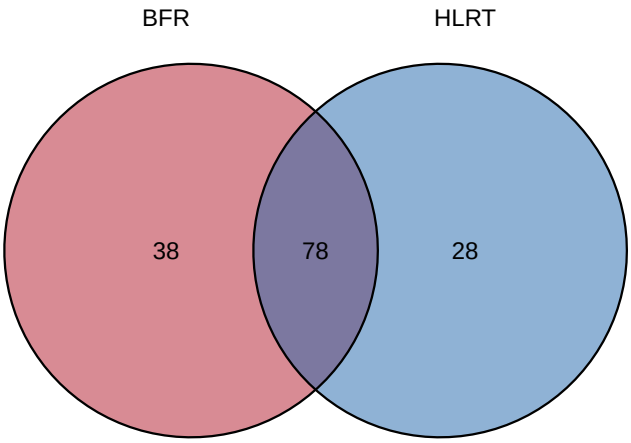
BFR: NES 2.24, padj 4e−05
HLRT: NES 1.86, padj 0.0036
shared 20, expected 13.7



Shared: AKT2, AUH, BDH2, CRAT, DECR1, DLAT, ECH1, ECI1, ECI2, ETFA, ETFB, HSD17B10, ILVBL, LDHD, LPIN1, NUDT19, PCCA, PCCB, SLC16A1, SLC16A3

Cellular Respiration

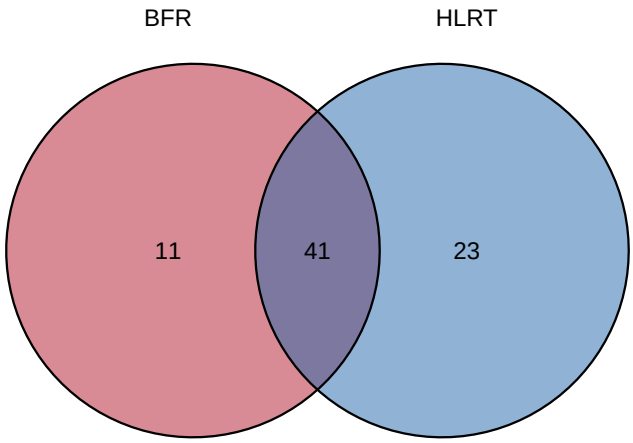
BFR: NES 2.72, padj 5.3e−18
HLRT: NES 2.59, padj 9.2e−16
shared 78, expected 61.2



Shared: ACO2, ATP5F1A, ATP5F1B, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, BCKDK, CHCHD2, CISD1, COQ10A, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CS, CYC1, CYCS, DLAT, DLST, ETFA, ETFB, FH, GHITM, HCCS, HK1, IDH3A, IDH3B, IDH3G, LYRM7, MDH2, MTCH2, NDUFA10, NDUFA4, NDUFA9, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OXA1L, PDHA1, PDHX, PDK2, PDK4, PDP1, PDPR, PFKFB1, PNPT1, PPIF, PRKAA1, SDHA, SHMT2, SIRT3, SOD2, STOML2, SUCLA2, SUCLG1, SUCLG2, TRAP1, UQCRC1, UQCRC2, UQCRH

Mitochondrial Translation

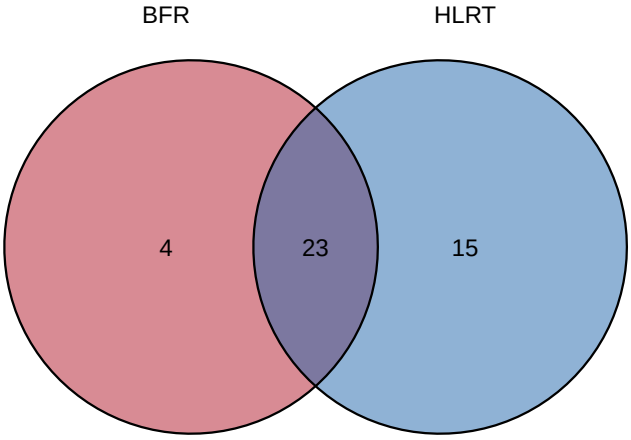
BFR: NES 2.80, padj 1.2e−12
HLRT: NES 2.65, padj 2.6e−11
shared 41, expected 40.1



Shared: DAP3, GFM1, MALSU1, MRPL1, MRPL10, MRPL11, MRPL12, MRPL13, MRPL14, MRPL15, MRPL2, MRPL21, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL48, MRPL50, MRPL53, MRPL9, MRPS10, MRPS11, MRPS18B, MRPS21, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS35, MRPS9, MRRF, MT−CO3, MT−CYB, MT−ND2, OXA1L, PTCD3, TSFM, TUFM

Mitotic Cell Cycle Phase Transition

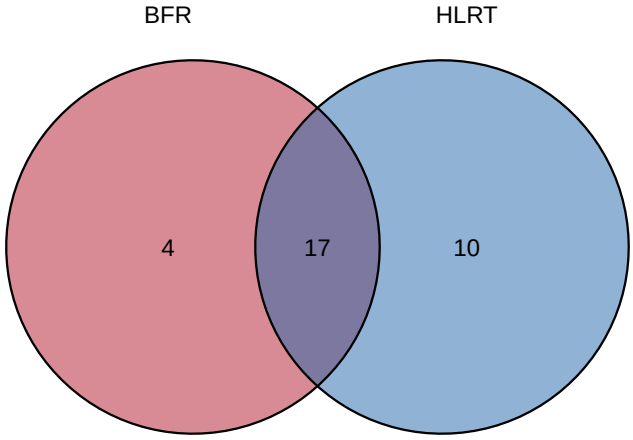
BFR: NES −1.76, padj 0.0026
HLRT: NES −1.76, padj 0.0042
shared 23, expected 11.5



Shared: BCL2, CAMK2A, CDK5RAP3, CDKN1B, CUL5, DGKZ, EIF4G1, HSPA2, MAP3K20, PPP2CA, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB6, RDX, RPL24, SKP1

Protein Modification By Small Protein Removal

BFR: NES −1.66, padj 0.036
HLRT: NES −1.63, padj 0.037
shared 17, expected 13.8

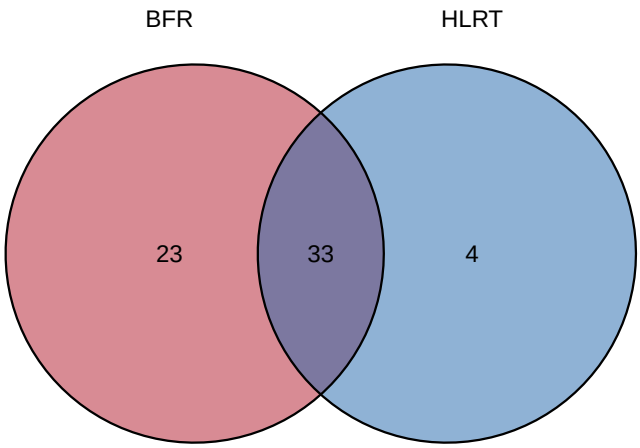


Shared: BRCC3, COPS2, COPS4, COPS6, COPS7A, COPS7B, COPS8, DNAJB2, HINT1, OTUB1, PARK7, PRKN, UCHL3, USP14, USP19, USP24, VCPIP1

S15. Leading edges by leg

Muscle Cell Development

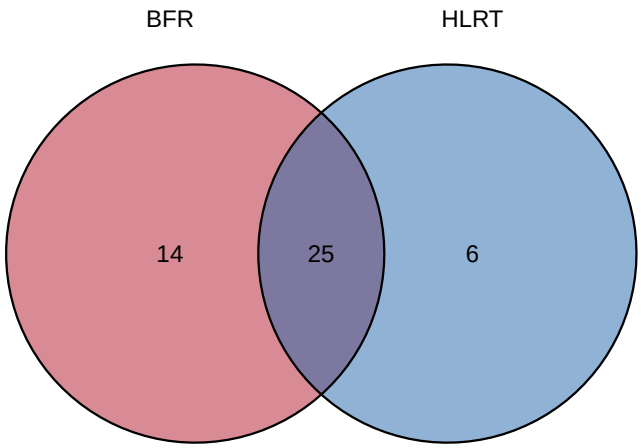
BFR: NES −2.06, padj 4.8e−05
HLRT: NES −1.98, padj 0.00018
shared 33, expected 23.0



Shared: ACTA1, ACTN3, ATP2A2, BCL2, CACNA1S, CASQ1, CFL2, DMD, GPX1, LMOD1, MYBPC1, MYBPC2, MYL2, MYLK3, MYOM2, MYOM3, NEB, PDLIM5, PGM5, PLEC, PPP3CB, RYR1, SGCB, SGCD, STAC3, SYNPO2L, SYPL2, TCAP, TMOD4, TNNT1, TNNT3, TPM1, TTN

Regulation Of Muscle Contraction

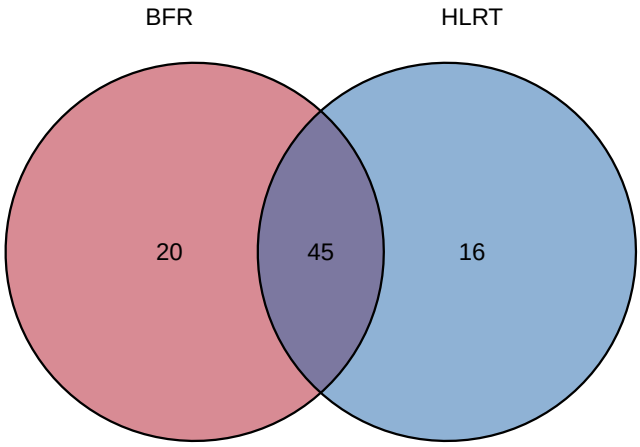
BFR: NES −2.03, padj 0.00021
HLRT: NES −2.30, padj 1.1e−05
shared 25, expected 18.6



Shared: ATP2A1, ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, DMD, ENO1, GSTM2, GSTO1, HRC, JPH1, MAP2K1, MYH7, MYL2, MYL3, MYLK2, RYR2, TMEM38B, TNNC1, TNNC2, TNNT1, TNNT3, TPM1

Myogenesis

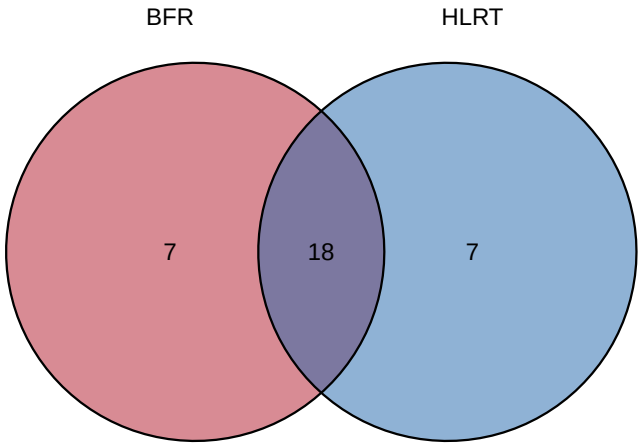
BFR: NES −1.63, padj 0.009
HLRT: NES −1.81, padj 0.00087
shared 45, expected 33.9



Shared: ACTA1, ATP2A1, CACNG1, CASQ1, CKB, COL15A1, COL4A2, COL6A2, CTF1, DMD, DTNA, GABARAPL2, HRC, HSPB8, ITGA7, LAMA2, LDB3, MAPK12, MYH1, MYH2, MYH4, MYH7, MYL1, MYL2, MYL3, MYL4, MYL6B, MYOM2, PVALB, RYR1, SGCA, SGCD, SGCG, SH3BGR, SSPN, SVIL, SYNGR2, TCAP, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TPD52L1

Regulation Of Monoatomic Ion Transmembrane Transport

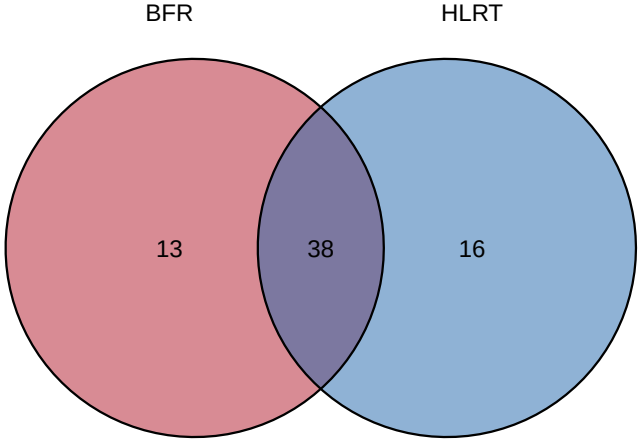
BFR: NES −1.96, padj 0.00017
HLRT: NES −1.71, padj 0.0096
shared 18, expected 7.5



Shared: ATP2A1, BCL2, CAB39, CACNA2D1, CACNB1, CACNG1, CASQ1, CHP1, DHRS7C, DMD, GSTM2, JPH1, PPP3CB, PPP3R1, SNTA1, STAC3, STIM1, UBQLN1

Mitochondrial Protein Degradation

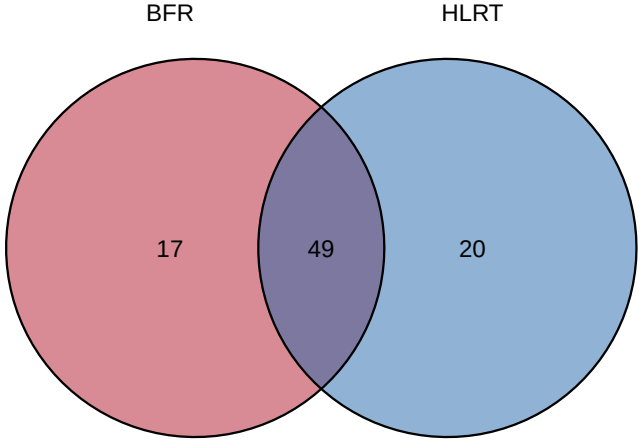
BFR: NES 3.08, padj 5e−19
HLRT: NES 2.70, padj 2.4e−12
shared 38, expected 32.4



Shared: ACO2, AFG3L2, ALDH1B1, ATP5F1A, ATP5F1B, ATP5MG, ATP5PD, CHCHD2, CLPP, COX5B, CS, ECH1, ECI1, FH, GLUD1, HSD17B10, HSPA9, HSPD1, HTRA2, IARS2, IDH3A, LDHD, LONP1, MDH2, ME2, NADK2, NDUFS1, NDUFS3, NDUFV1, OGDH, OPA1, PCCB, PDHA1, PMPCA, SHMT2, SSBP1, SUCLG2, UQCRC2

Mitochondrial Gene Expression

BFR: NES 3.02, padj 8.6e−19
HLRT: NES 2.88, padj 1.3e−15
shared 49, expected 47.4

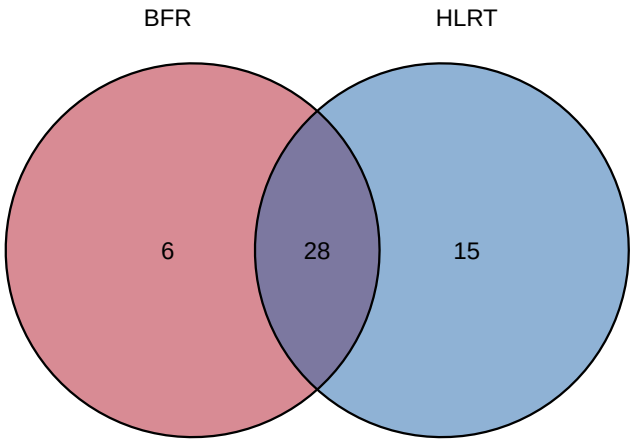


Shared: DAP3, DARS2, GFM1, HSD17B10, IARS2, LARS2, LRPPRC, MALSU1, MRPL1, MRPL10, MRPL11, MRPL12, MRPL13, MRPL14, MRPL2, MRPL21, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL48, MRPL50, MRPL53, MRPL9, MRPS11, MRPS18B, MRPS21, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS35, MRPS9, OXA1L, PNPT1, PRKAA1, PTCD3, SARS2, SHMT2, SLIRP, SUPV3L1, TACO1, TBRG4, TRMT10C, TSFM, TUFM

S15. Leading edges by leg

Fatty Acid Metabolism

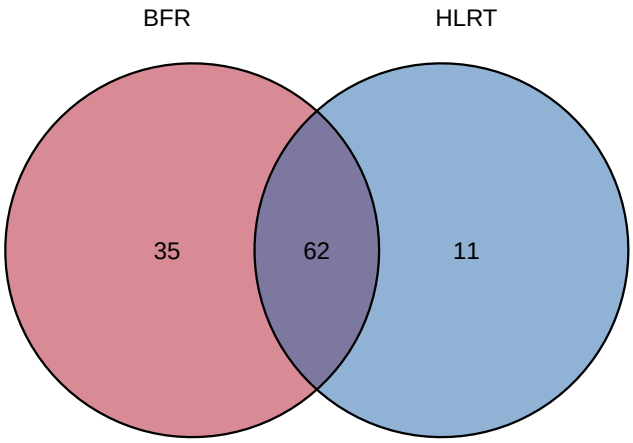
BFR: NES 1.99, padj 7.2e−05
HLRT: NES 1.61, padj 0.017
shared 28, expected 14.6



Shared: ACO2, ACSL1, ACSS1, AUH, BPHL, CPOX, CRAT, DECR1, DLST, ECH1, ECI1, ECI2, FH, HCCS, HIBCH, HMGCL, HSD17B10, HSDL2, IDH3B, IDH3G, MDH2, PDHA1, PDHB, SDHA, SUCLA2, SUCLG1, SUCLG2, UROD

Muscle Contraction

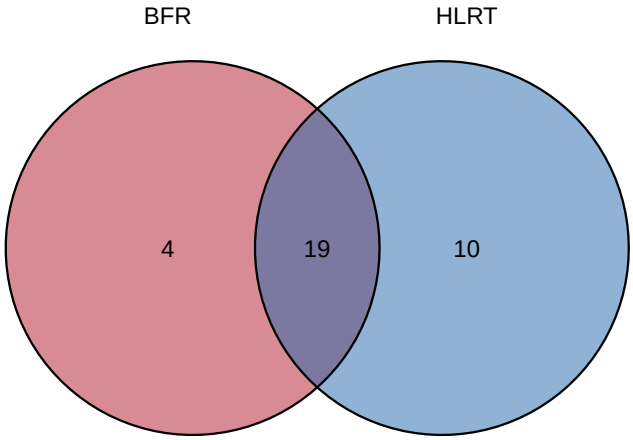
BFR: NES −1.83, padj 8.9e−05
HLRT: NES −1.92, padj 3.7e−05
shared 62, expected 38.5



Shared: ACTA1, ACTN3, ATP2A1, ATP2A2, CA3, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, CD38, CERT1, DMD, DTNA, EMD, ENO1, FKBP1A, GSTM2, GSTO1, HRC, JPH1, JSRP1, LMOD1, MAP2K1, MYH1, MYH13, MYH14, MYH4, MYH7, MYH8, MYL1, MYL2, MYL3, MYL4, MYL5, MYL6B, MYLK2, MYOM2, PLN, RPL24, RYR1, RYR2, SCN4A, SGCA, SGCD, SNTA1, SSPN, STAC3, SYNM, TCAP, TMEM38B, TMOD1, TMOD4, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TPM1, TPM3, TRDN, TTN

Cell Cycle G1 S Phase Transition

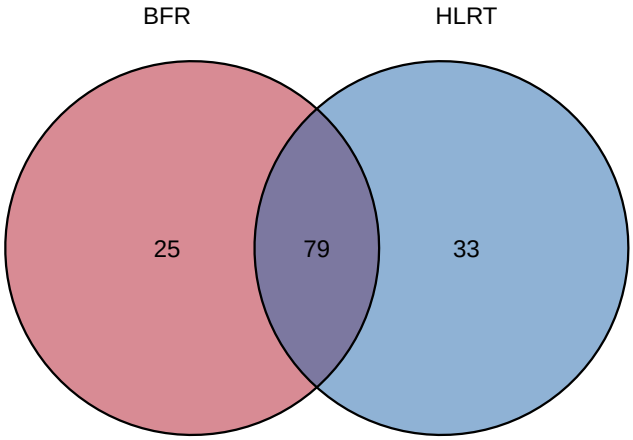
BFR: NES −1.98, padj 0.00052
HLRT: NES −1.97, padj 0.0011
shared 19, expected 12.4



Shared: BCL2, CAMK2A, CDKN1B, CUL5, DGKZ, EIF4G1, PPP2CA, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB6, RDX, SKP1

OXPHOS

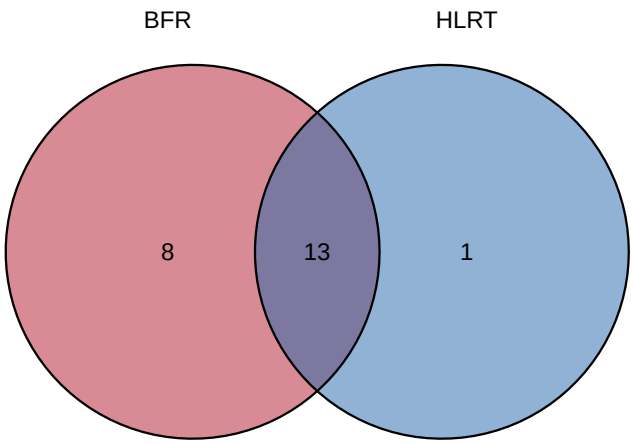
BFR: NES 3.11, padj 6.4e−26
HLRT: NES 2.84, padj 6.5e−20
shared 79, expected 66.9



Shared: ABCB7, ACO2, AFG3L2, AIFM1, ATP5F1A, ATP5F1B, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, BDH2, COX17, COX5B, COX6B1, COX7A2, COX7B, CS, CYC1, CYCS, DECR1, DLAT, DLST, ECH1, ECI1, ETFA, ETFB, FH, GLUD1, GOT2, GRPEL1, HCCS, HSD17B10, HSPA9, HTRA2, IDH3A, IDH3B, IDH3G, LRPPRC, MDH2, MPC1, MRPL11, MRPS11, NDUFA4, NDUFA9, NDUFB5, NDUFB7, NDUFC2, NDUFS1, NDUFS3, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, OGDH, OPA1, OXA1L, PDHA1, PDHB, PDHX, PDK4, PDP1, PHB2, PMPCA, PRDX3, SDHA, SLC25A12, SLC25A3, SLC25A4, SUCLA2, SUCLG1, SUPV3L1, TIMM13, TIMM50, UQCRC1, UQCRC2, UQCRFS1, UQCRH, VDAC2

Tricarboxylic Acid Cycle

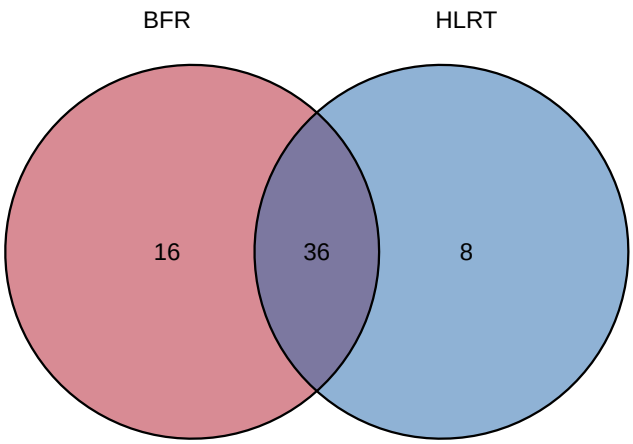
BFR: NES 2.69, padj 5.6e−08
HLRT: NES 2.08, padj 0.0009
shared 13, expected 10.5



Shared: ACO2, CS, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, NNT, OGDH, PDHA1, SUCLA2, SUCLG1

Fatty Acid Metabolic Process

BFR: NES 1.87, padj 0.00026
HLRT: NES 1.66, padj 0.0048
shared 36, expected 16.7

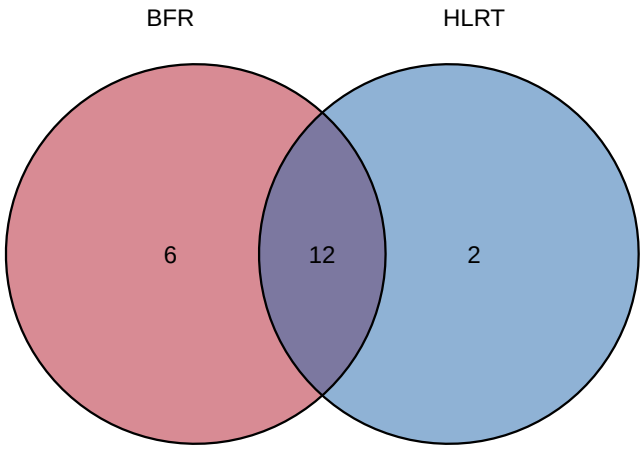


Shared: ACAD9, ACOT9, ACSF3, ACSL1, ACSS1, AKR1C3, AKT2, AUH, BCKDK, BDH2, CBR1, CBR4, CCDC93, CES2, CRAT, DECR1, ECH1, ECI2, ETFA, ETFB, HSD17B10, HSD17B12, HSD17B8, ILVBL, MECR, MMUT, NDUFS6, NUDT19, PCCA, PCCB, PDK2, PDK4, PDP1, PDPR, PRKAA1, PTGES2

S15. Leading edges by leg

Citrate Cycle TCA Cycle

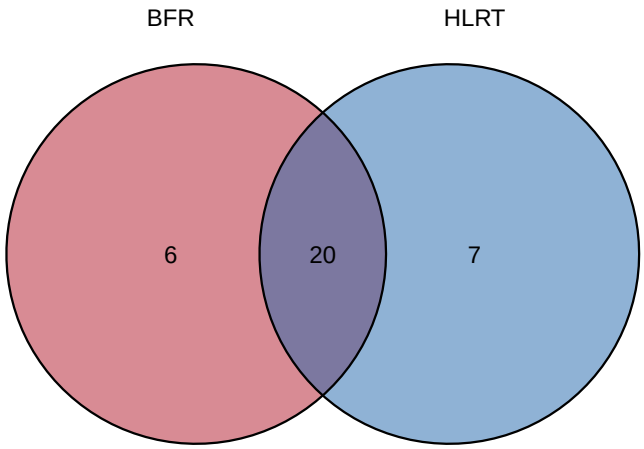
BFR: NES 2.64, padj 5.7e−08
HLRT: NES 2.09, padj 0.00063
shared 12, expected 9.7



Shared: ACO2, CS, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, OGDH, PDHA1, SUCLA2, SUCLG1

Signaling By The B Cell Receptor Bcr

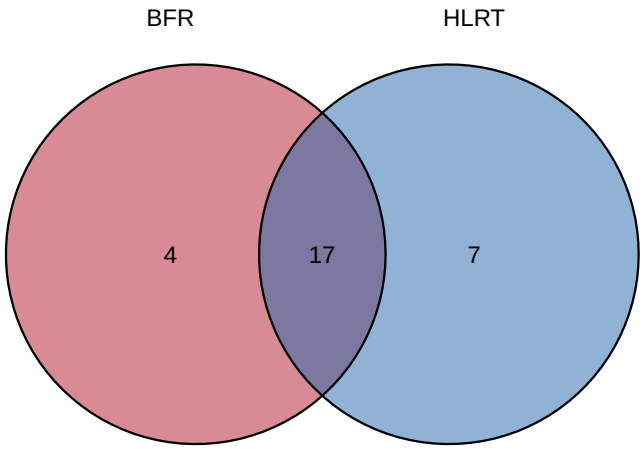
BFR: NES −2.25, padj 4.6e−05
HLRT: NES −2.27, padj 8.5e−05
shared 20, expected 14.6



Shared: GRB2, PIK3R1, PPP3CB, PPP3R1, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SKP1, STIM1

Dilated Cardiomyopathy

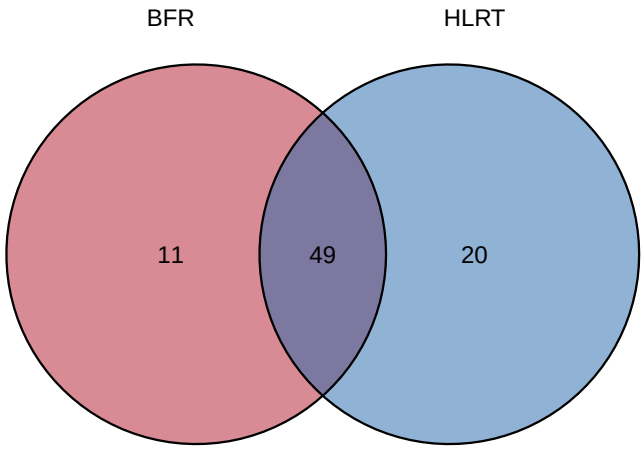
BFR: NES −2.00, padj 0.00094
HLRT: NES −2.51, padj 1.8e−07
shared 17, expected 14.0



Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, DAG1, DMD, EMD, ITGA7, MYH7, MYL2, MYL3, RYR2, SGCA, SGCB, SGCD, SGCG

Huntingtons Disease

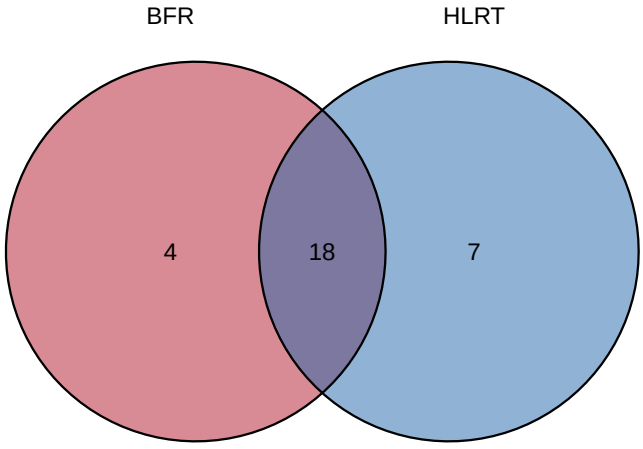
BFR: NES 2.48, padj 4.2e−09
HLRT: NES 2.54, padj 3.6e−10
shared 49, expected 40.6



Shared: AP2B1, ATP5F1A, ATP5F1B, ATP5F1C, ATP5F1D, ATP5PB, ATP5PD, ATP5PO, BAX, CASP3, COX4I1, COX5A, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, CYCS, MT−CO3, NDUFA10, NDUFA4, NDUFA9, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, SDHA, SLC25A4, SOD2, UQCRC1, UQCRC2, UQCRFS1, UQCRH, VDAC2

Hypertrophic Cardiomyopathy Hcm

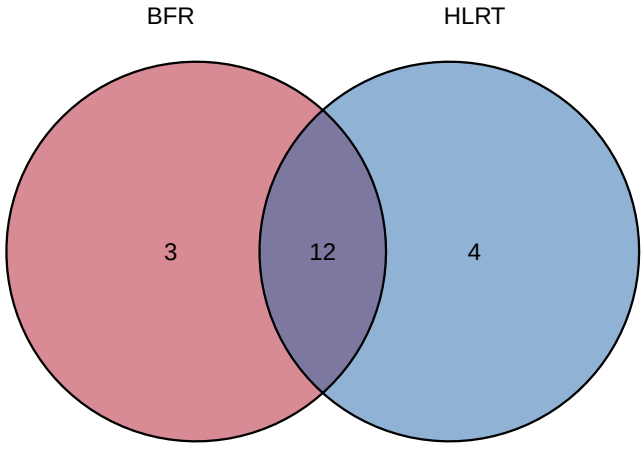
BFR: NES −1.92, padj 0.0012
HLRT: NES −2.41, padj 7.3e−06
shared 18, expected 14.5



Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, DAG1, DMD, EMD, ITGA7, MYH7, MYL2, MYL3, PRKAG3, RYR2, SGCA, SGCB, SGCD, SGCG

Arrhythmogenic Right Ventricular Cardiomyopathy Arvc

BFR: NES −2.16, padj 0.00021
HLRT: NES −2.29, padj 8.5e−05
shared 12, expected 8.3

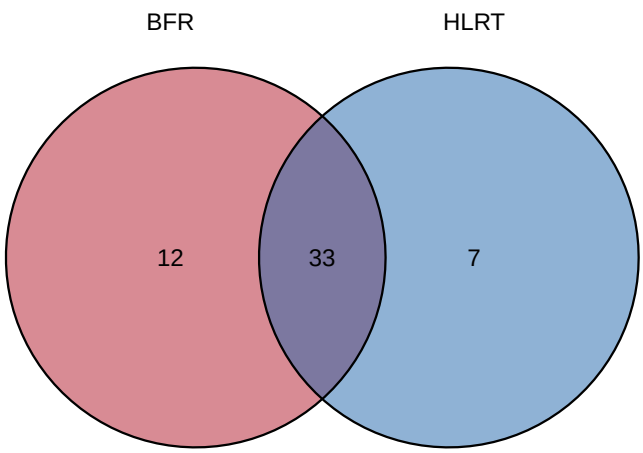


Shared: ATP2A2, CACNA1S, CACNA2D1, CACNB1, CACNG1, DAG1, DMD, ITGA7, SGCA, SGCB, SGCD, SGCG

S15. Leading edges by leg

Protein Localization

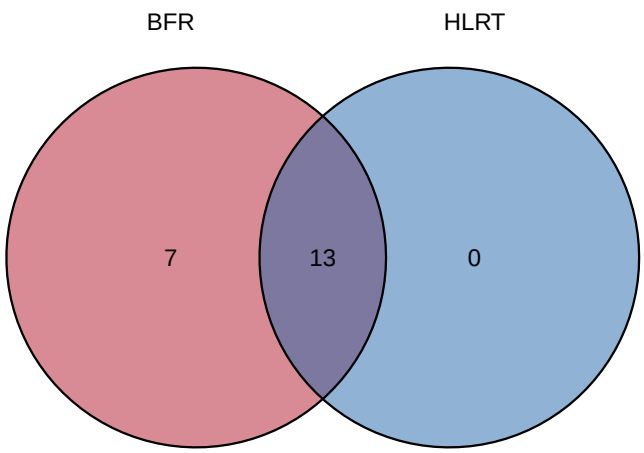
BFR: NES 2.65, padj 1.6e−11
HLRT: NES 2.02, padj 0.00012
shared 33, expected 19.6



Shared: ACO2, ATP5F1A, ATP5F1B, CHCHD10, CHCHD2, CHCHD3, CMC2, COA4, COA6, COX17, CRAT, CS, CYC1, ECH1, ECI2, EPHX2, GSTK1, HMGCL, HSPA9, HSPD1, IDH3G, LDHD, MTX1, NUDT19, PITRM1, PMPCA, SLC25A12, SLC25A4, TIMM13, TIMM21, TIMM44, TIMM50, TIMM8A

Citric Acid Cycle TCA Cycle

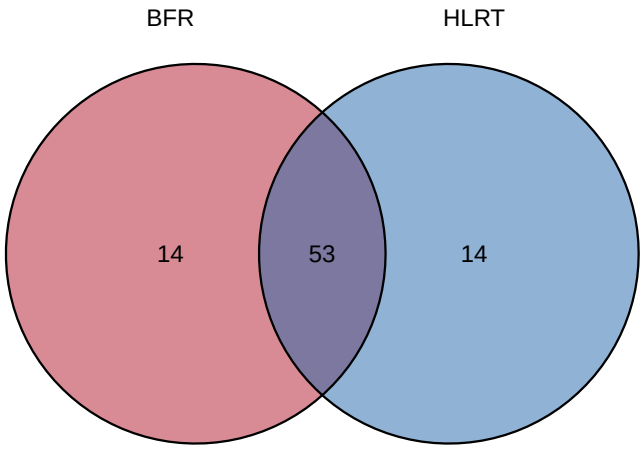
BFR: NES 2.72, padj 5.9e−09
HLRT: NES 2.32, padj 2.2e−05
shared 13, expected 9.0



Shared: ACO2, CS, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, NNT, OGDH, SUCLA2, SUCLG1, TRAP1

Proton Transmembrane Transport

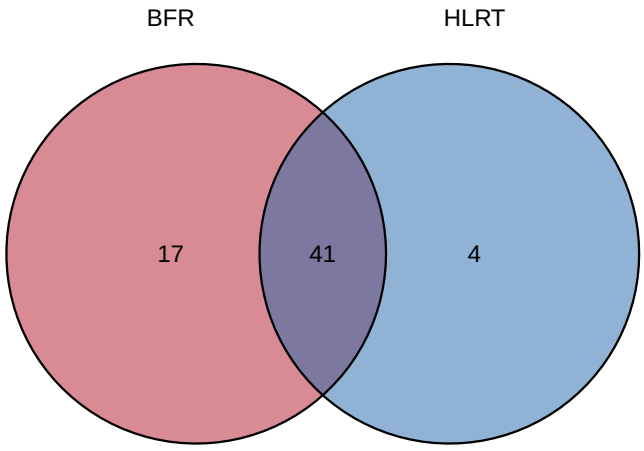
BFR: NES 2.69, padj 2.1e−12
HLRT: NES 2.59, padj 8.4e−11
shared 53, expected 48.8



Shared: ATP1A1, ATP5F1A, ATP5F1B, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, ATP5PO, COX4I1, COX5A, COX5B, COX6B1, COX7A1, COX7A2L, COX7B, CYC1, GHITM, LETM1, MT−CO3, MT−ND2, NDUFA10, NDUFA12, NDUFA4, NDUFA8, NDUFA9, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, NNT, PHB2, SLC16A1, SLC25A12, SLC25A3, SLC25A4, UQCRC1, UQCRFS1, UQCRH

Muscle Contraction

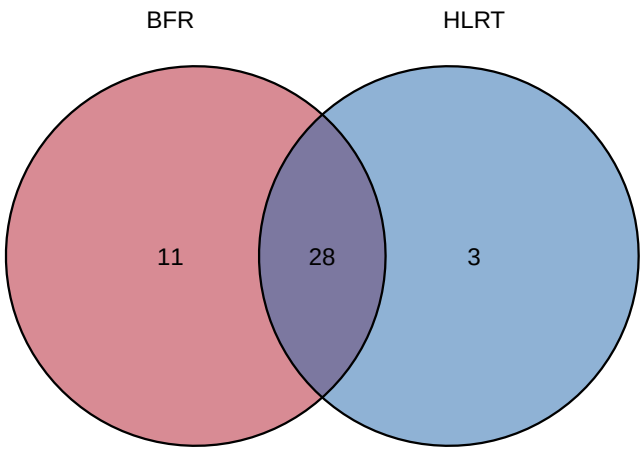
BFR: NES −2.04, padj 7.8e−05
HLRT: NES −1.81, padj 0.0022
shared 41, expected 30.3



Shared: ABCC9, ACTA1, ACTN3, ALDH2, ATP2A1, ATP2A2, ATP2B2, CACNB1, CAMK2A, CASQ1, DMD, LMOD1, MYBPC1, MYBPC2, MYH8, MYL1, MYL2, MYL3, MYL4, MYL5, MYL6B, NEB, PLN, PXN, RYR1, RYR2, SCN4A, STIM1, TCAP, TMOD1, TMOD4, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TPM1, TPM3, TRDN, TTN

Cellular Component Assembly Involved In Morphogenesis

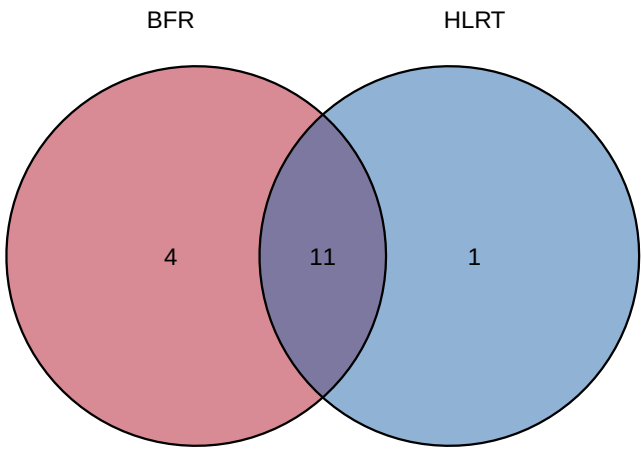
BFR: NES −1.99, padj 0.00045
HLRT: NES −1.97, padj 0.0021
shared 28, expected 20.8



Shared: ACTA1, AGFG1, CASQ1, CFL2, GNPAT, GPC1, LDB3, LMOD1, MYBPC1, MYBPC2, MYL2, MYLK3, MYOM2, MYOM3, MYOZ2, MYPN, NEB, PGM5, PLEC, PLN, SYNPO2L, TCAP, TMOD1, TMOD4, TNNT1, TNNT3, TPM1, TTN

T Cell Homeostasis

BFR: NES −2.22, padj 0.00019
HLRT: NES −2.05, padj 0.0035
shared 11, expected 8.2

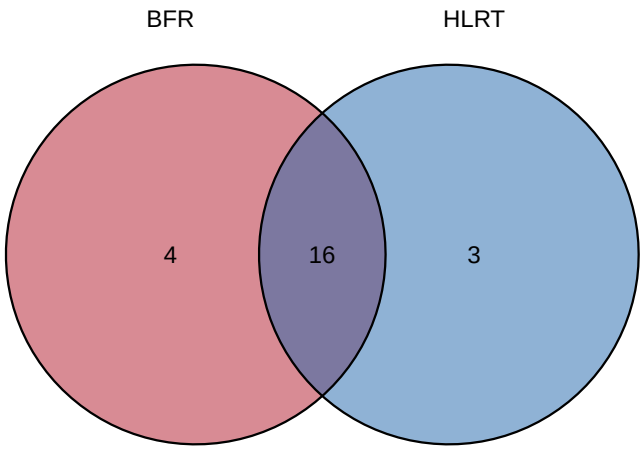


Shared: BCL2, PPP2CA, PPP3CB, PSMA3, PSMA4, PSMA5, PSMB1, PSMB2, PSMB3, PSMB4, STAT5B

S15. Leading edges by leg

Meiotic Cell Cycle

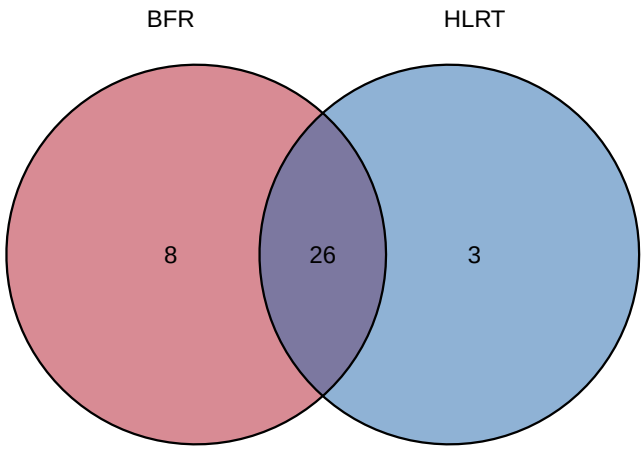
BFR: NES −1.79, padj 0.0058
HLRT: NES −1.76, padj 0.027
shared 16, expected 10.0



Shared: HSPA2, PPP2CA, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, RPA1, SUN2

Mitochondrial Protein Import

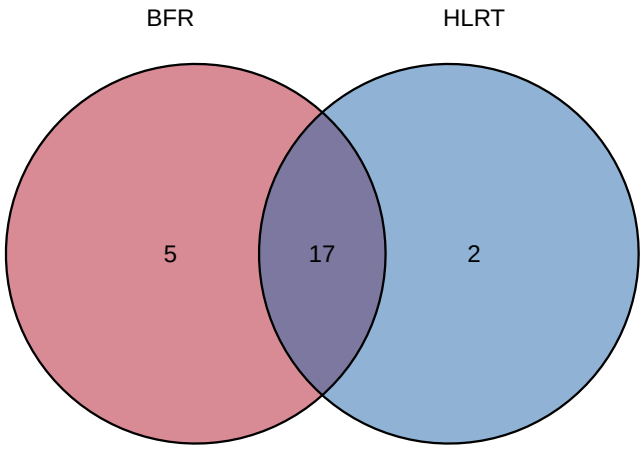
BFR: NES 2.92, padj 2.6e−13
HLRT: NES 2.39, padj 8.5e−07
shared 26, expected 19.0



Shared: ACO2, ATP5F1A, ATP5F1B, CHCHD10, CHCHD2, CHCHD3, CMC2, COA4, COA6, COX17, CS, CYC1, HSPA9, HSPD1, IDH3G, LDHD, MTX1, PITRM1, PMPCA, SLC25A12, SLC25A4, TIMM13, TIMM21, TIMM44, TIMM50, TIMM8A

Cristae Formation

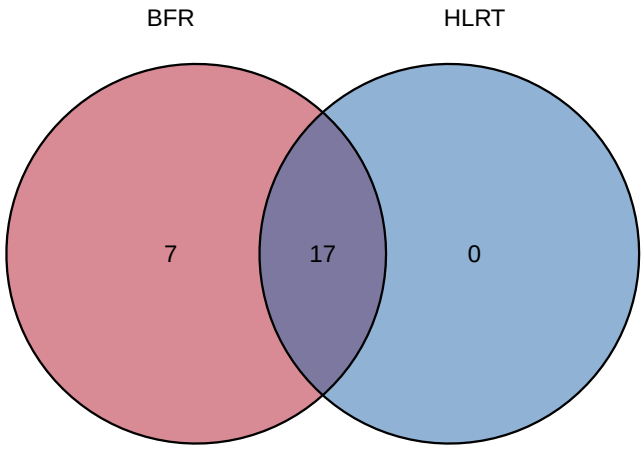
BFR: NES 2.31, padj 7.1e−05
HLRT: NES 1.80, padj 0.014
shared 17, expected 16.1



Shared: APOO, ATP5F1A, ATP5F1B, ATP5F1C, ATP5F1D, ATP5ME, ATP5MF, ATP5MG, ATP5PB, ATP5PD, ATP5PO, CHCHD3, HSPA9, MICOS13, MTX1, SAMM50, TMEM11

Non Integrin Membrane ECM Interactions

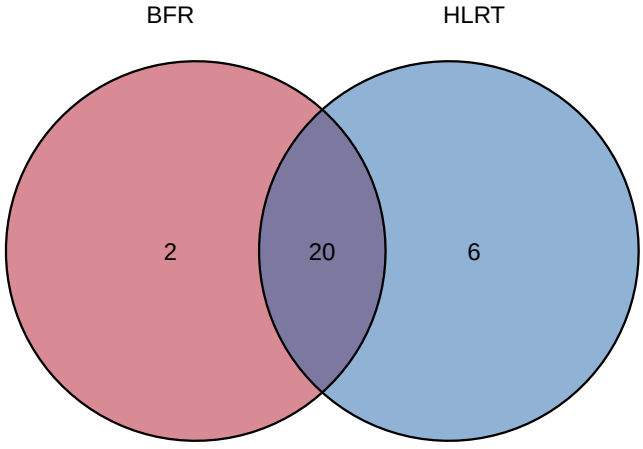
BFR: NES −2.04, padj 0.001
HLRT: NES −2.39, padj 1.6e−05
shared 17, expected 14.1



Shared: ACTA1, COL4A1, COL4A2, DAG1, DMD, DTNA, HSPG2, LAMA2, LAMB2, LAMC1, SGCA, SGCB, SGCD, SGCG, SNTA1, SNTB2, SSPN

Acyl Coa Metabolic Process

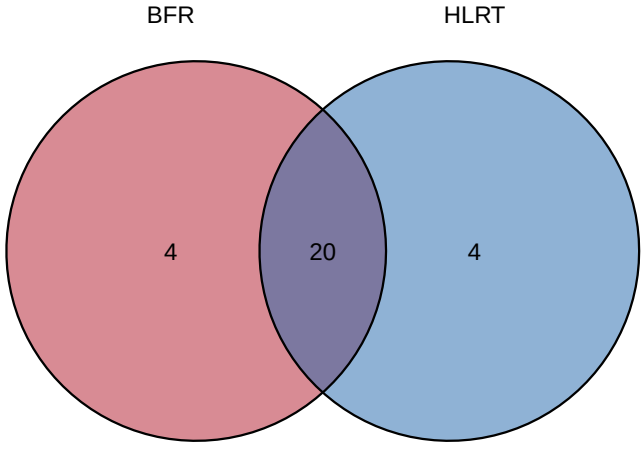
BFR: NES 2.32, padj 1e−05
HLRT: NES 2.08, padj 0.00047
shared 20, expected 10.6



Shared: ACOT9, ACSF3, ACSL1, ACSS1, BCKDK, CBR4, DLAT, DLST, HSD17B12, MMUT, NUDT19, OGDH, PCCA, PDHA1, PDK2, PDK4, PDP1, SUCLA2, SUCLG1, SUCLG2

Amide Biosynthetic Process

BFR: NES 2.23, padj 3.5e−05
HLRT: NES 1.75, padj 0.012
shared 20, expected 9.8

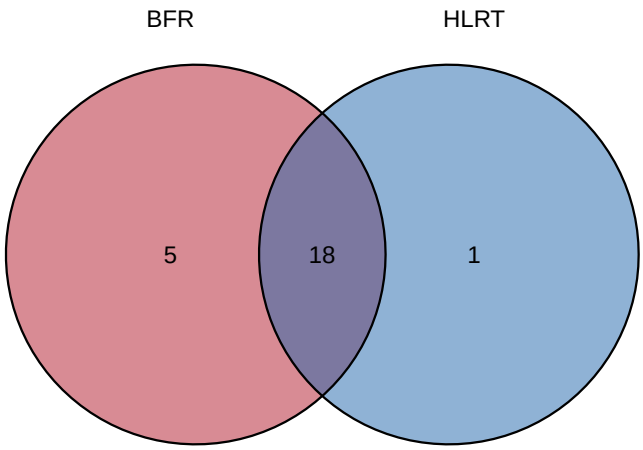


Shared: ACSF3, ACSL1, ACSS1, BCKDK, CARNS1, CBR4, DLAT, DLST, FH, HSD17B12, MECR, MMUT, OGDH, PCCA, PDHA1, PDK2, PDK4, PDP1, PRKAA1, SIRT3

S15. Leading edges by leg

Regulation Of Cell Cycle G1 S Phase Transition

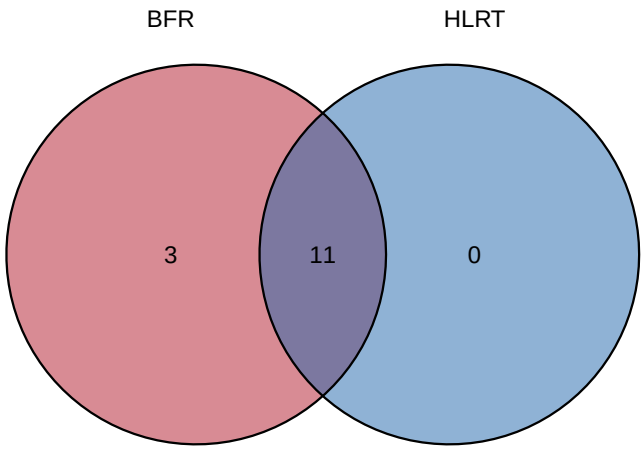
BFR: NES −2.12, padj 0.00013
HLRT: NES −1.69, padj 0.05
shared 18, expected 11.5



Shared: BCL2, CDKN1B, DGKZ, EIF4G1, PPP2CA, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, RDX

Proteasomal Ubiquitin Independent Protein Catabolic Process

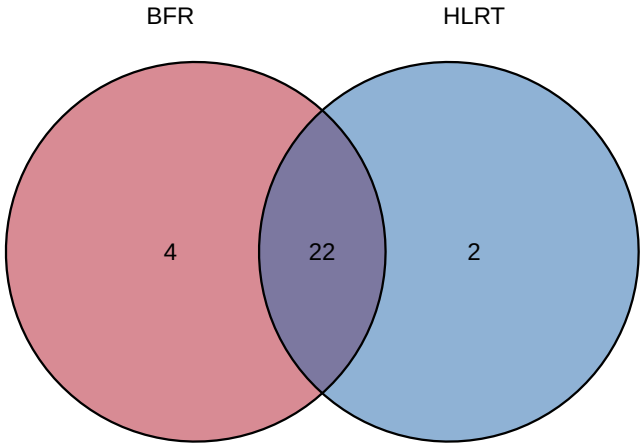
BFR: NES −2.19, padj 0.00014
HLRT: NES −2.10, padj 0.0014
shared 11, expected 10.3



Shared: PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6

Skeletal Muscle Contraction

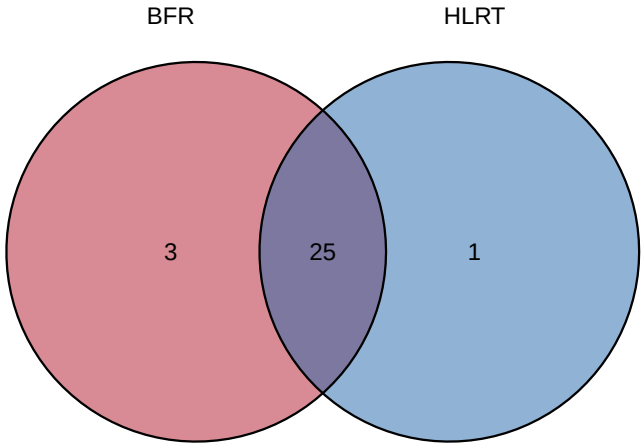
BFR: NES −1.71, padj 0.03
HLRT: NES −2.35, padj 2.2e−05
shared 22, expected 19.5



Shared: ACTN3, ATP2A1, CA3, CASQ1, DMD, FKBP1A, GSTM2, GSTO1, JSRP1, MYH14, MYH7, MYH8, SCN4A, STAC3, SYNM, TCAP, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TTN

Striated Muscle Contraction

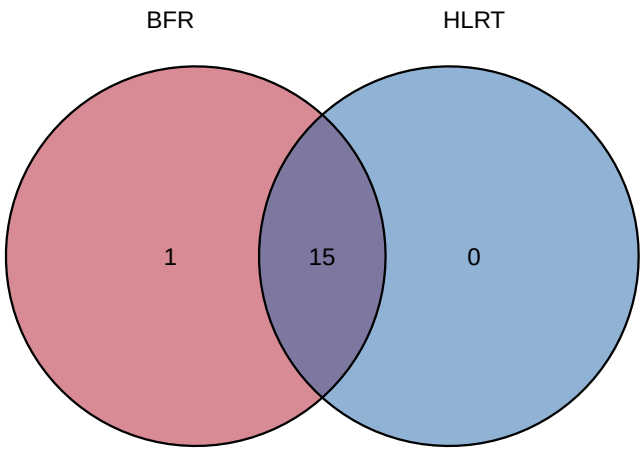
BFR: NES −1.80, padj 0.014
HLRT: NES −2.29, padj 0.00017
shared 25, expected 24.3



Shared: ACTA1, ACTN2, ACTN3, DMD, MYBPC1, MYBPC2, MYH8, MYL1, MYL2, MYL3, MYL4, NEB, TCAP, TMOD1, TMOD4, TNNC1, TNNC2, TNNI1, TNNI2, TNNT1, TNNT3, TPM1, TPM2, TPM3, TTN

Formation Of The Dystrophin Glycoprotein Complex DGC

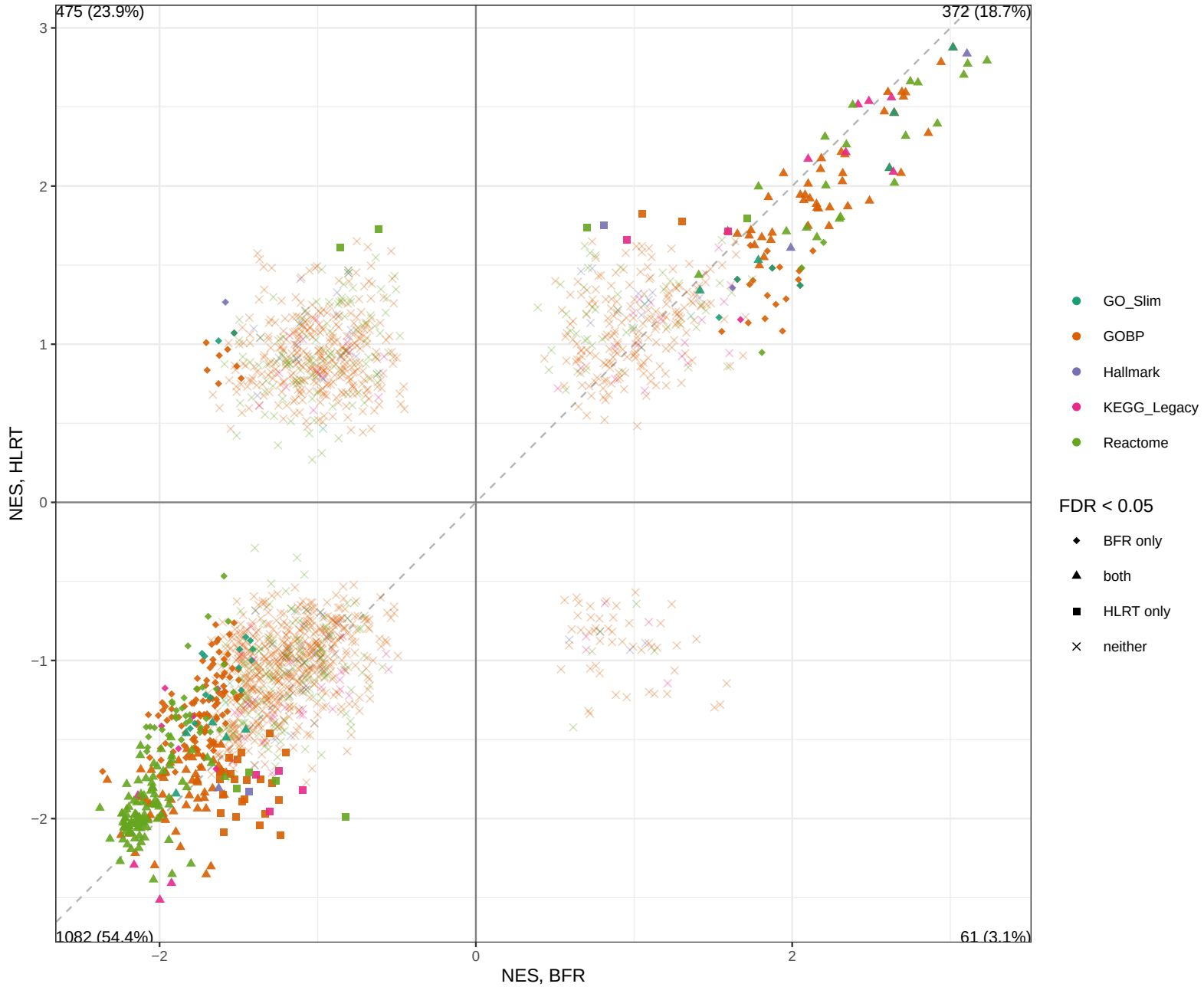
BFR: NES −1.92, padj 0.0078
HLRT: NES −2.35, padj 3.3e−05
shared 15, expected 13.3



Shared: ACTA1, DAG1, DMD, DTNA, HSPG2, LAMA2, LAMB2, LAMC1, SGCA, SGCB, SGCD, SGCG, SNTA1, SNTB2, SSPN

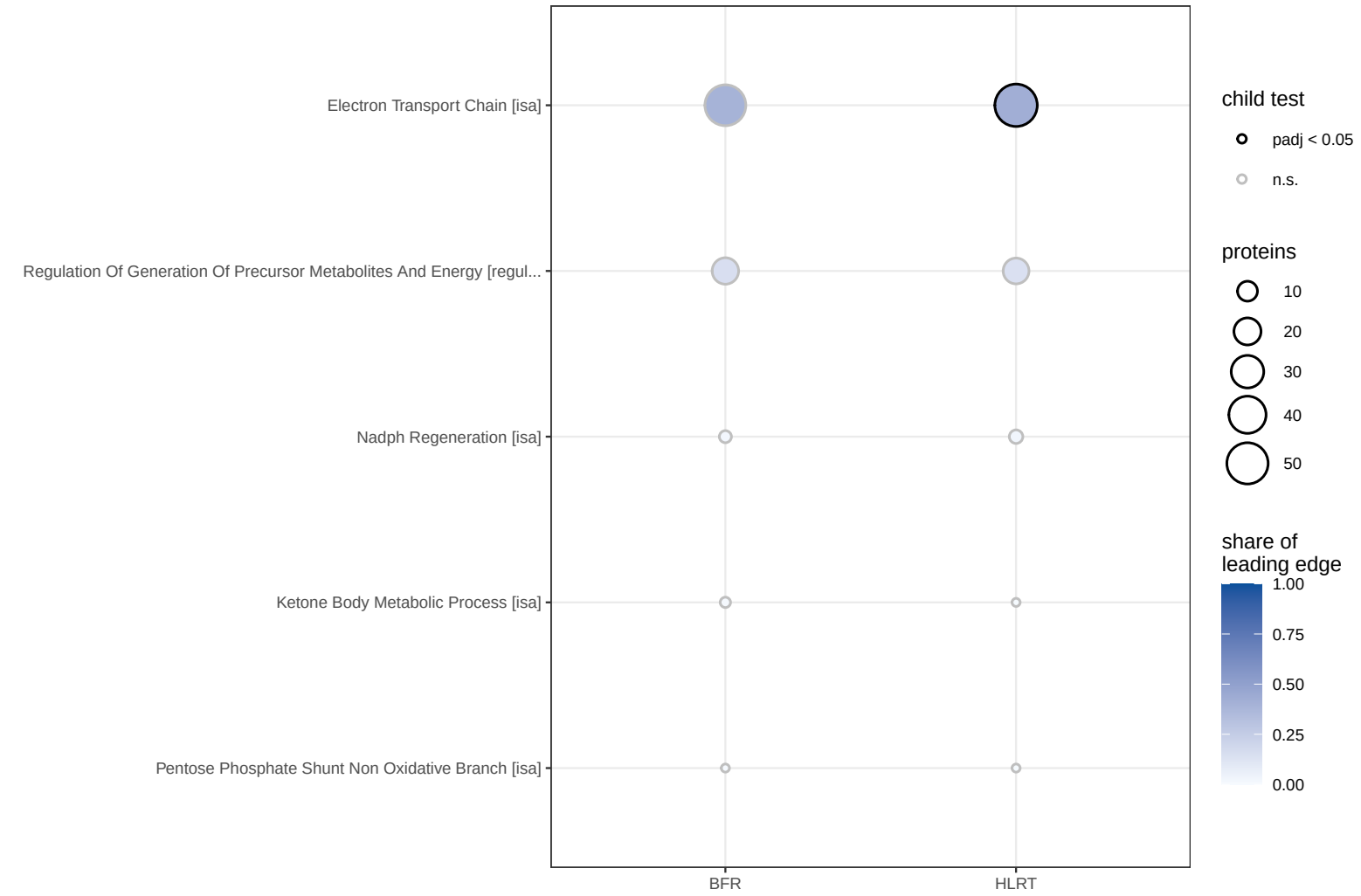
S16. Pathway NES, BFR against HLRT

1990 pathways, Spearman rho 0.76



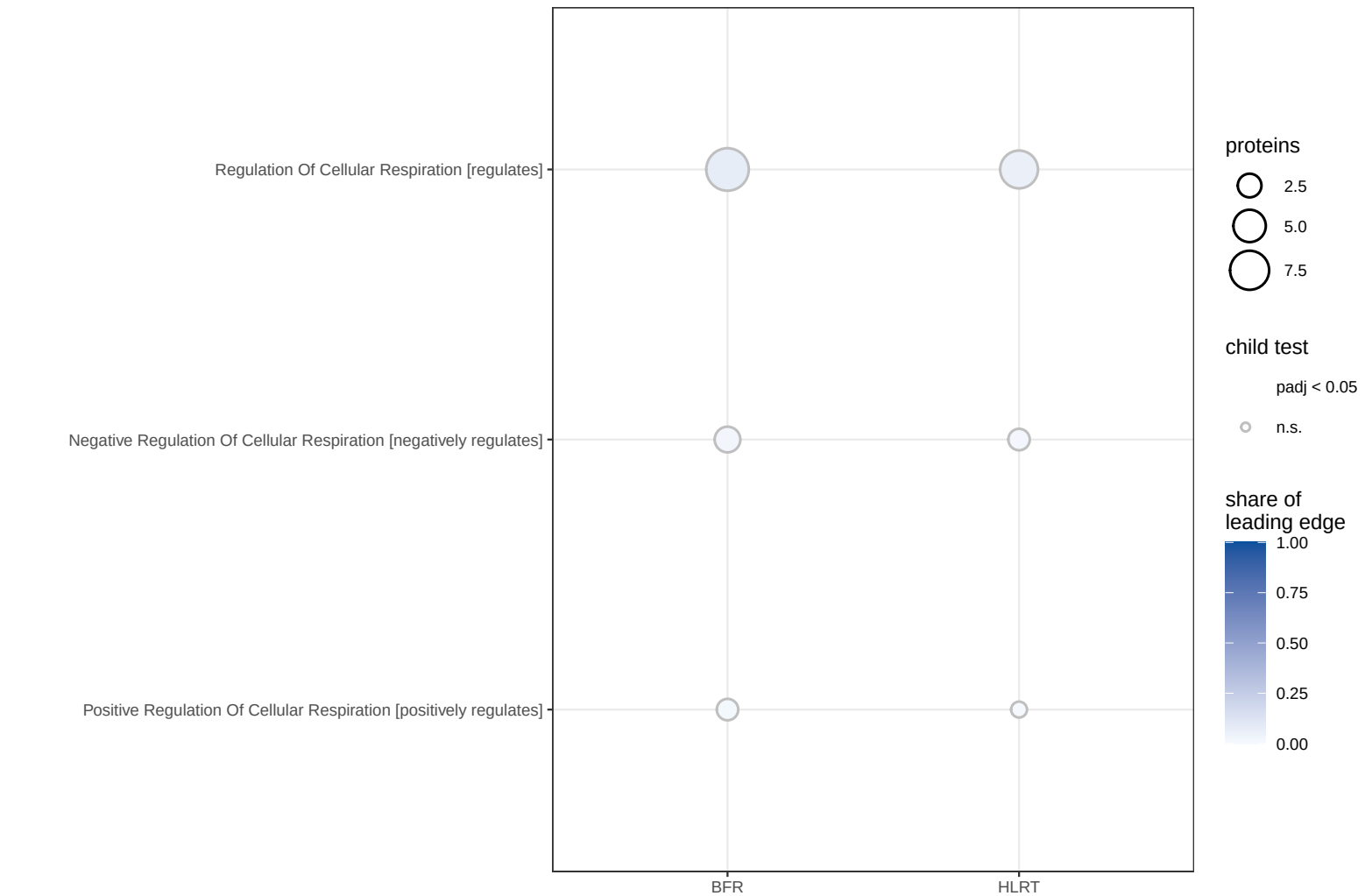
S17. Generation Of Precursor Metabolites And Energy

leading edge: BFR 123, HLRT 125; in significant descendants: BFR 85%, HLRT 85%



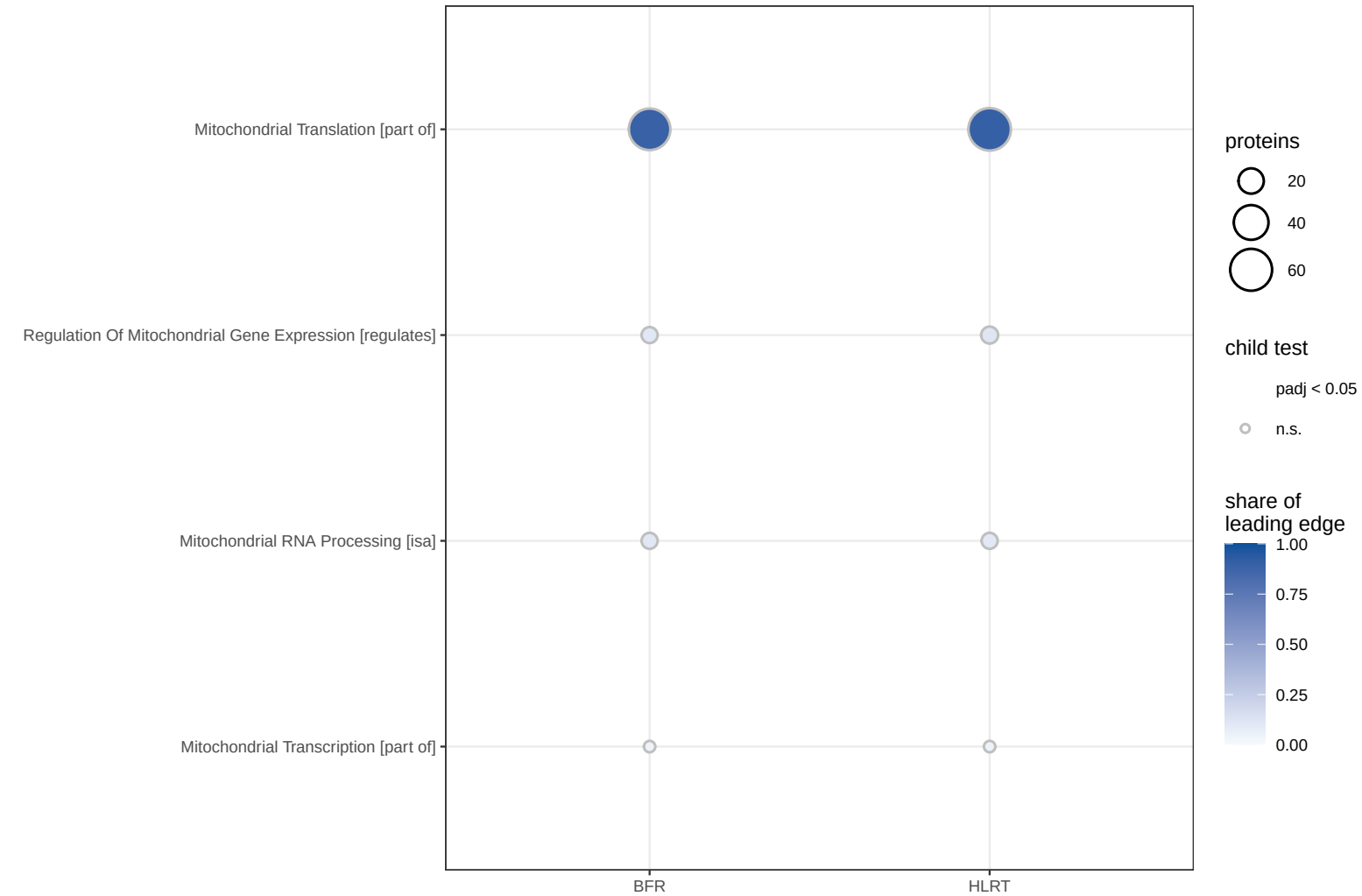
S17. Cellular Respiration

leading edge: BFR 116, HLRT 106; in significant descendants: BFR 74%, HLRT 74%



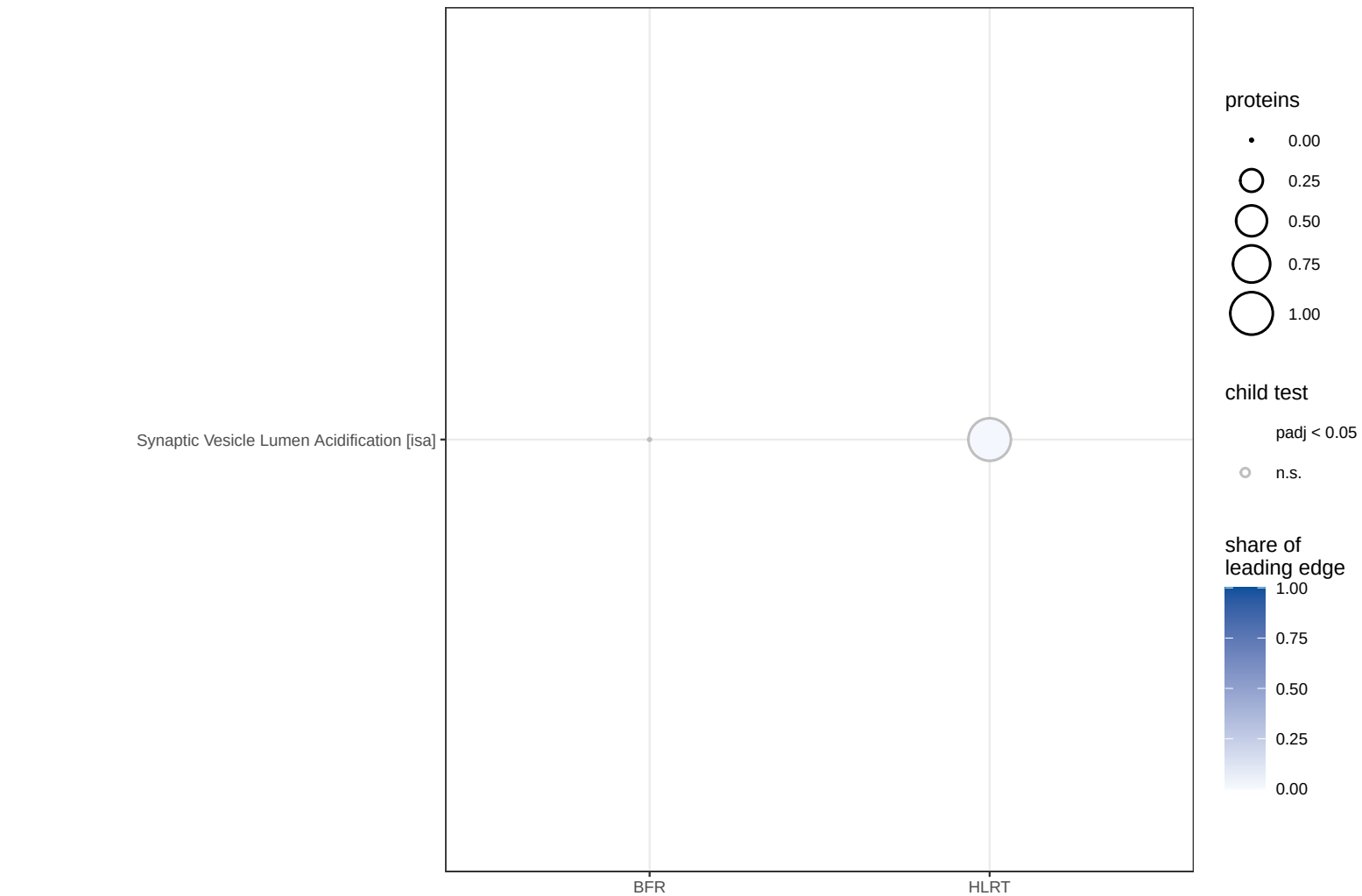
S17. Mitochondrial Gene Expression

leading edge: BFR 66, HLRT 69; in significant descendants: BFR 89%, HLRT 90%



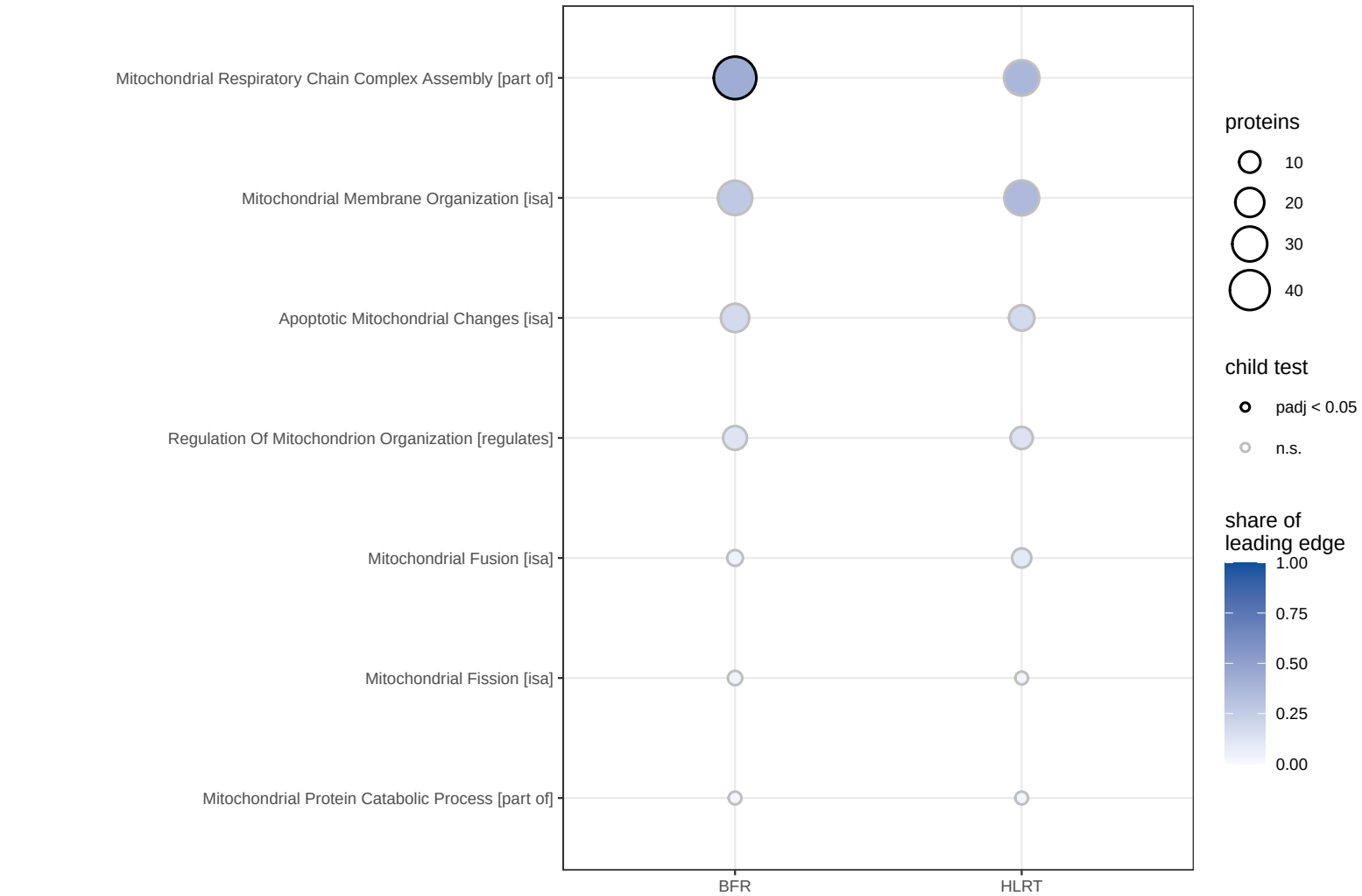
S17. Proton Transmembrane Transport

leading edge: BFR 67, HLRT 67; in significant descendants: BFR 0%, HLRT 0%



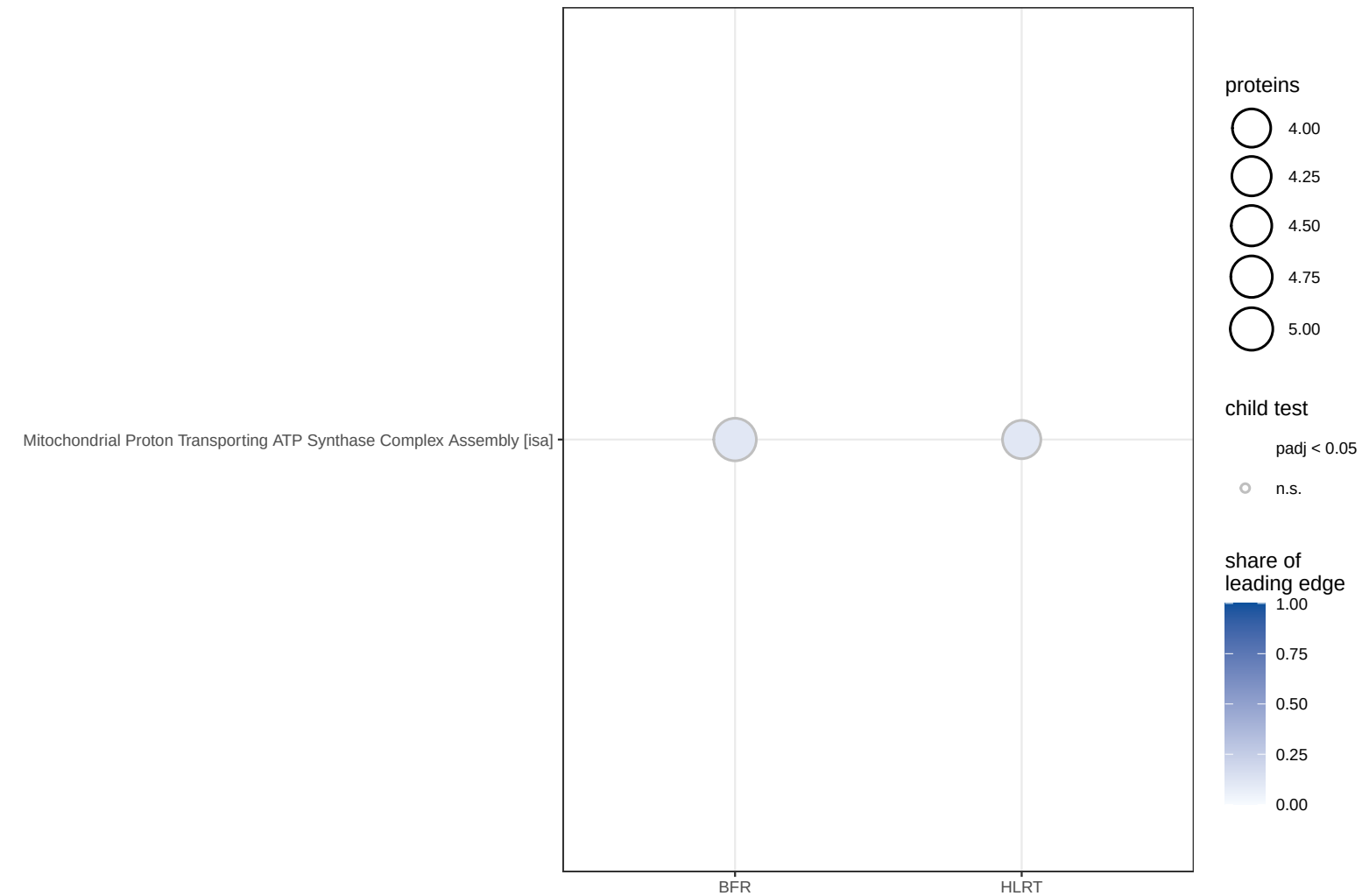
S17. Mitochondrion Organization

leading edge: BFR 106, HLRT 83; in significant descendants: BFR 69%, HLRT 70%



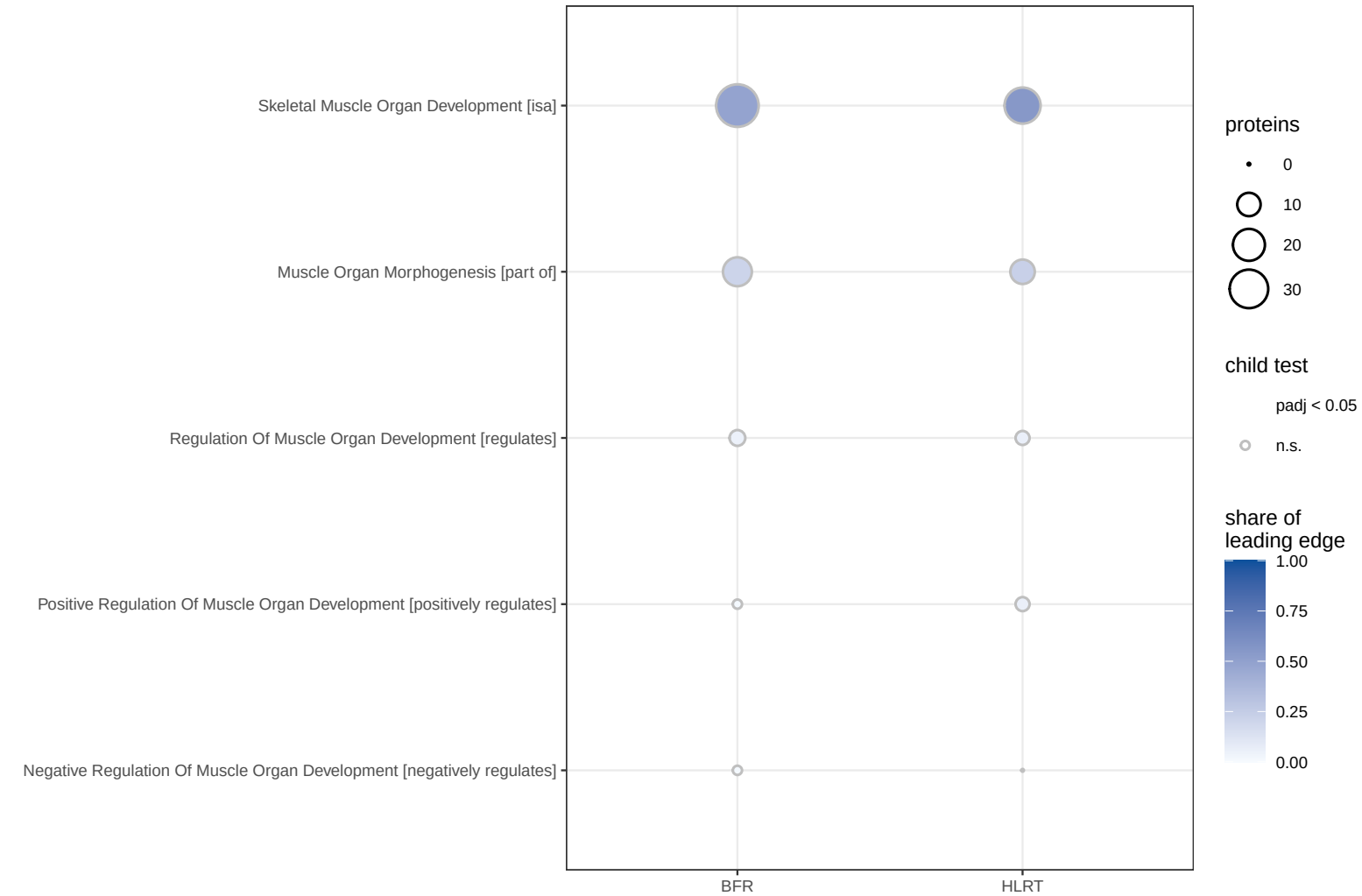
S17. Mitochondrial Respiratory Chain Complex Assembly

leading edge: BFR 46, HLRT 37; in significant descendants: BFR 0%, HLRT 0%



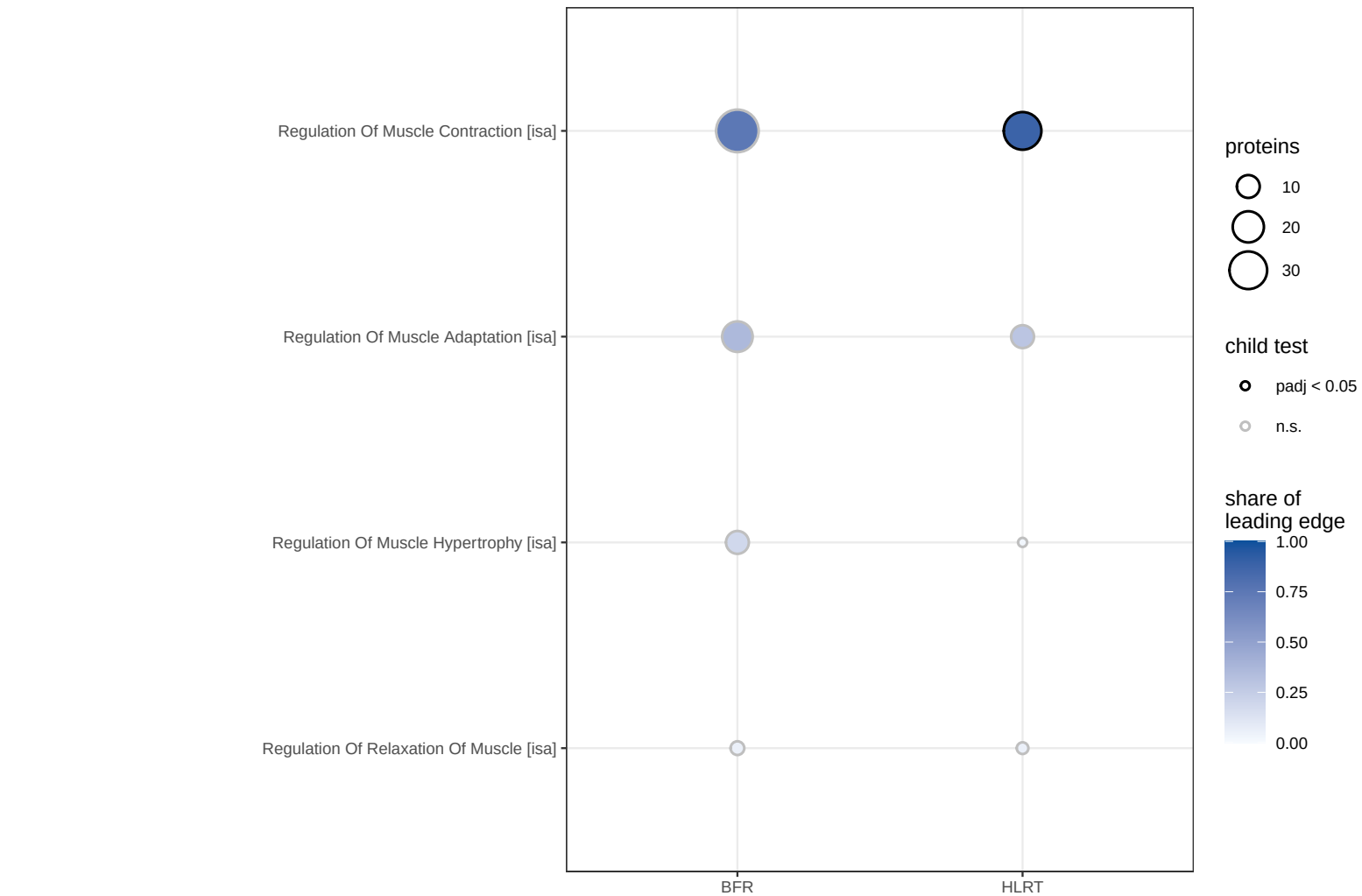
S17. Muscle Organ Development

leading edge: BFR 76, HLRT 47; in significant descendants: BFR 57%, HLRT 74%



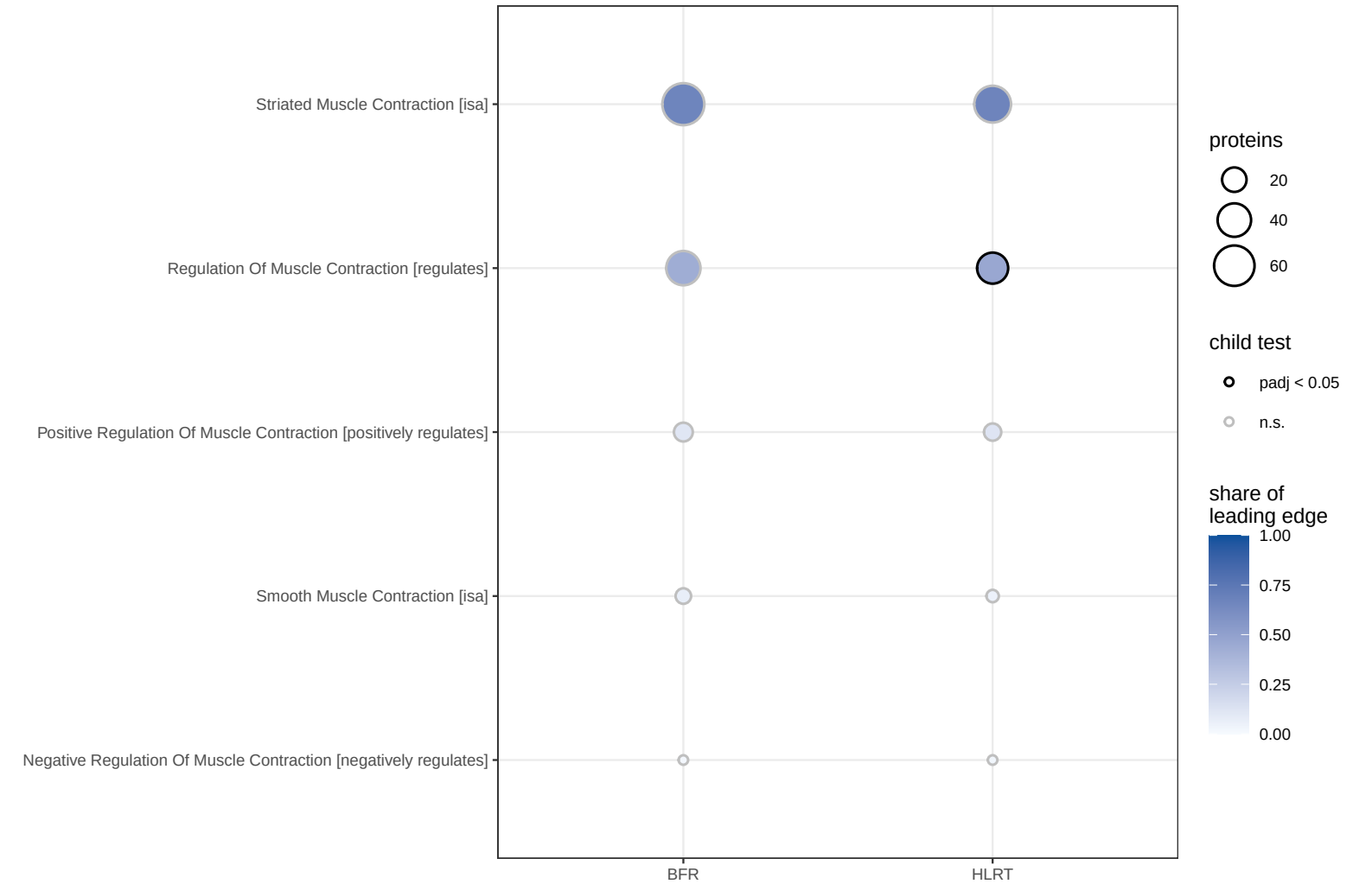
S17. Regulation Of Muscle System Process

leading edge: BFR 52, HLRT 34; in significant descendants: BFR 75%, HLRT 88%



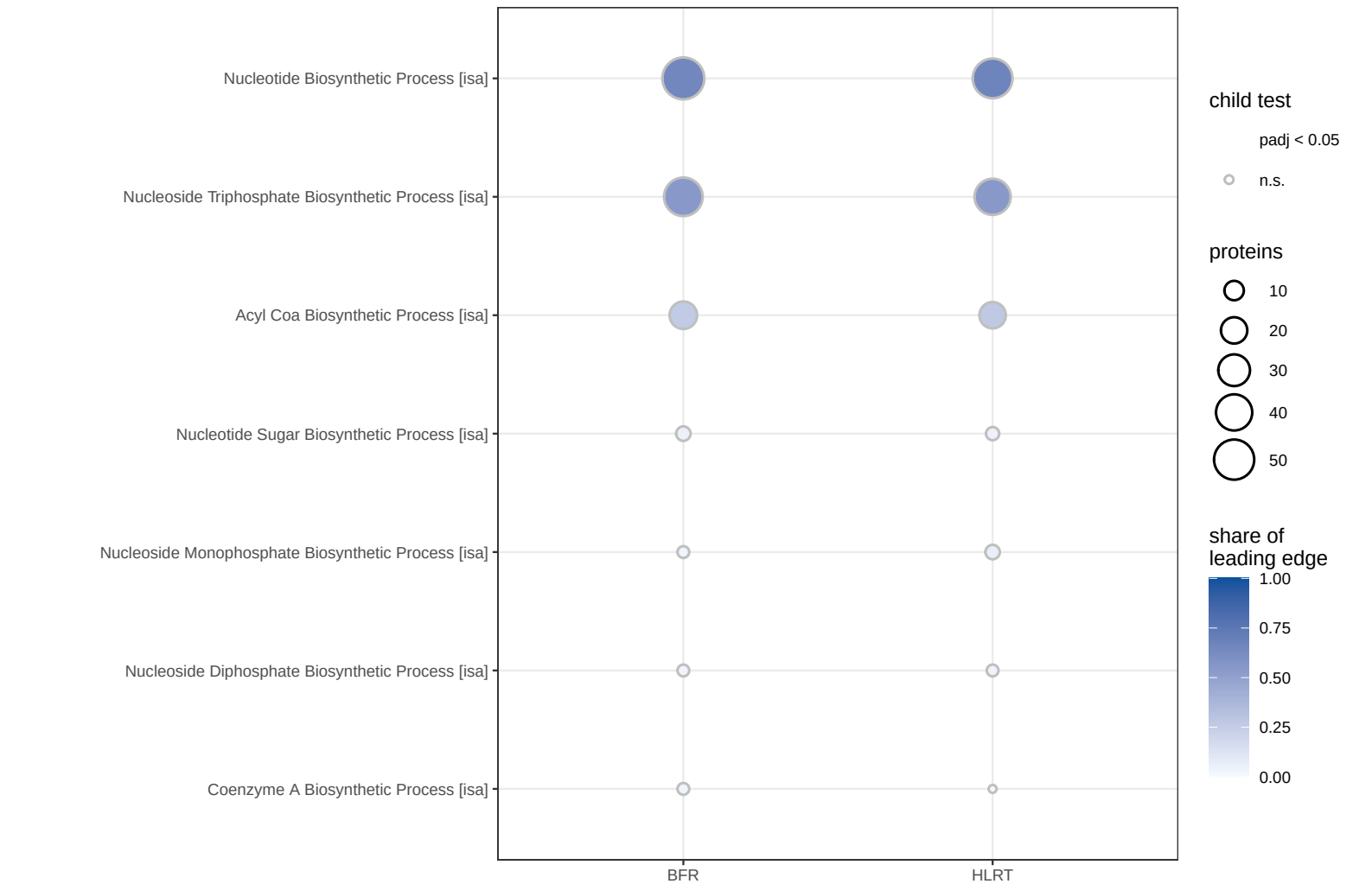
S17. Muscle Contraction

leading edge: BFR 97, HLRT 73; in significant descendants: BFR 73%, HLRT 74%



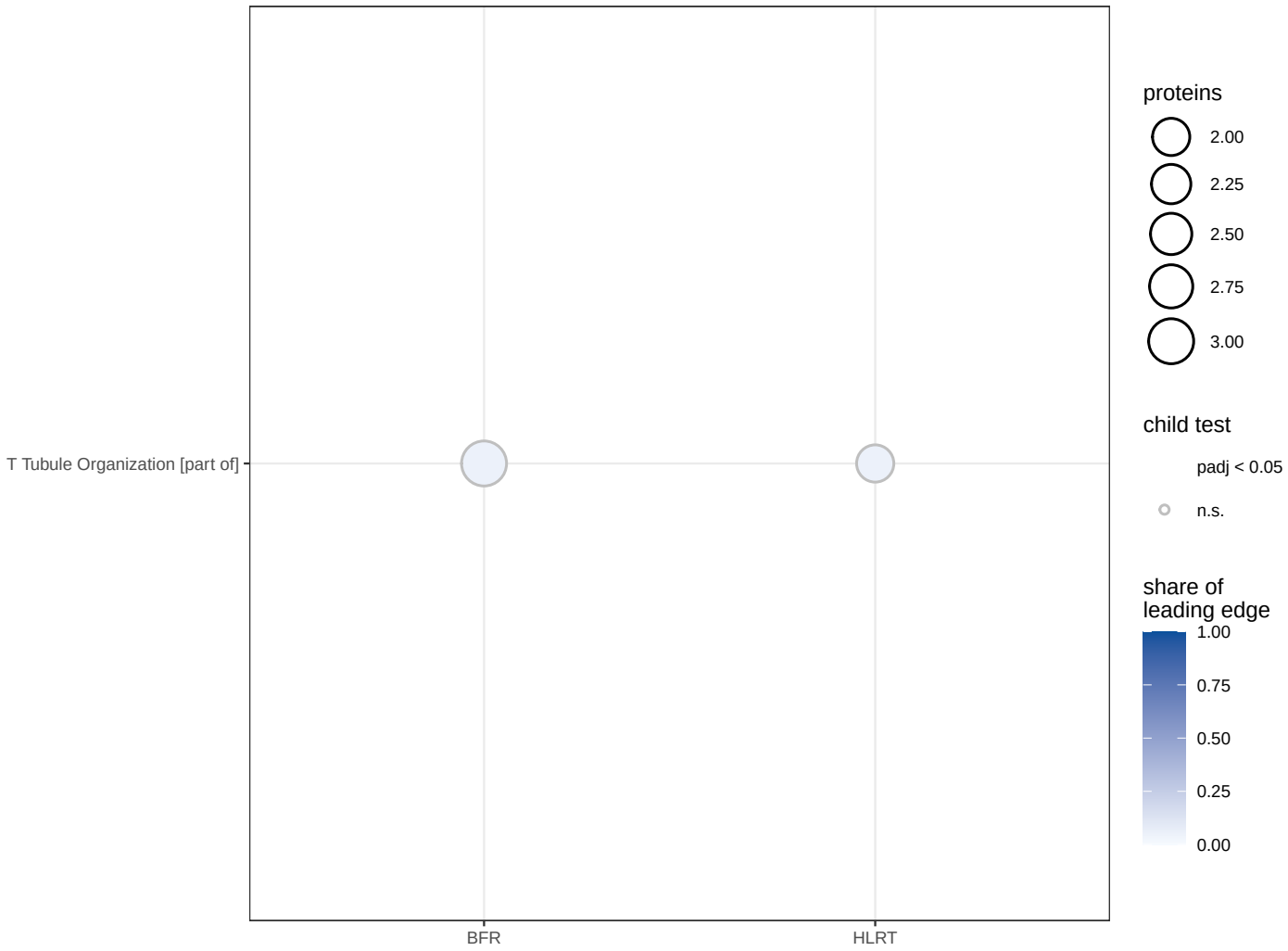
S17. Nucleoside Phosphate Biosynthetic Process

leading edge: BFR 84, HLRT 73; in significant descendants: BFR 89%, HLRT 93%



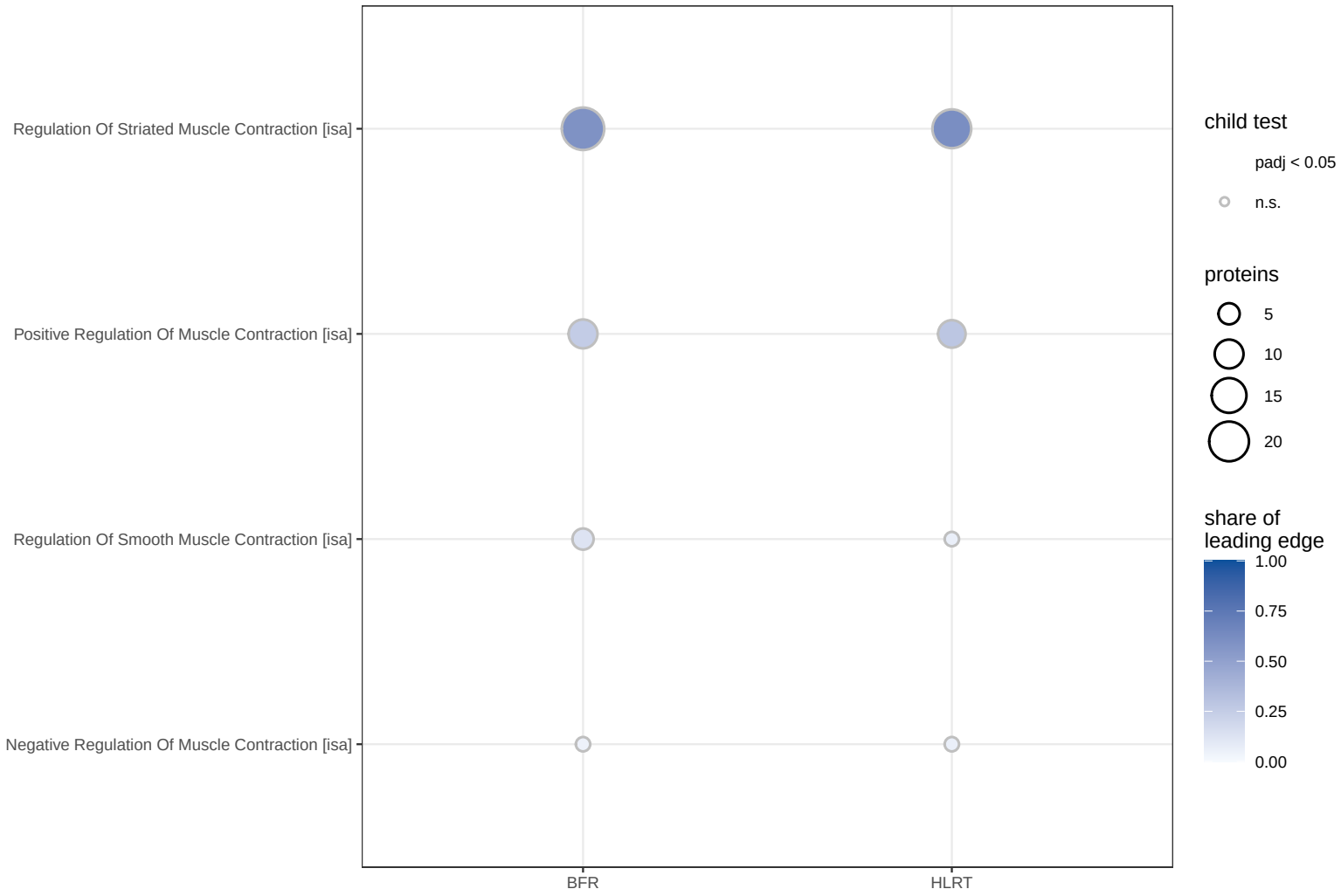
S17. Muscle Cell Development

leading edge: BFR 56, HLRT 37; in significant descendants: BFR 70%, HLRT 81%



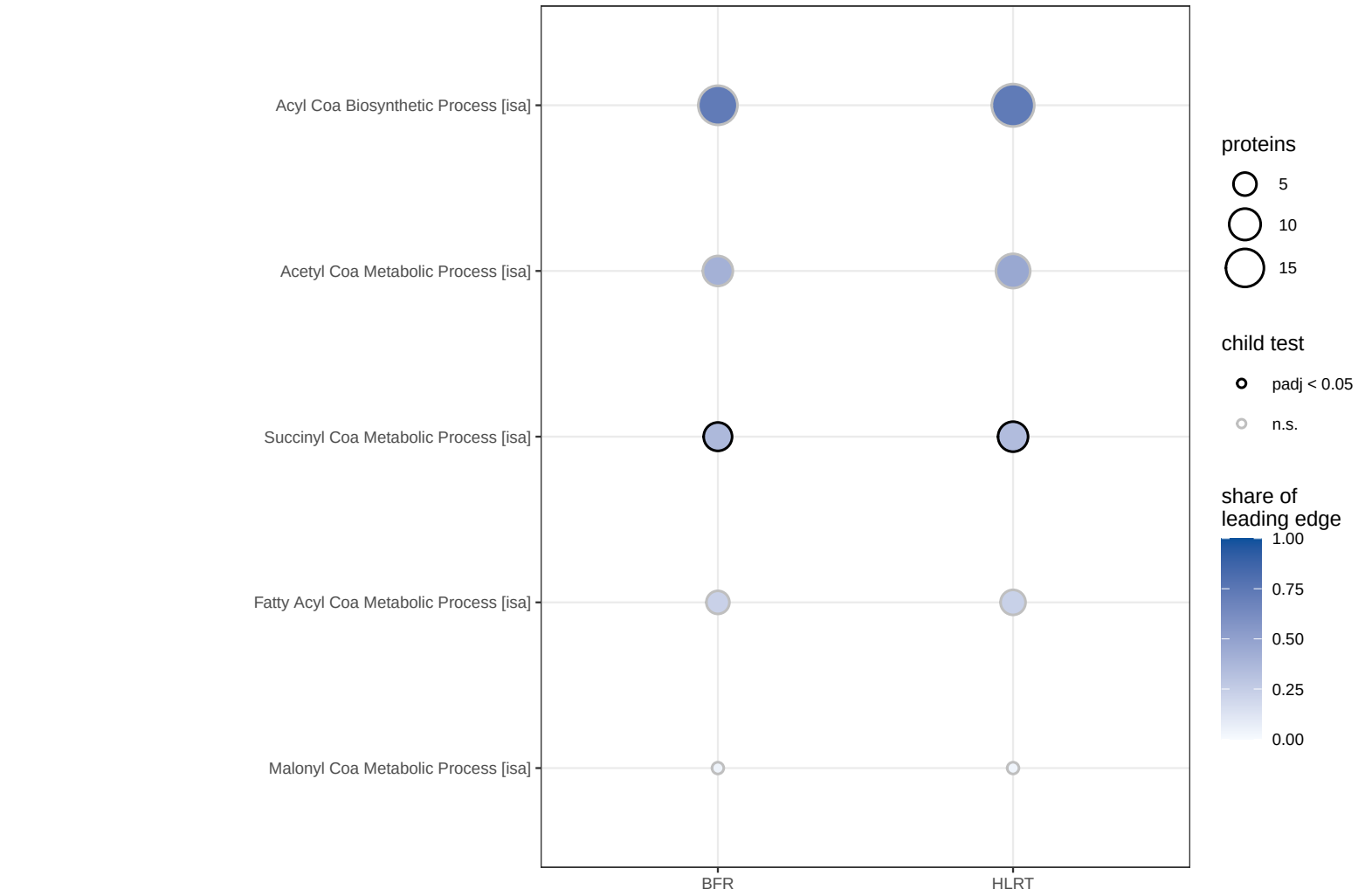
S17. Regulation Of Muscle Contraction

leading edge: BFR 39, HLRT 31; in significant descendants: BFR 0%, HLRT 61%



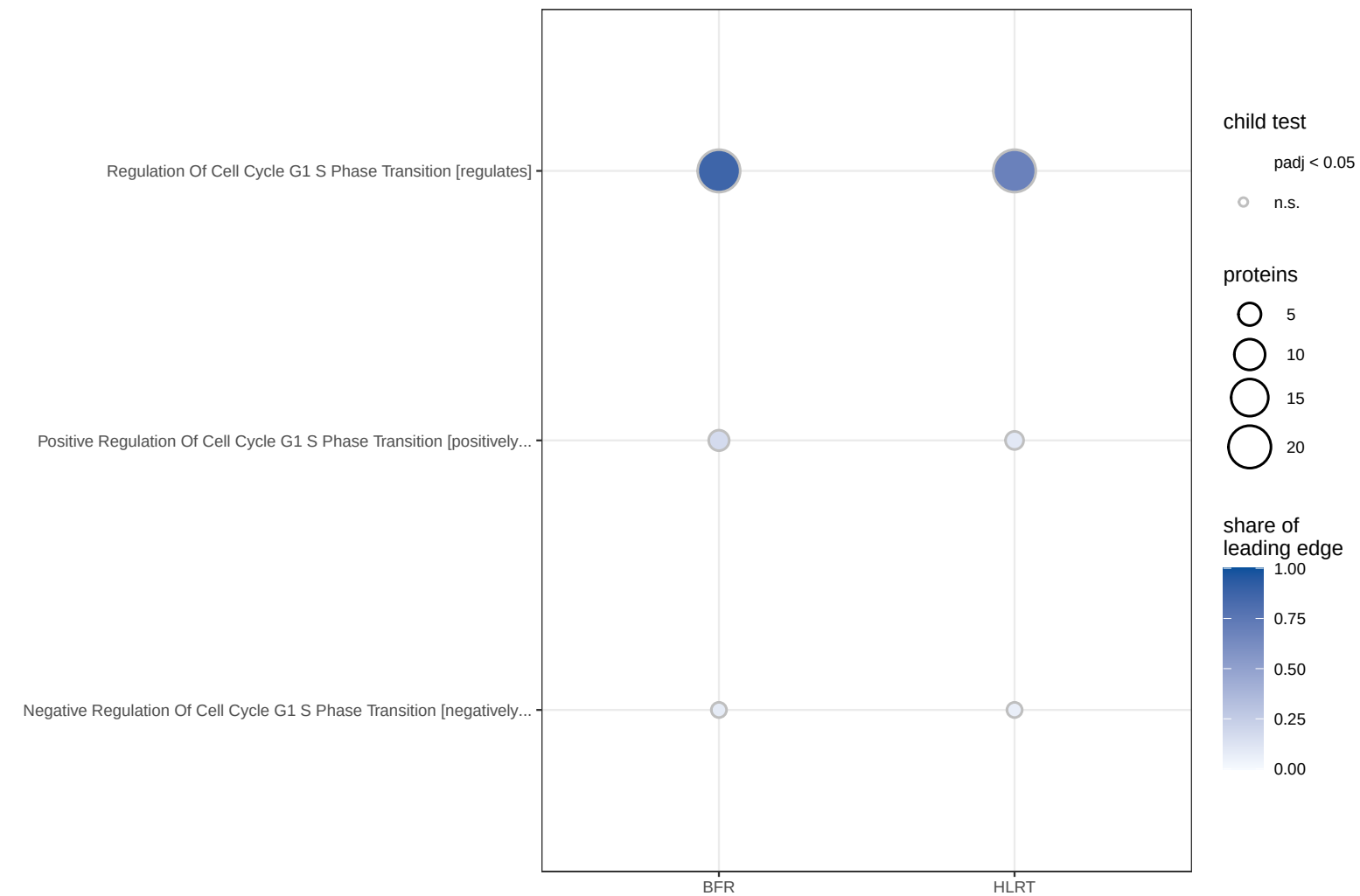
S17. Acyl Coa Metabolic Process

leading edge: BFR 22, HLRT 26; in significant descendants: BFR 73%, HLRT 77%



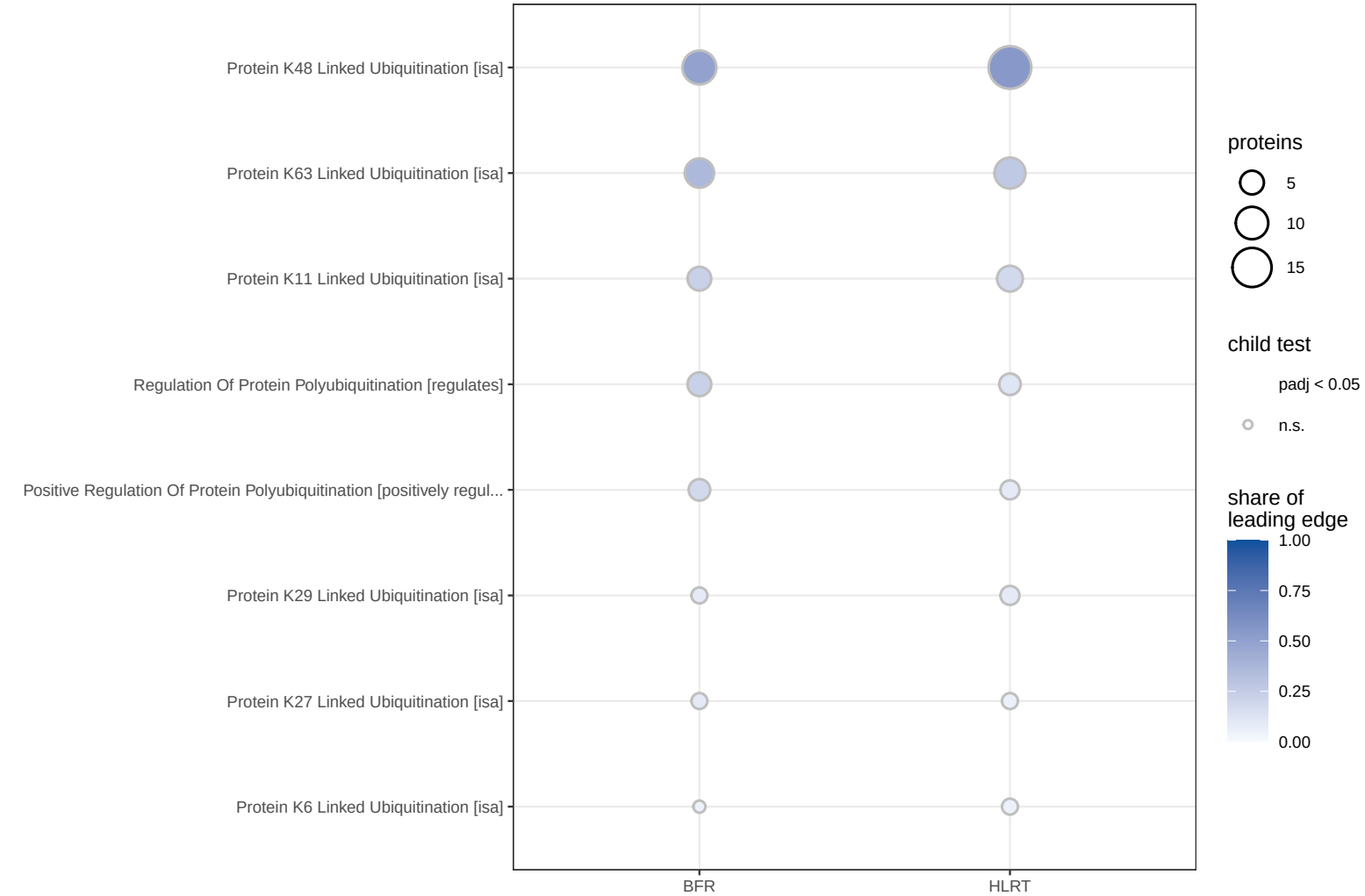
S17. Cell Cycle G1 S Phase Transition

leading edge: BFR 23, HLRT 29; in significant descendants: BFR 87%, HLRT 66%



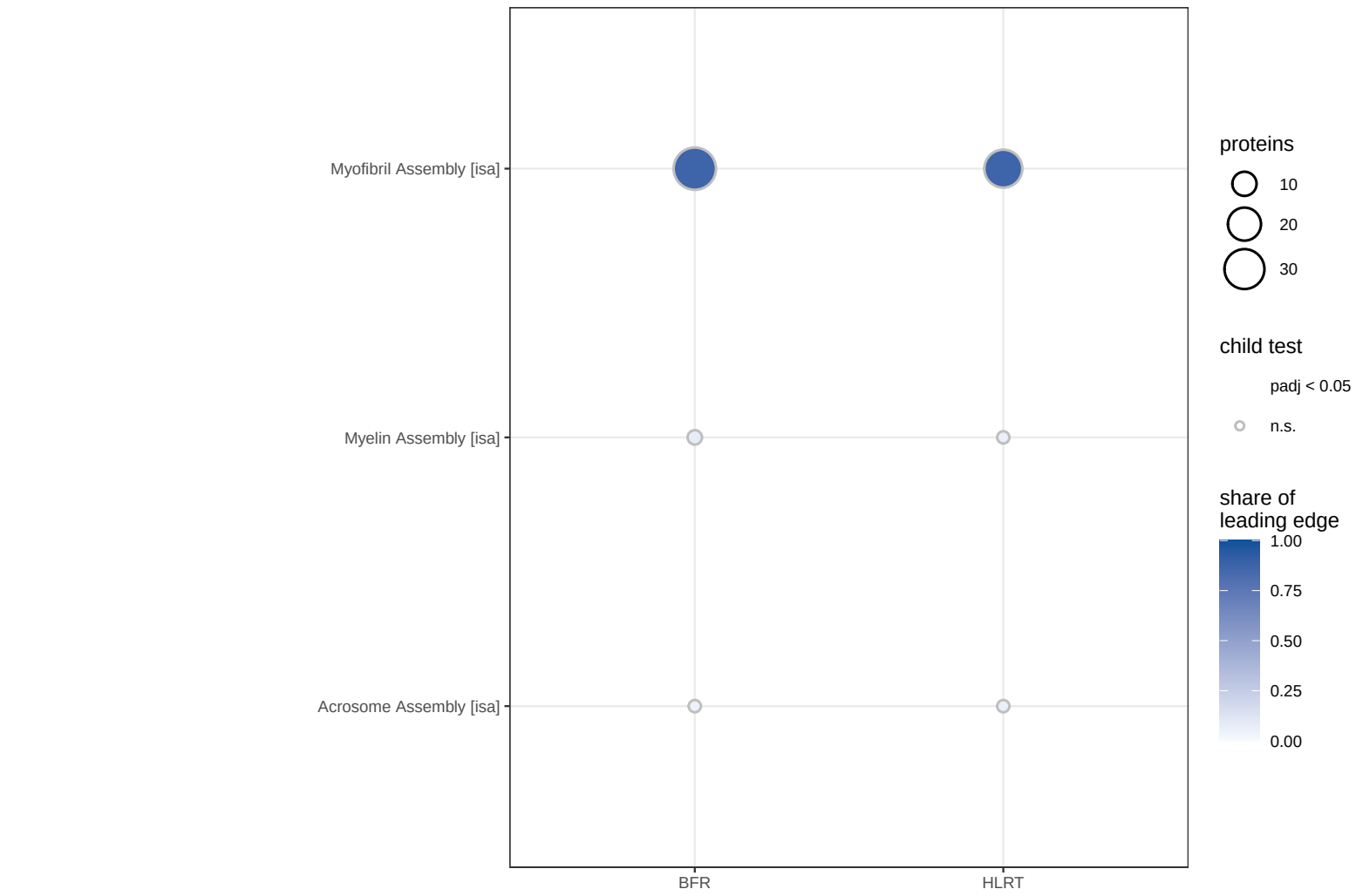
S17. Protein Polyubiquitination

leading edge: BFR 22, HLRT 33; in significant descendants: BFR 64%, HLRT 55%



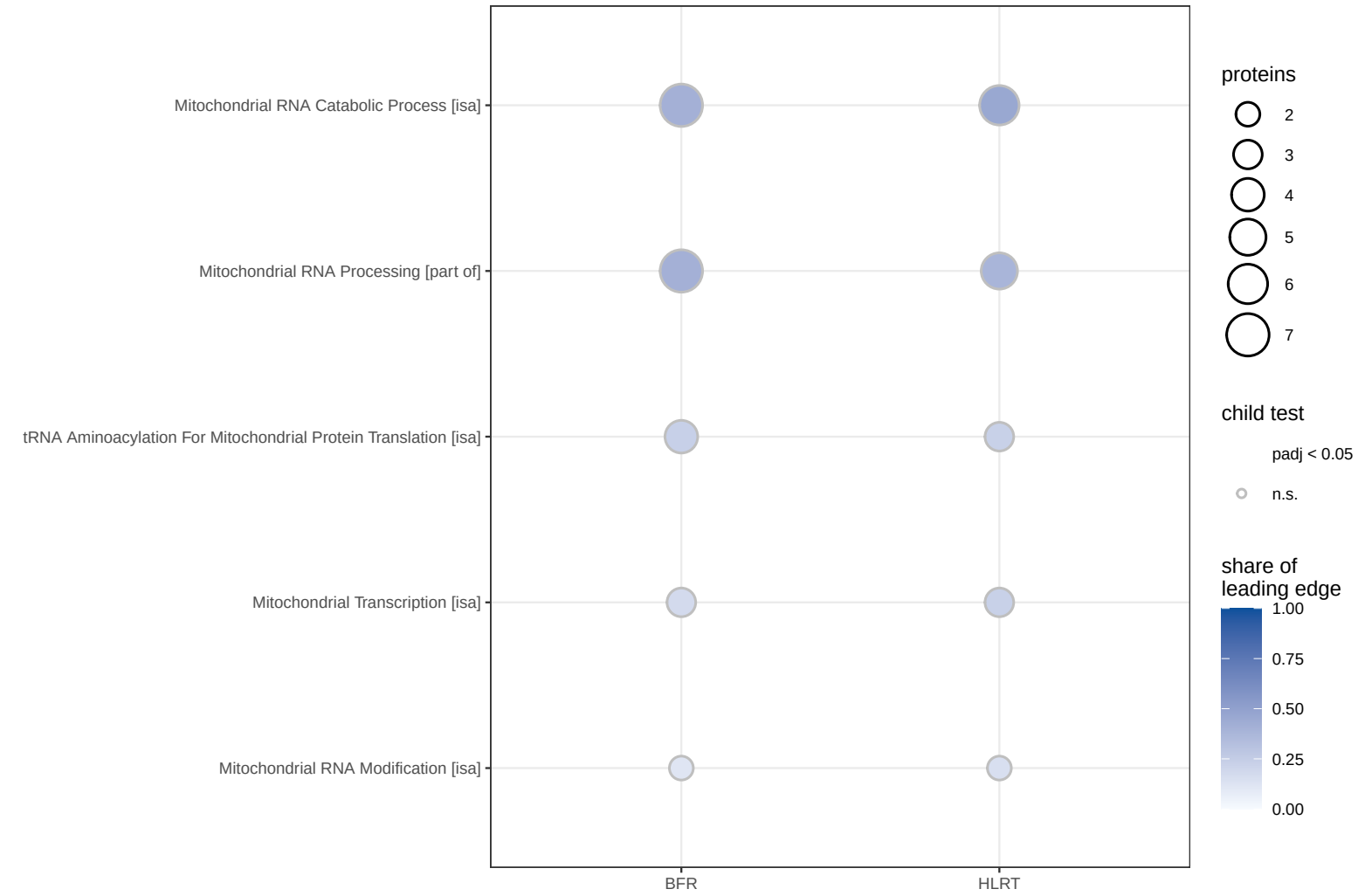
S17. Cellular Component Assembly Involved In Morphogenesis

leading edge: BFR 39, HLRT 31; in significant descendants: BFR 87%, HLRT 87%



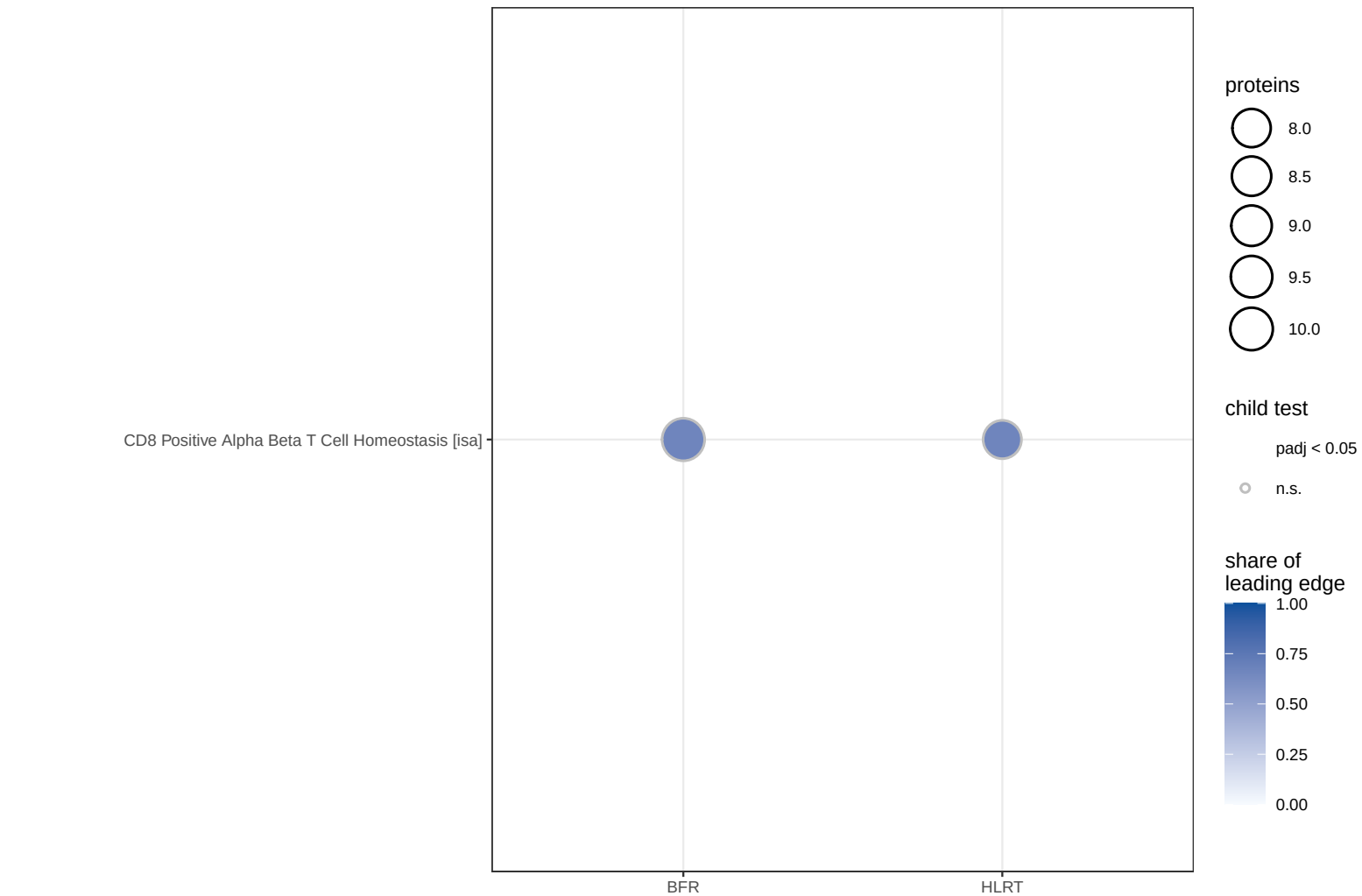
S17. Mitochondrial RNA Metabolic Process

leading edge: BFR 17, HLRT 13; in significant descendants: BFR 0%, HLRT 0%



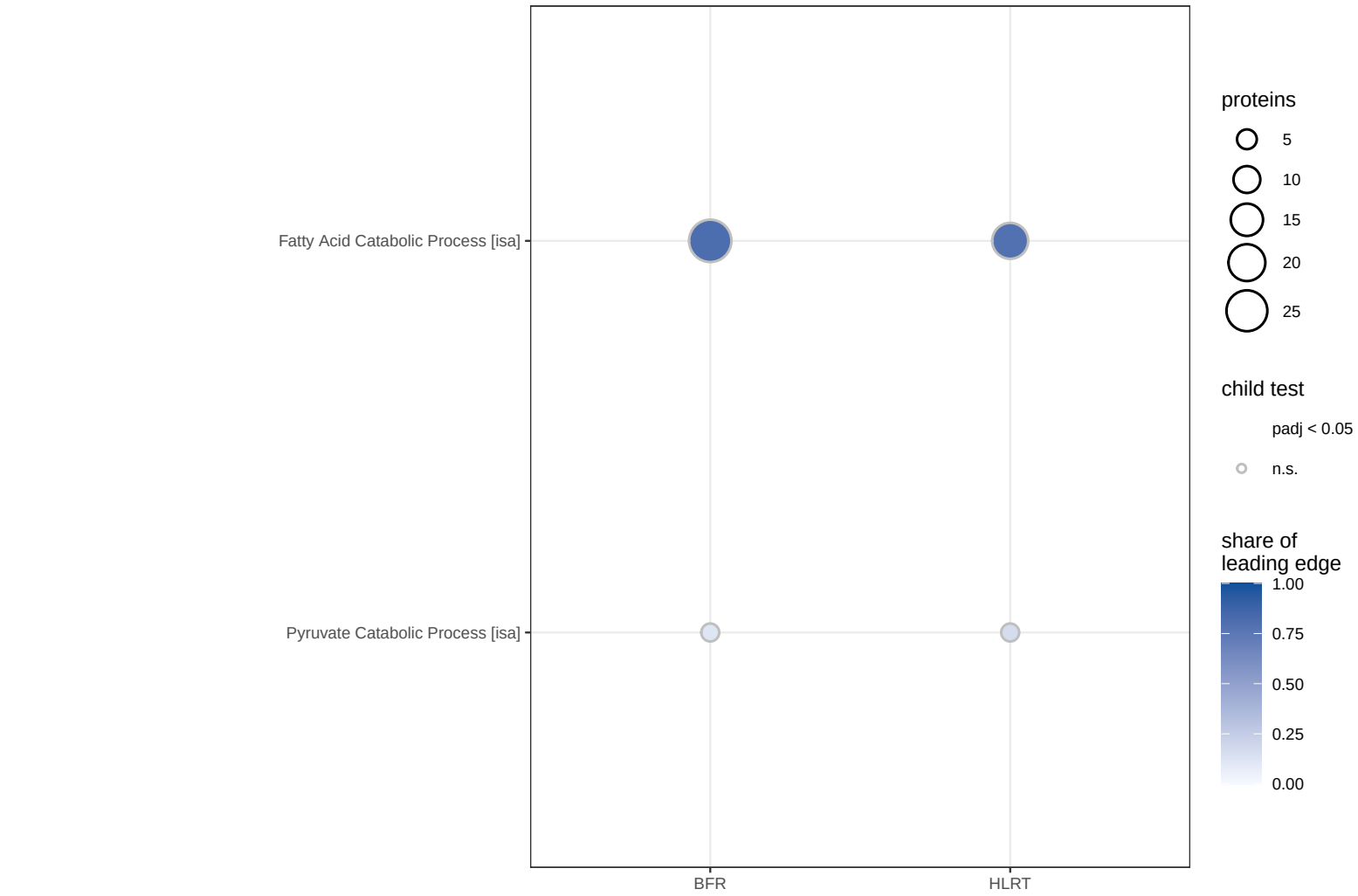
S17. T Cell Homeostasis

leading edge: BFR 15, HLRT 12; in significant descendants: BFR 0%, HLRT 0%



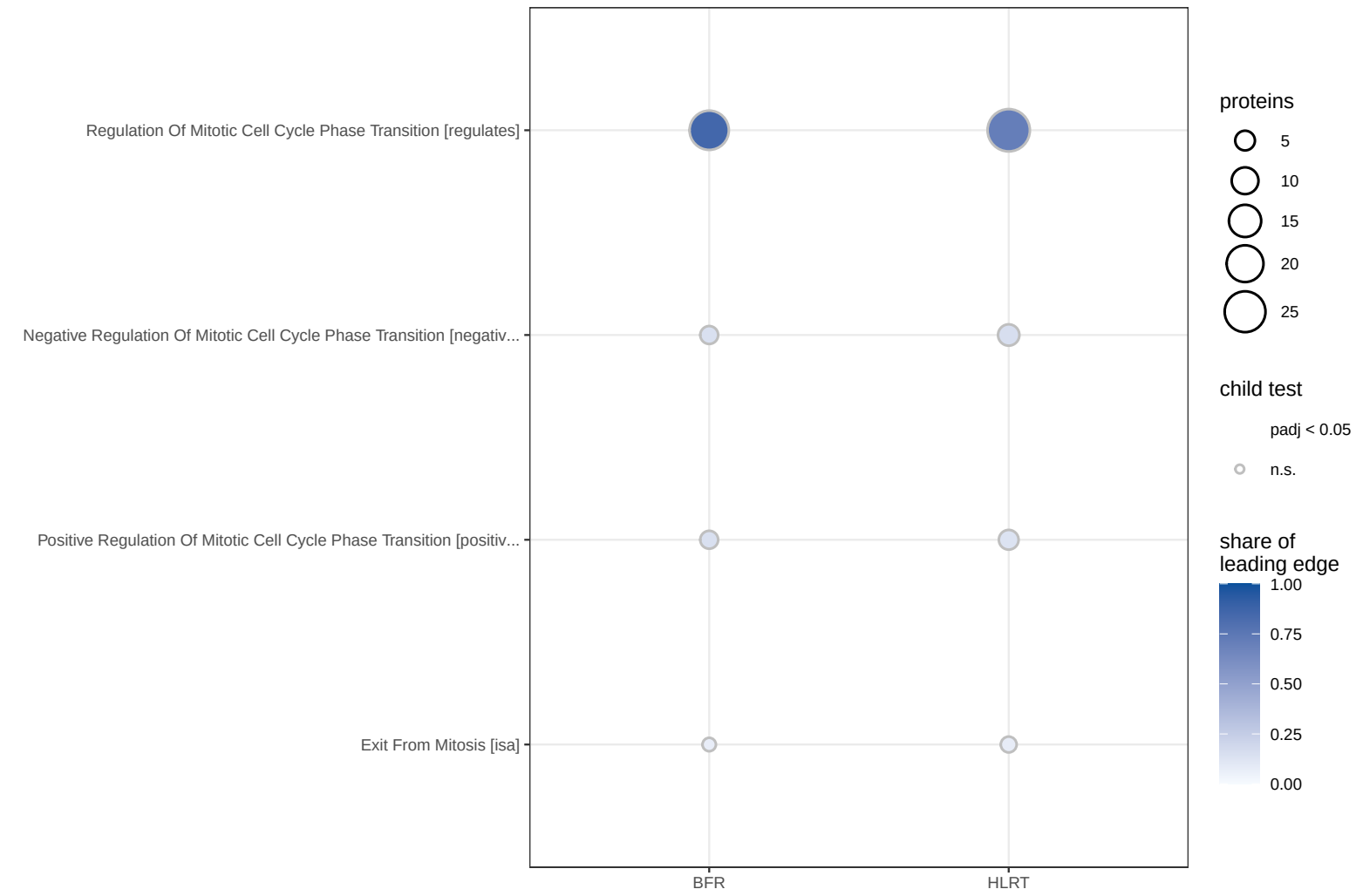
S17. Monocarboxylic Acid Catabolic Process

leading edge: BFR 33, HLRT 24; in significant descendants: BFR 82%, HLRT 71%



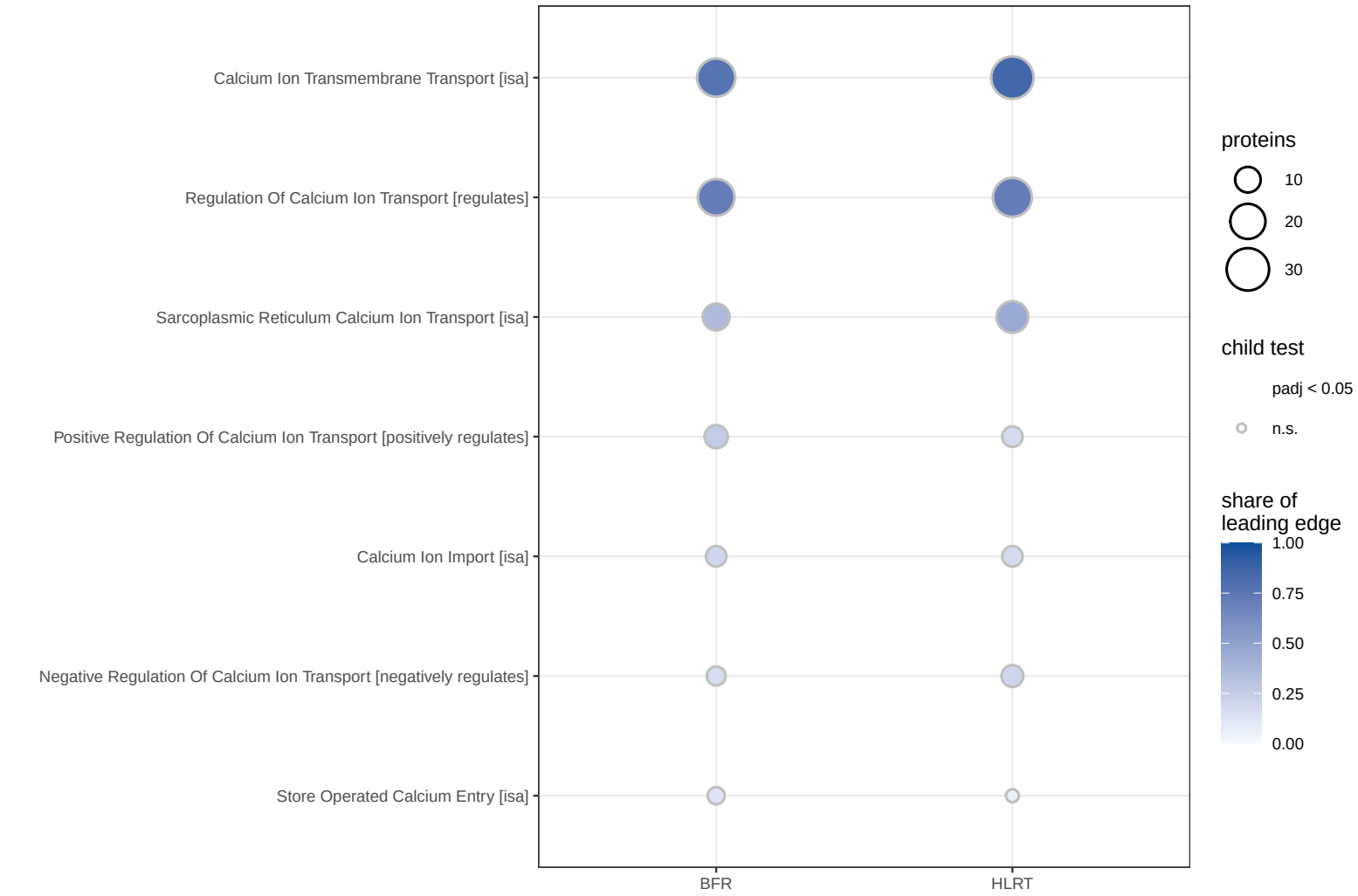
S17. Mitotic Cell Cycle Phase Transition

leading edge: BFR 27, HLRT 38; in significant descendants: BFR 85%, HLRT 71%



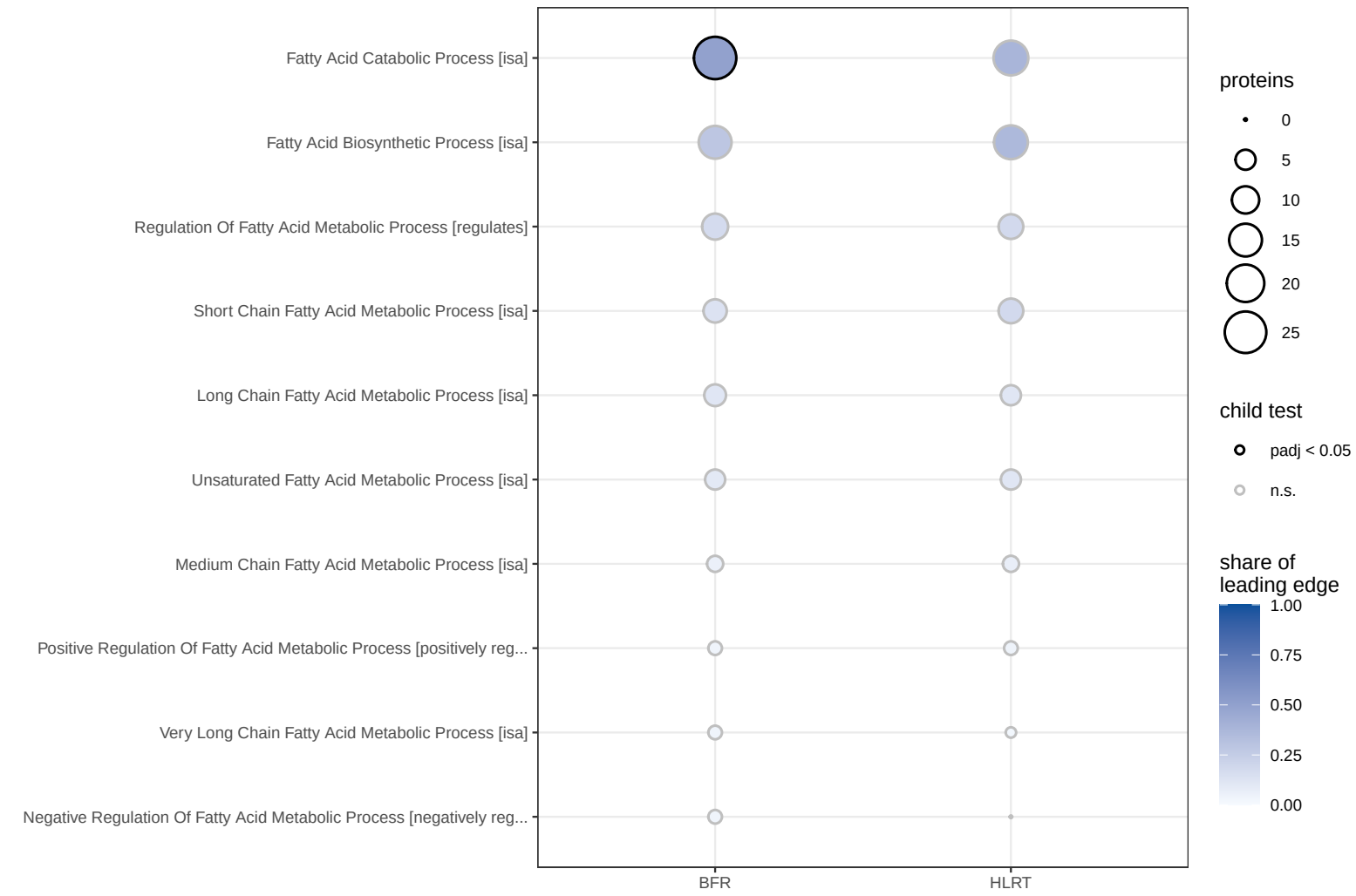
S17. Calcium Ion Transport

leading edge: BFR 31, HLRT 35; in significant descendants: BFR 94%, HLRT 97%



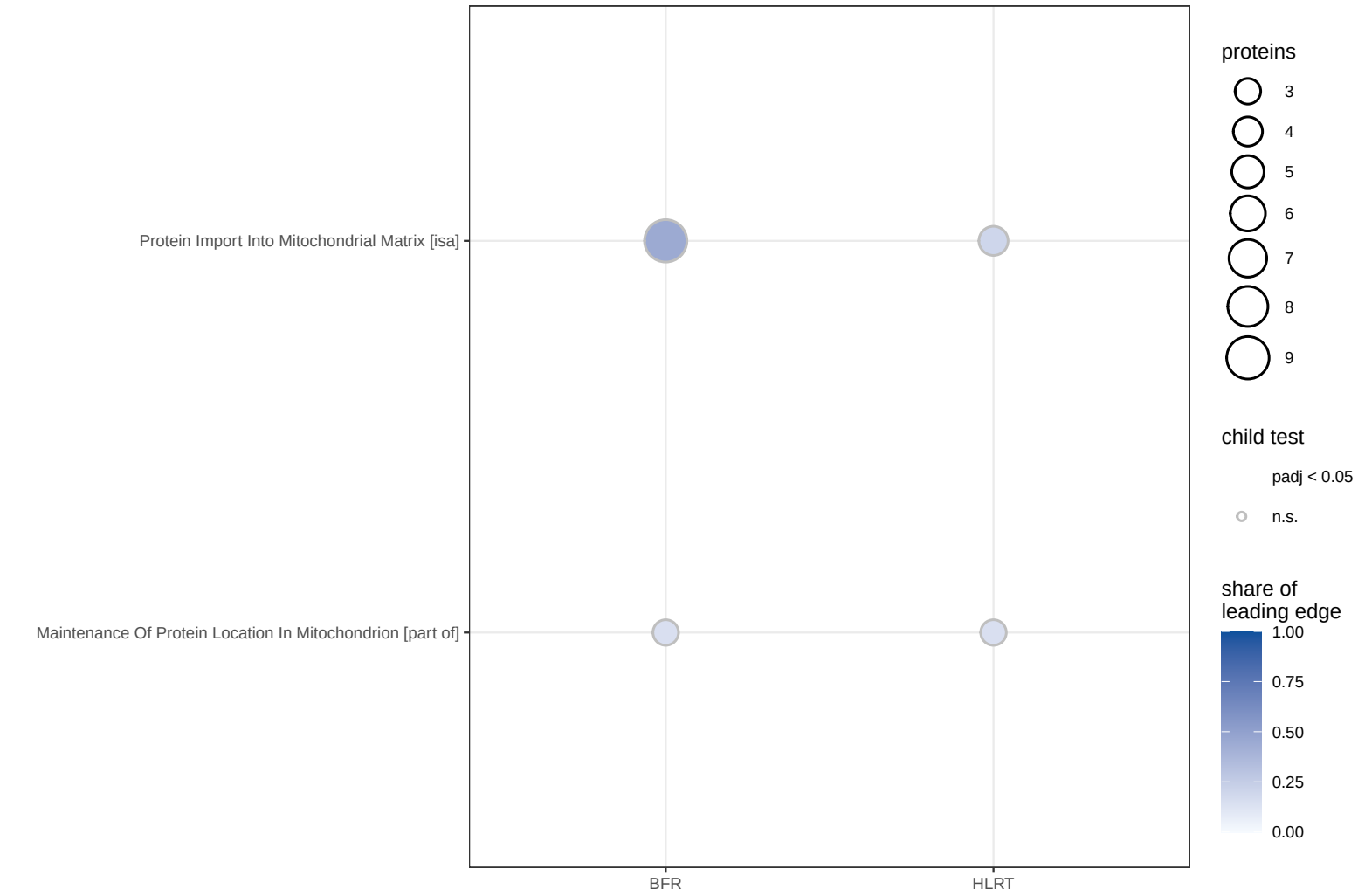
S17. Fatty Acid Metabolic Process

leading edge: BFR 52, HLRT 44; in significant descendants: BFR 50%, HLRT 39%



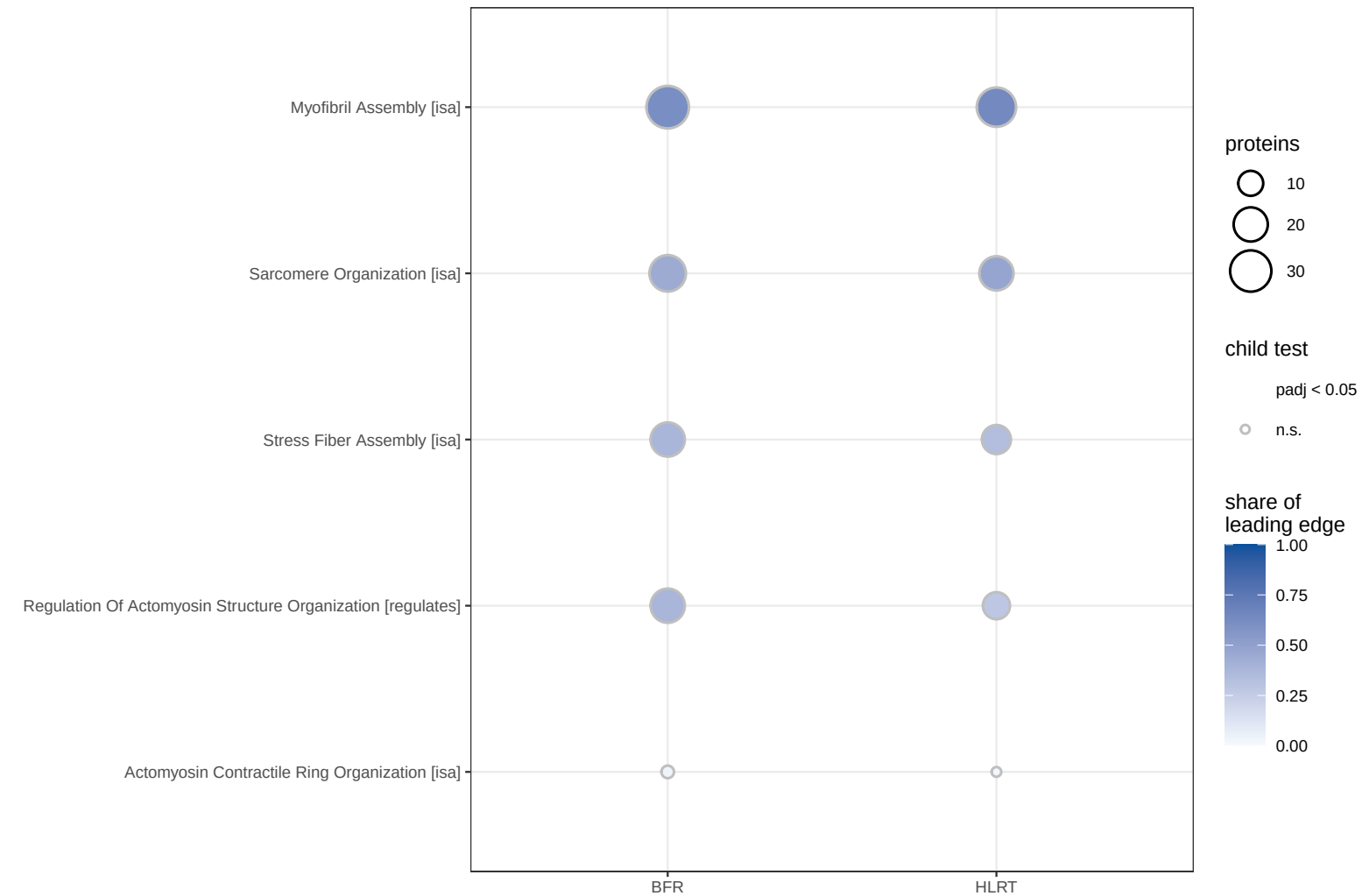
S17. Protein Localization To Mitochondrion

leading edge: BFR 20, HLRT 20; in significant descendants: BFR 45%, HLRT 0%



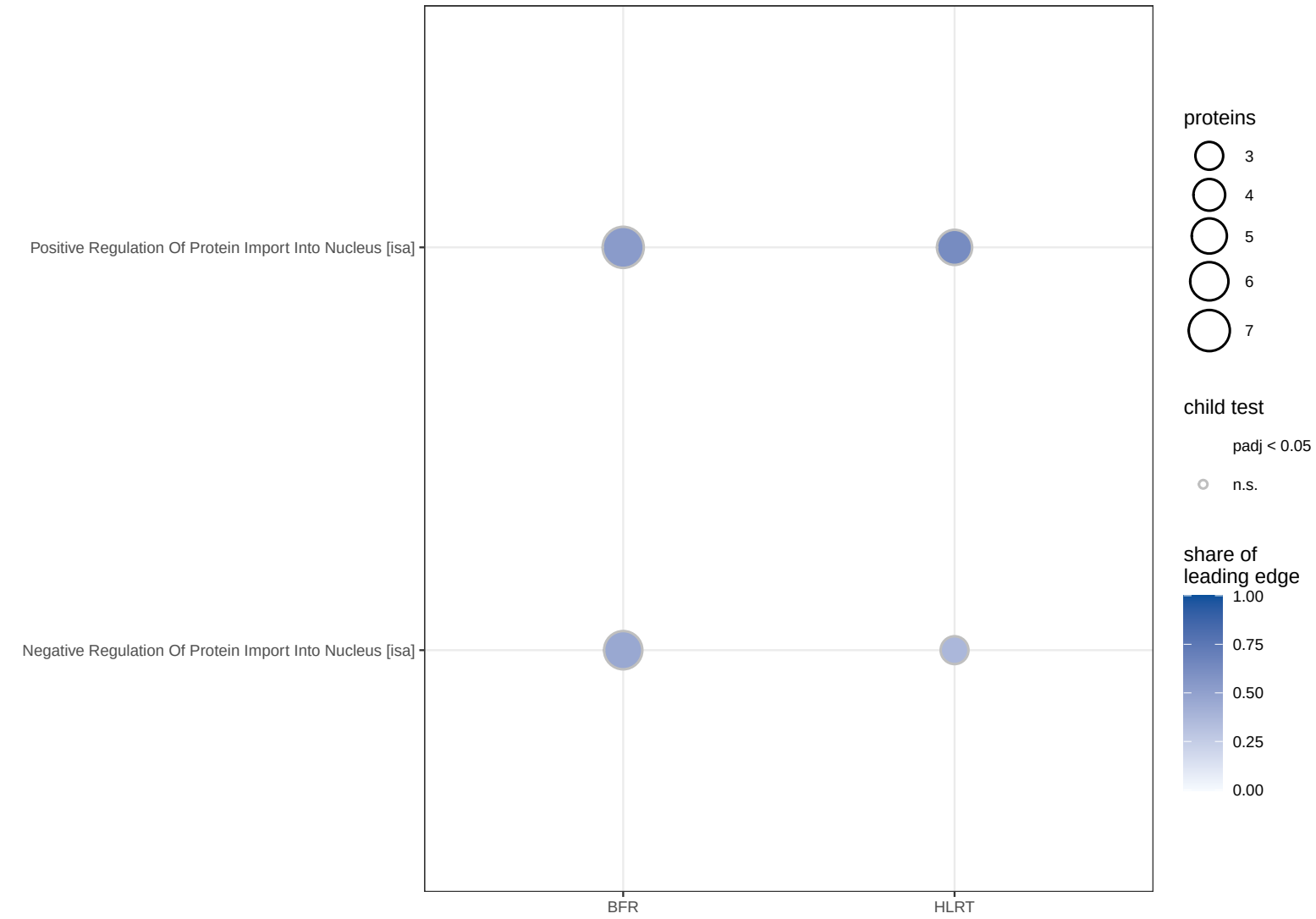
S17. Actomyosin Structure Organization

leading edge: BFR 52, HLRT 42; in significant descendants: BFR 62%, HLRT 64%



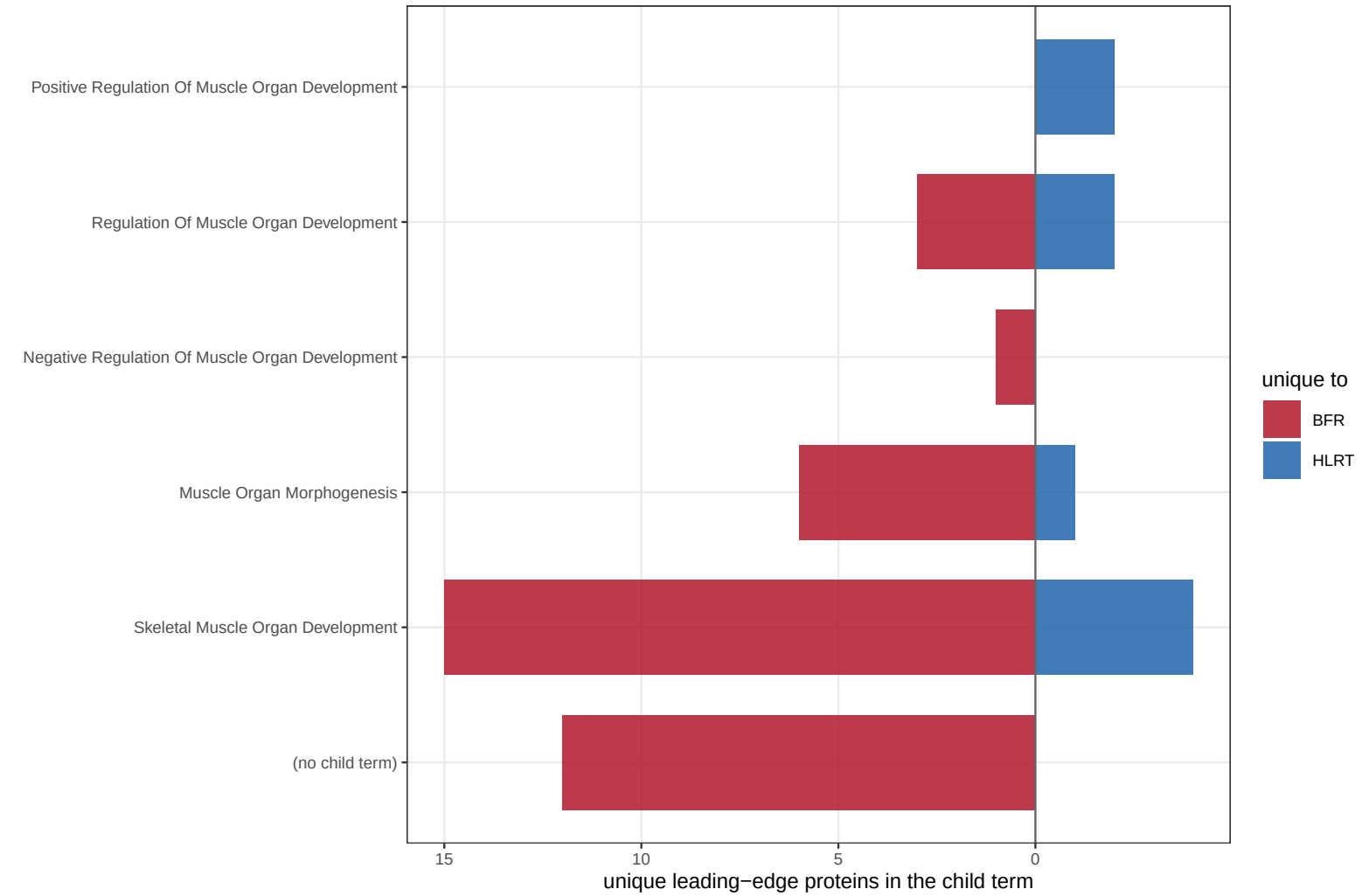
S17. Regulation Of Protein Import Into Nucleus

leading edge: BFR 13, HLRT 8; in significant descendants: BFR 0%, HLRT 0%



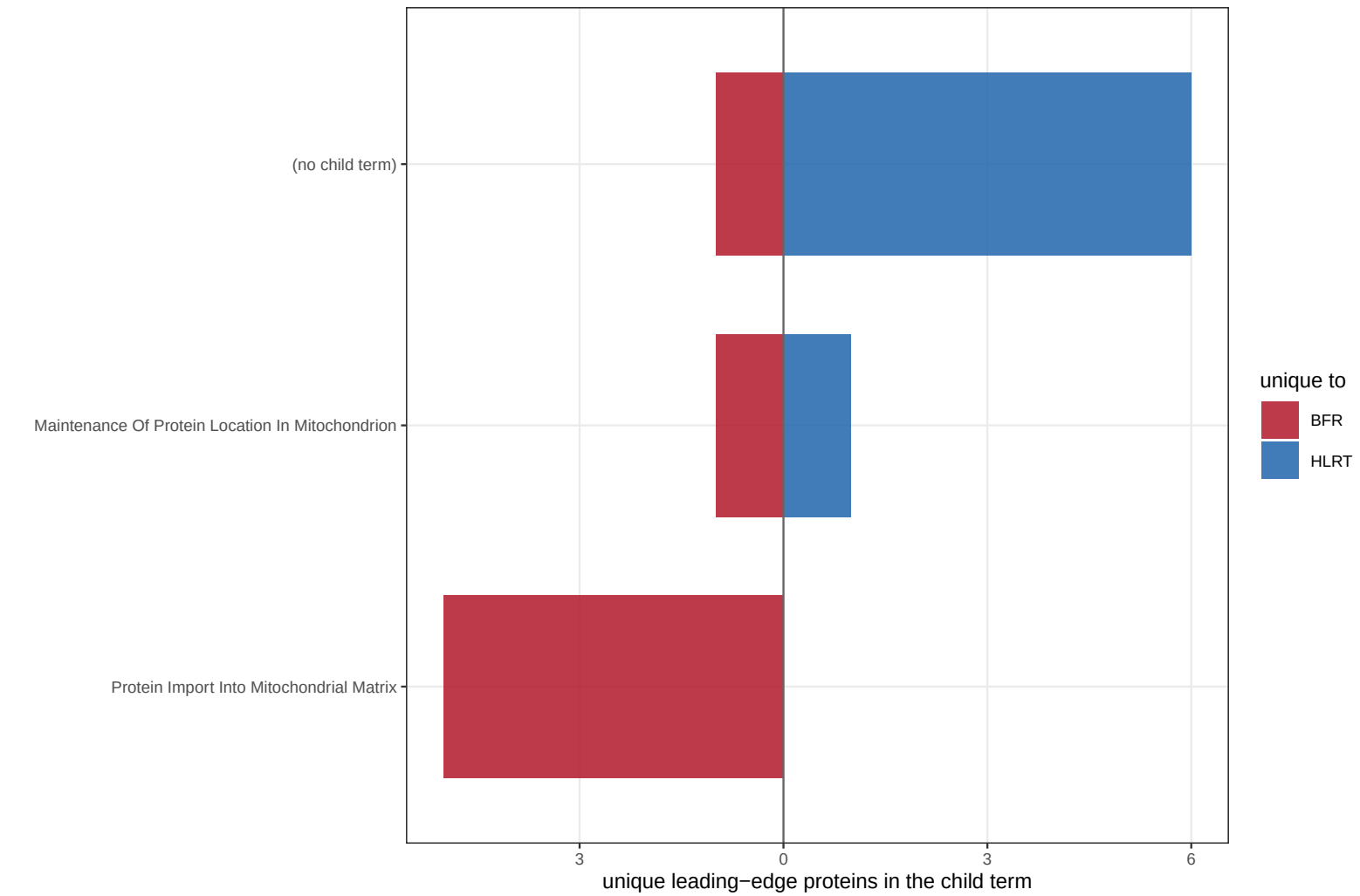
S18. Muscle Organ Development

only BFR 34, only HLRT 5, shared 42; divergence 1.48, permutation p 0.017 (BH 0.471)



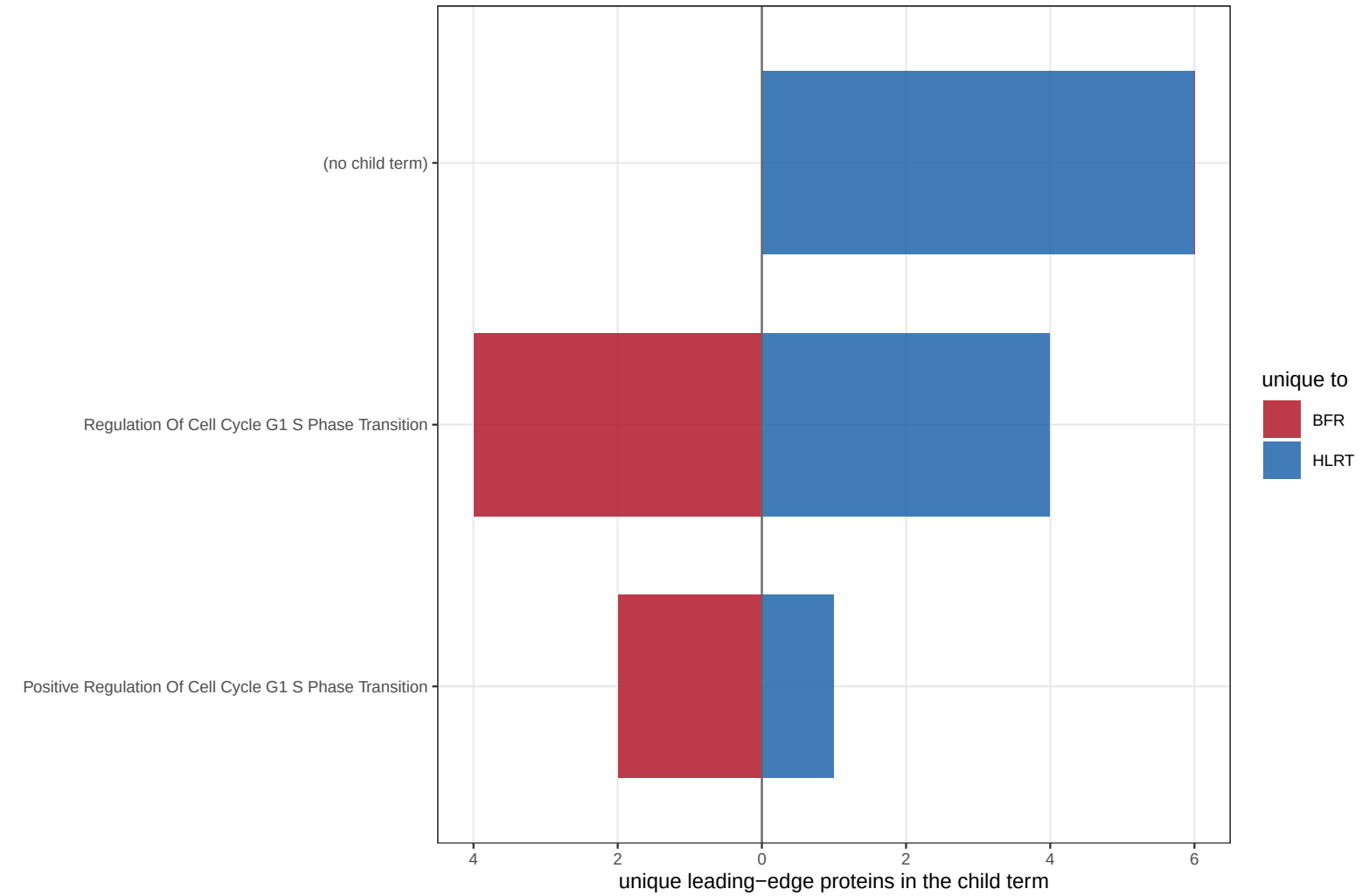
S18. Protein Localization To Mitochondrion

only BFR 7, only HLRT 7, shared 13; divergence 1.43, permutation p 0.0365 (BH 0.471)



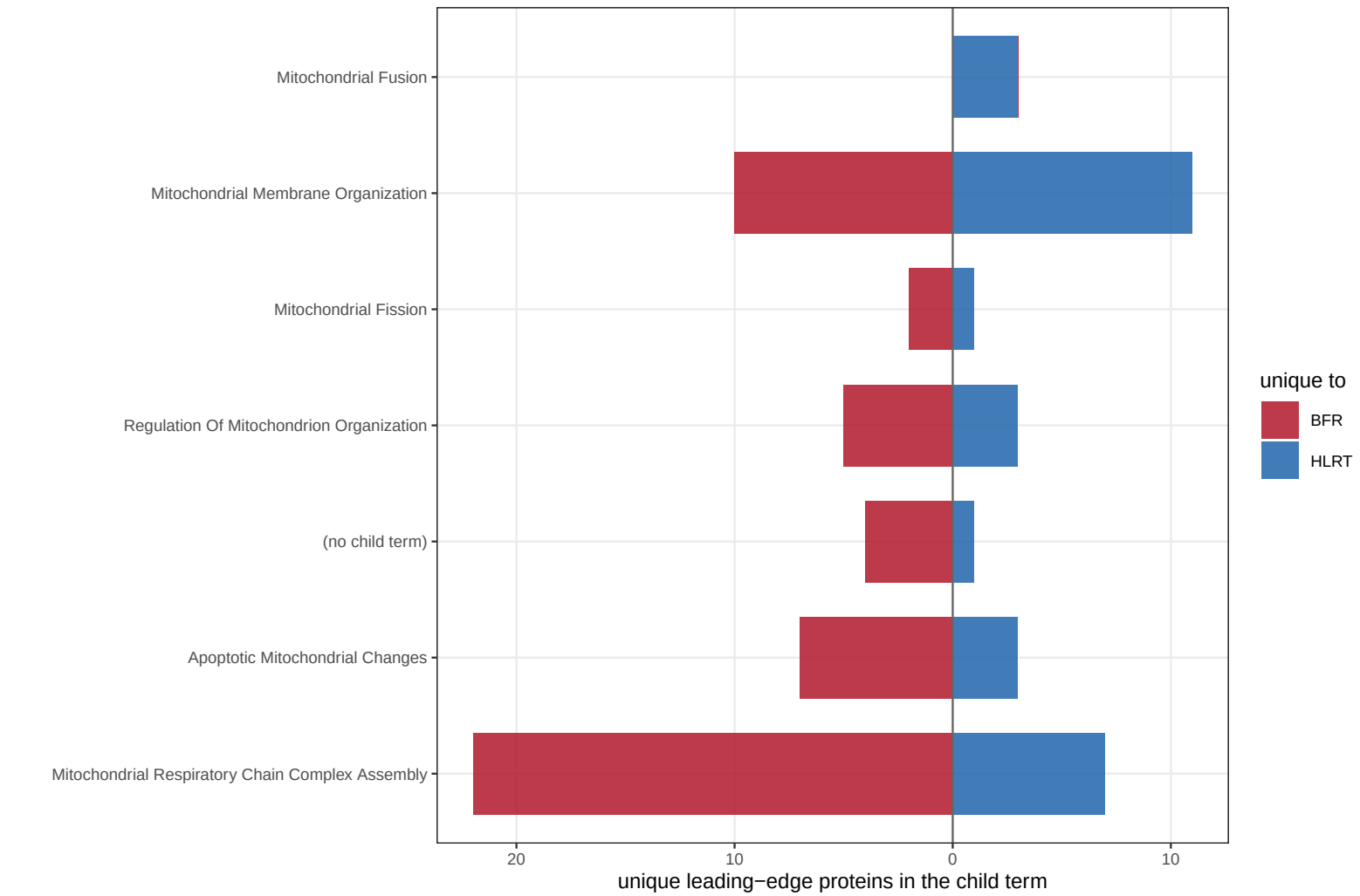
S18. Cell Cycle G1 S Phase Transition

only BFR 4, only HLRT 10, shared 19; divergence 1.60, permutation p 0.0505 (BH 0.471)



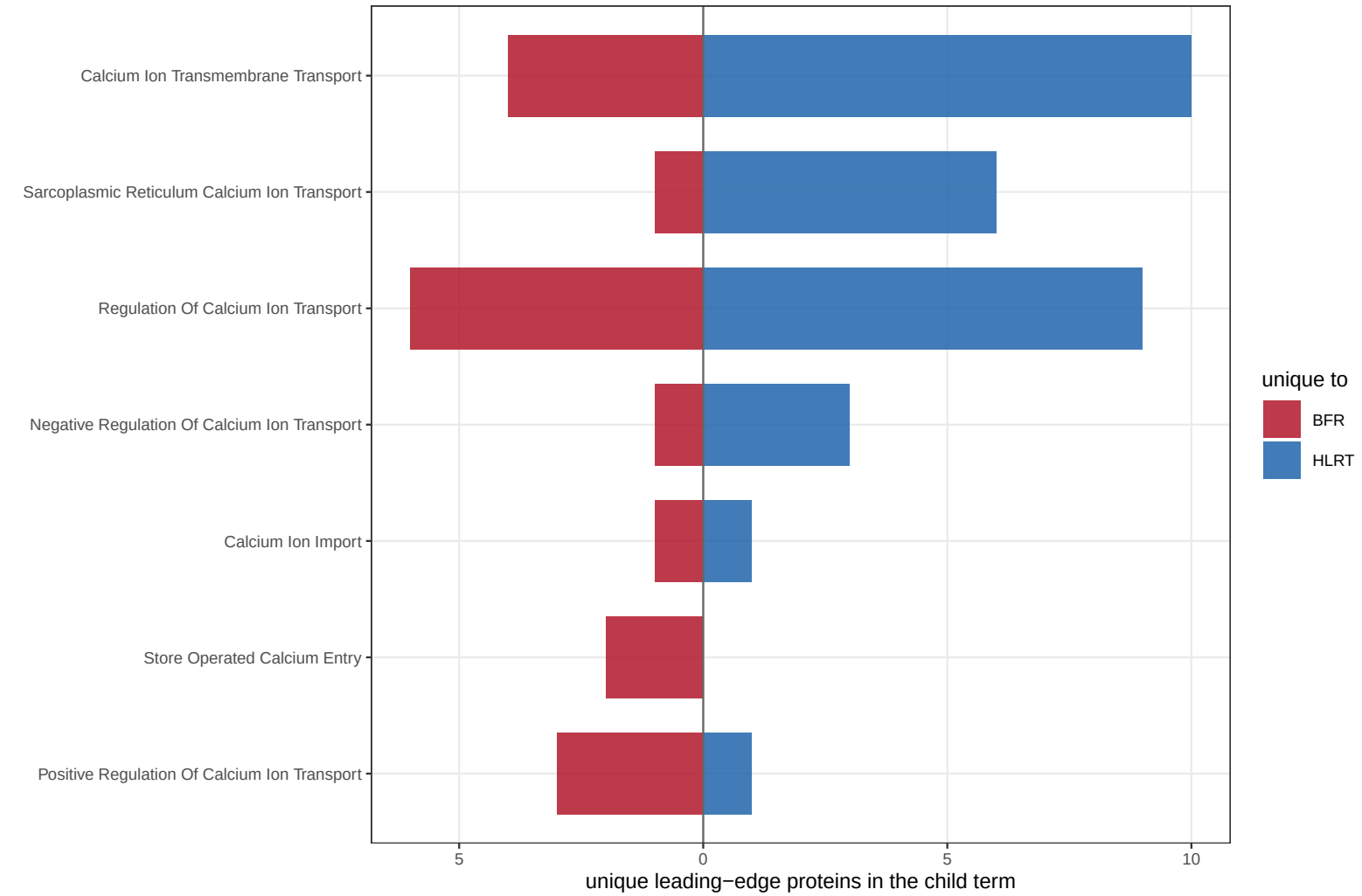
S18. Mitochondrion Organization

only BFR 41, only HLRT 18, shared 65; divergence 0.78, permutation p 0.109 (BH 0.628)



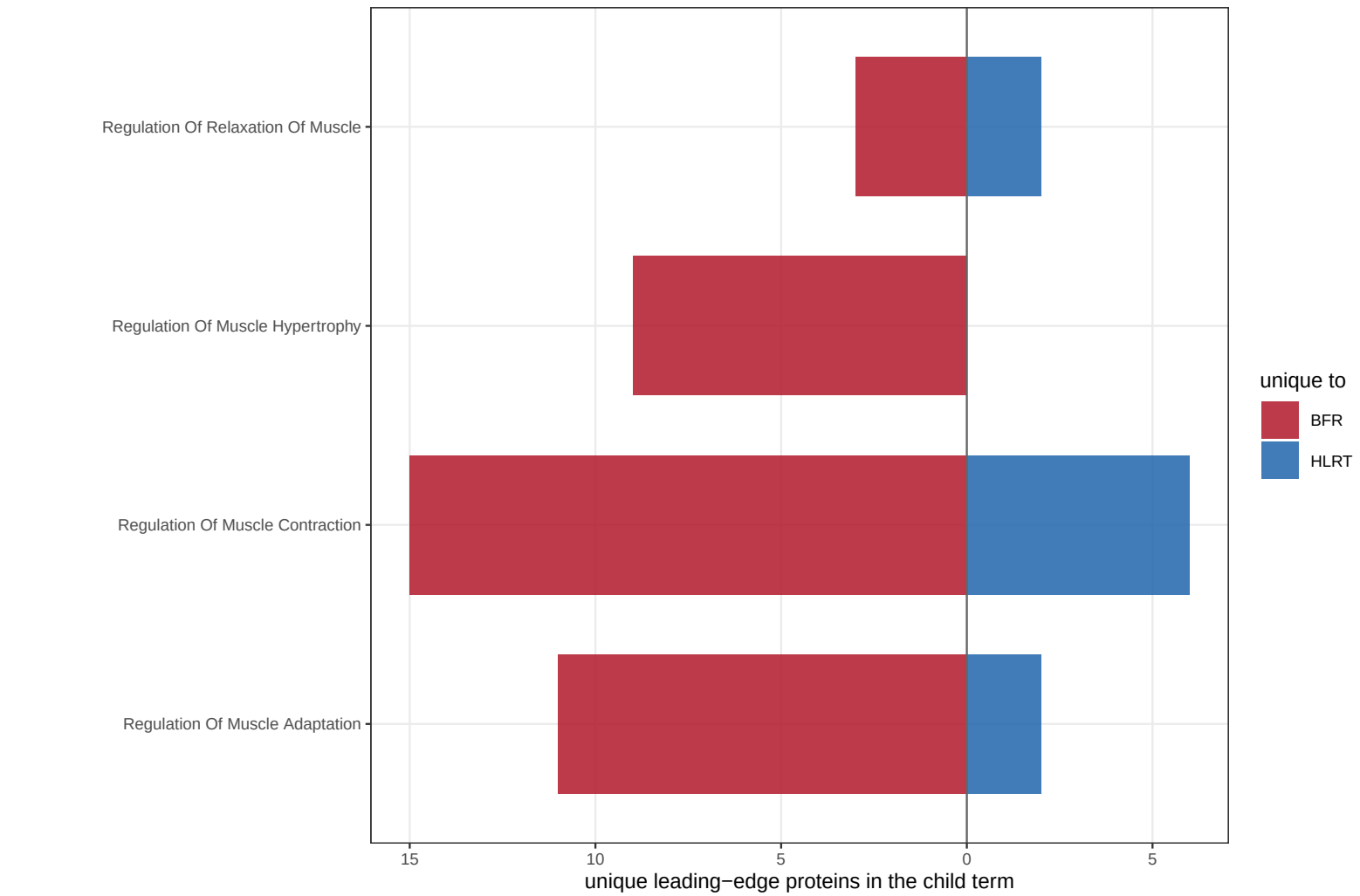
S18. Calcium Ion Transport

only BFR 8, only HLRT 12, shared 23; divergence 1.42, permutation p 0.12 (BH 0.628)



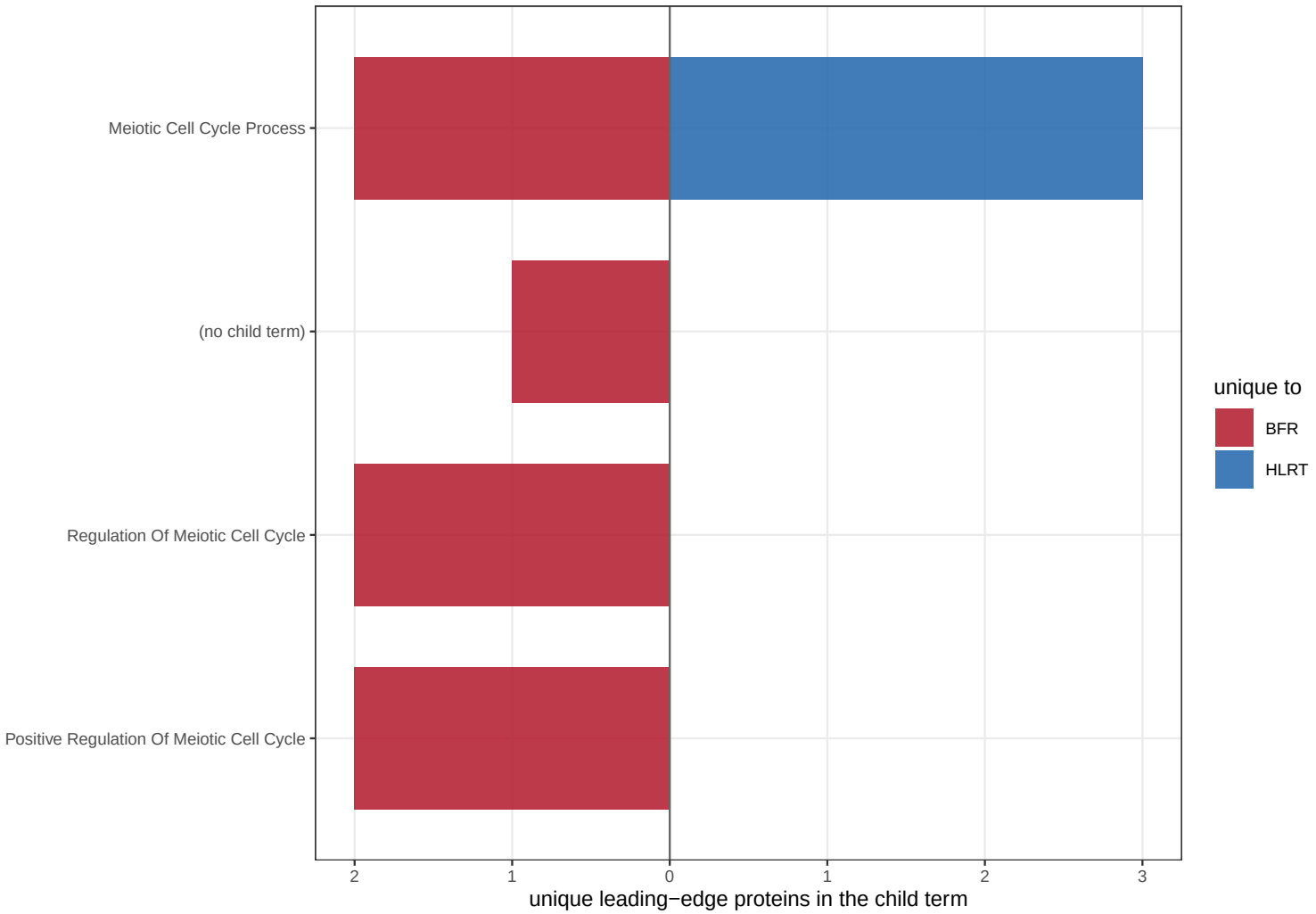
S18. Regulation Of Muscle System Process

only BFR 25, only HLRT 7, shared 27; divergence 0.94, permutation p 0.155 (BH 0.628)



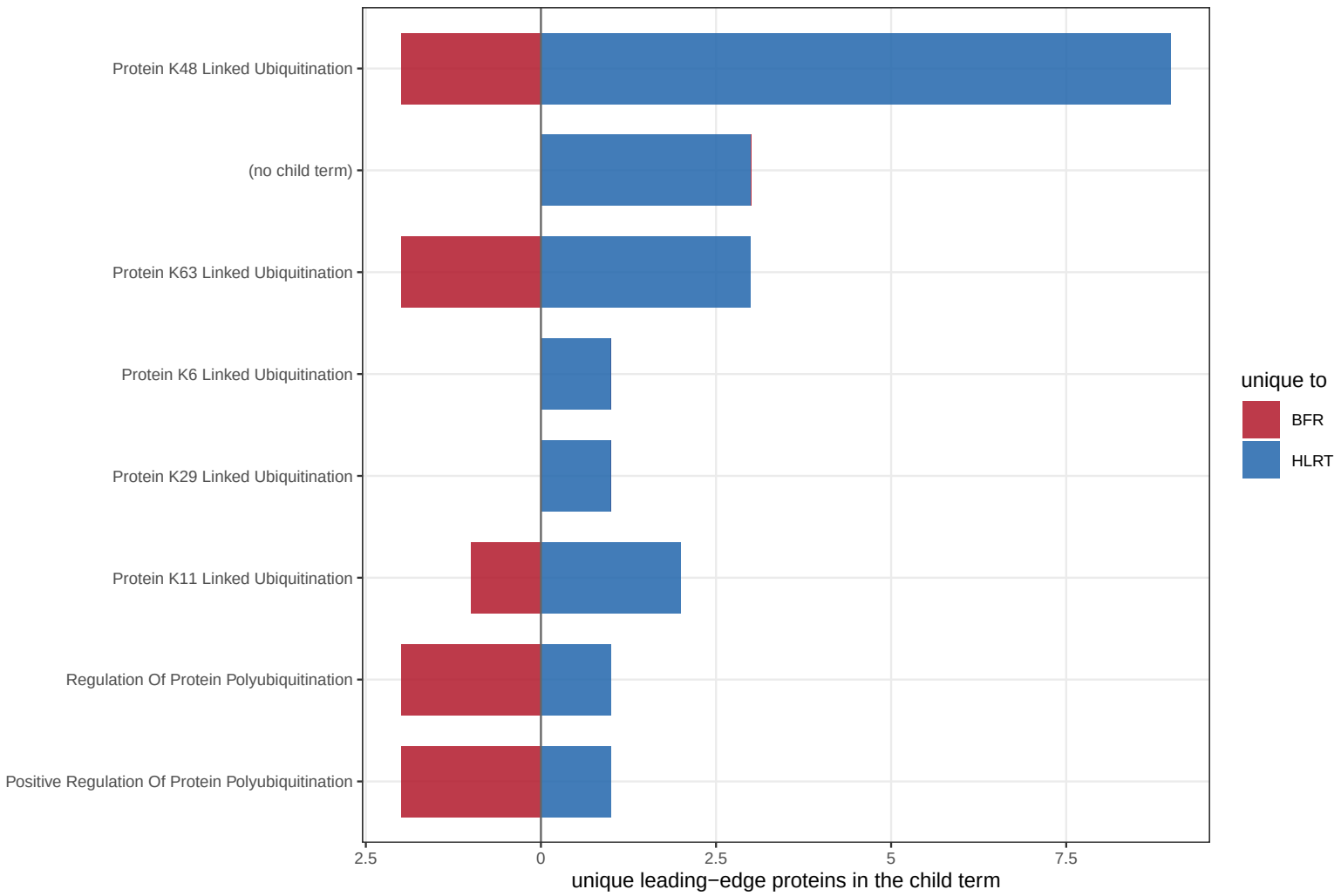
S18. Meiotic Cell Cycle

only BFR 4, only HLRT 3, shared 16; divergence 1.75, permutation p 0.157 (BH 0.628)



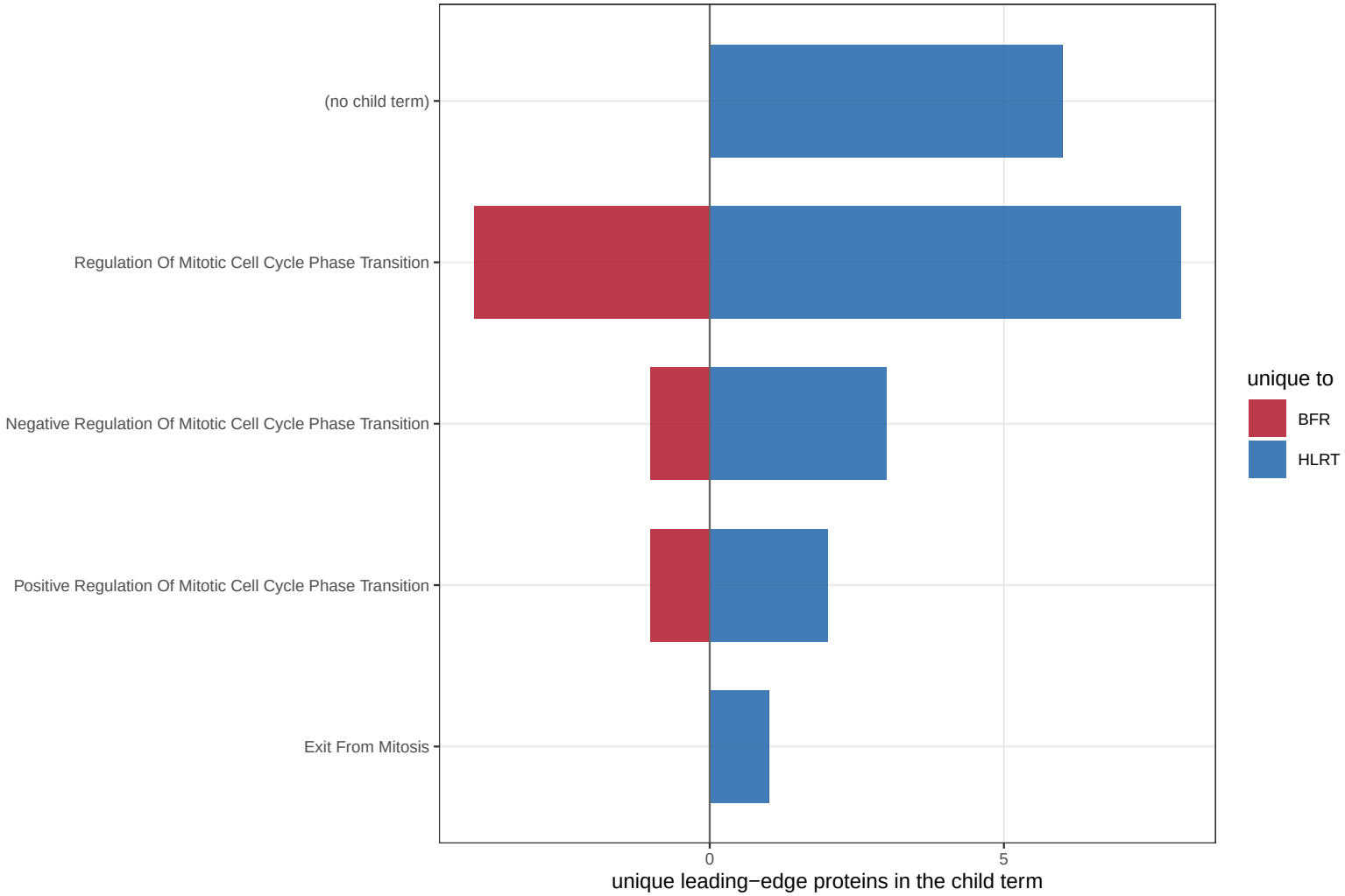
S18. Protein Polyubiquitination

only BFR 4, only HLRT 15, shared 18; divergence 1.72, permutation p 0.188 (BH 0.659)



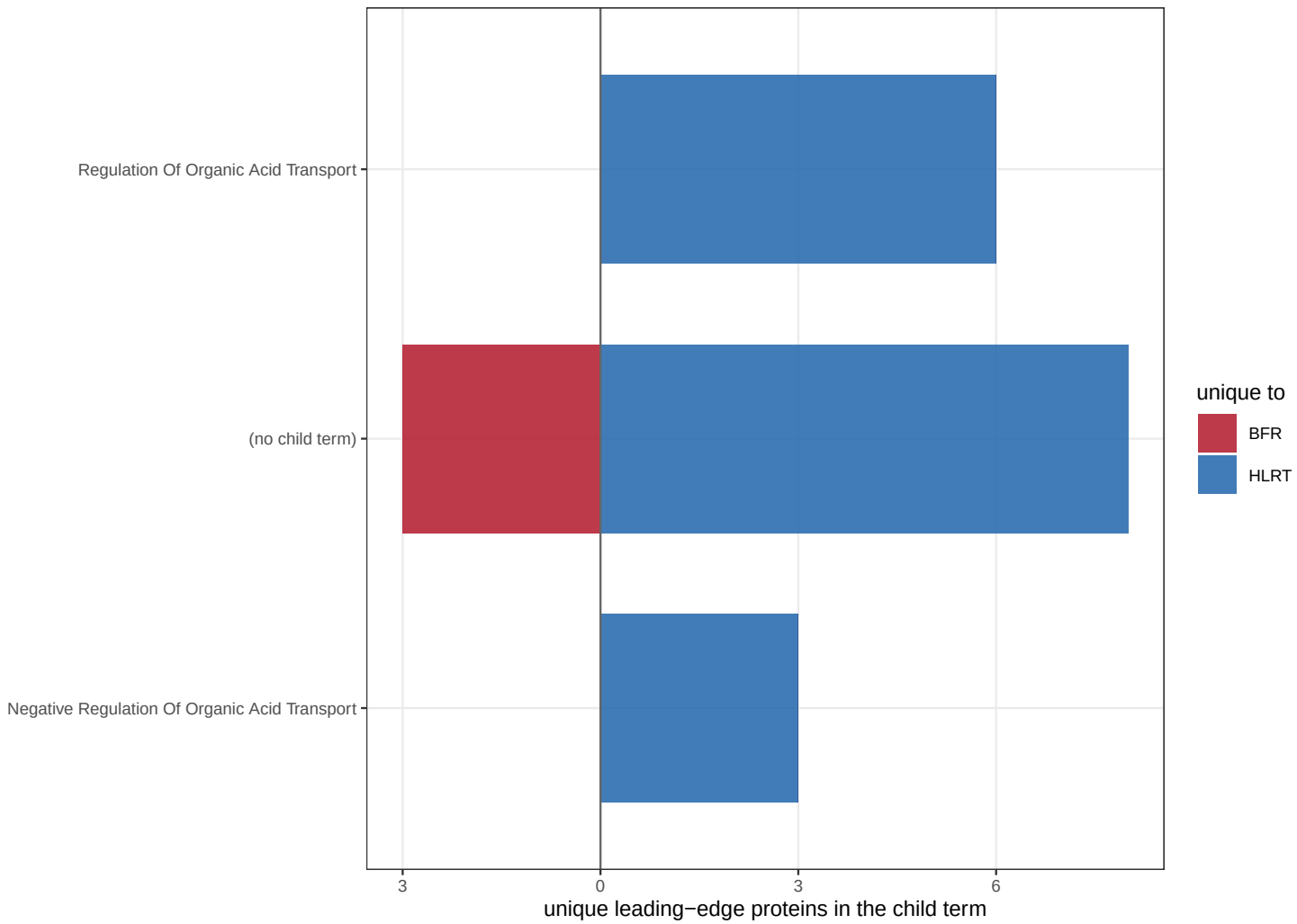
S18. Mitotic Cell Cycle Phase Transition

only BFR 4, only HLRT 15, shared 23; divergence 1.10, permutation p 0.285 (BH 0.828)



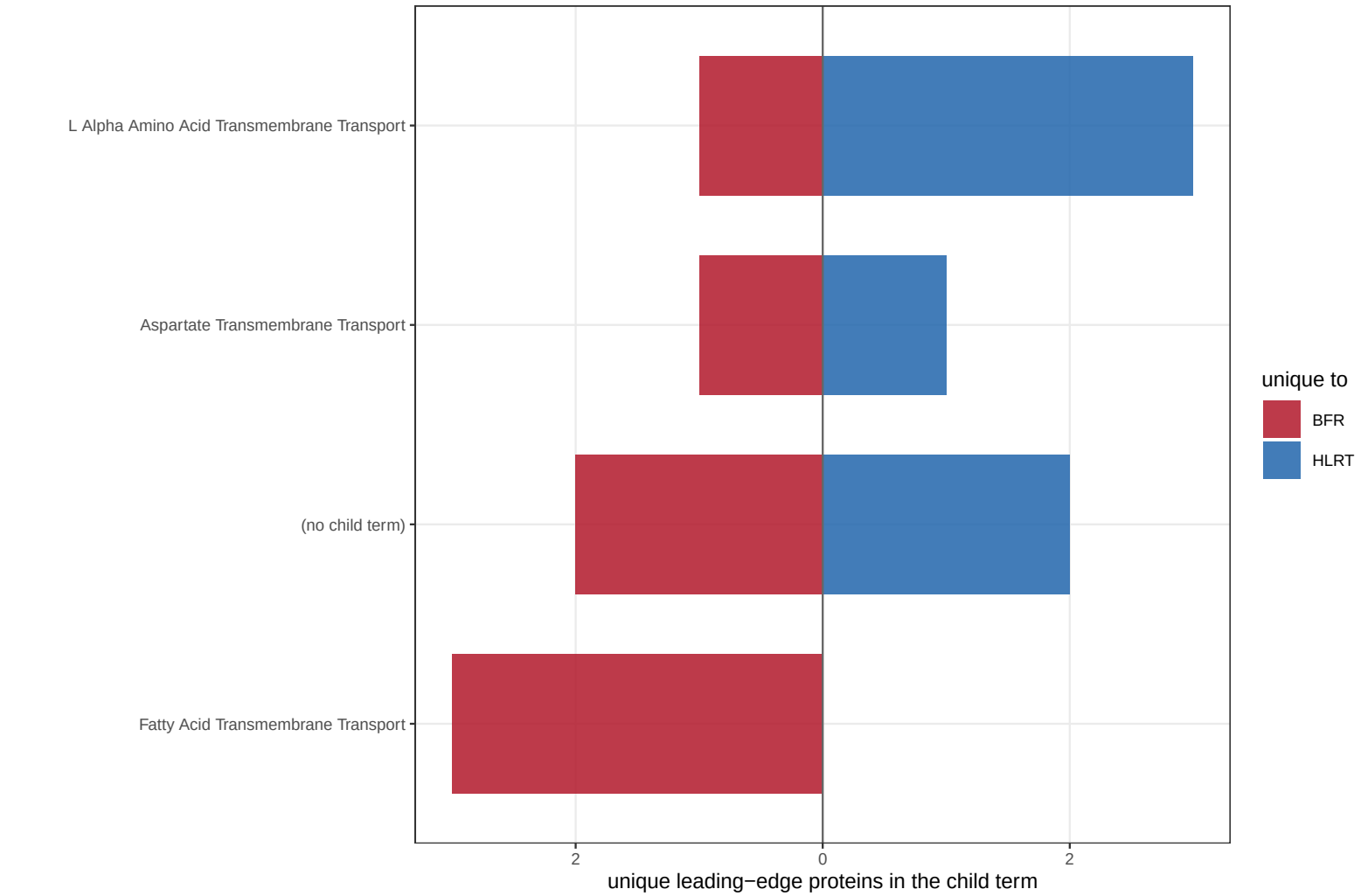
S18. Organic Acid Transport

only BFR 3, only HLRT 14, shared 8; divergence 1.07, permutation p 0.316 (BH 0.828)



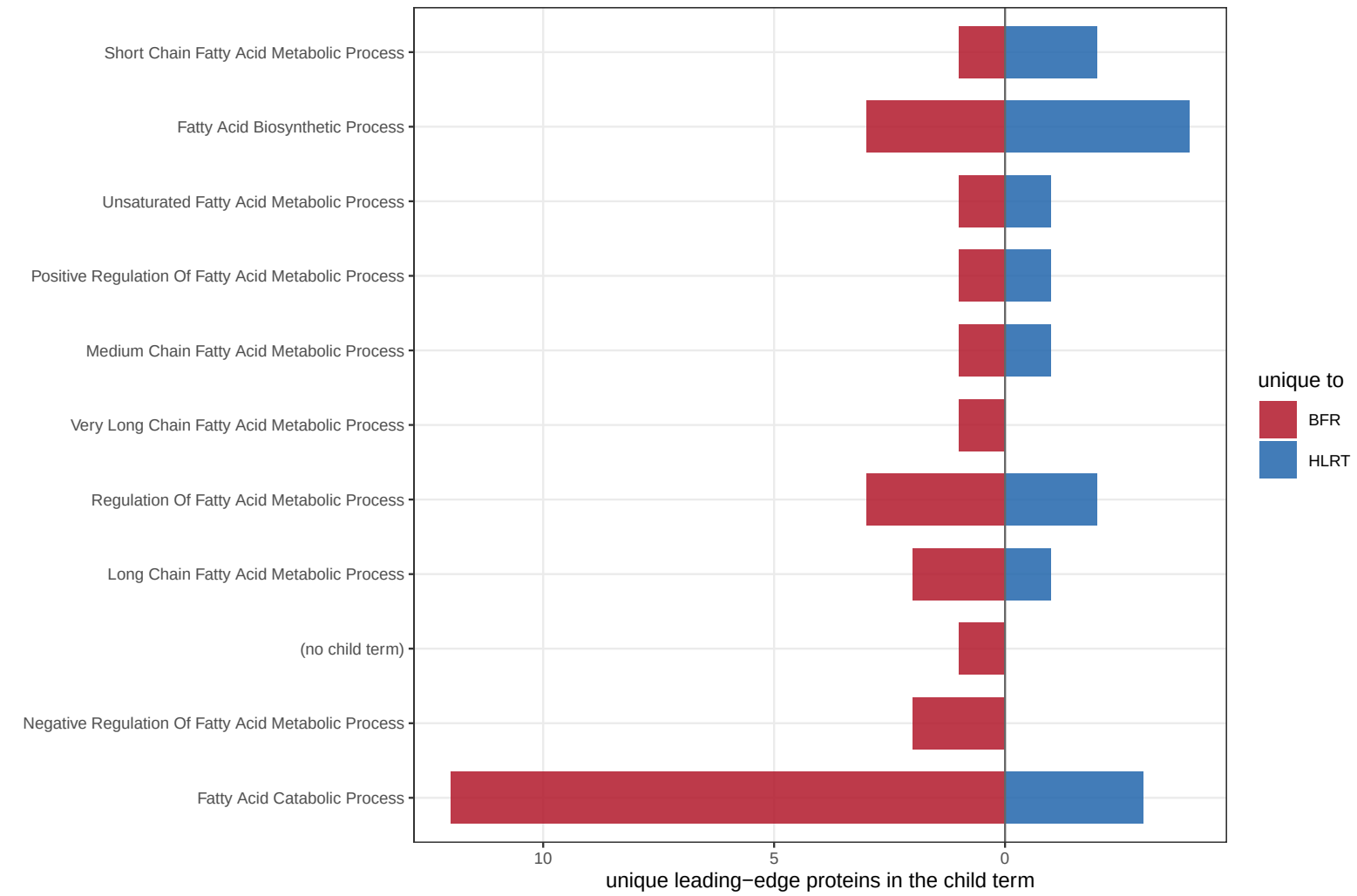
S18. Carboxylic Acid Transmembrane Transport

only BFR 6, only HLRT 5, shared 7; divergence 1.03, permutation p 0.325 (BH 0.828)



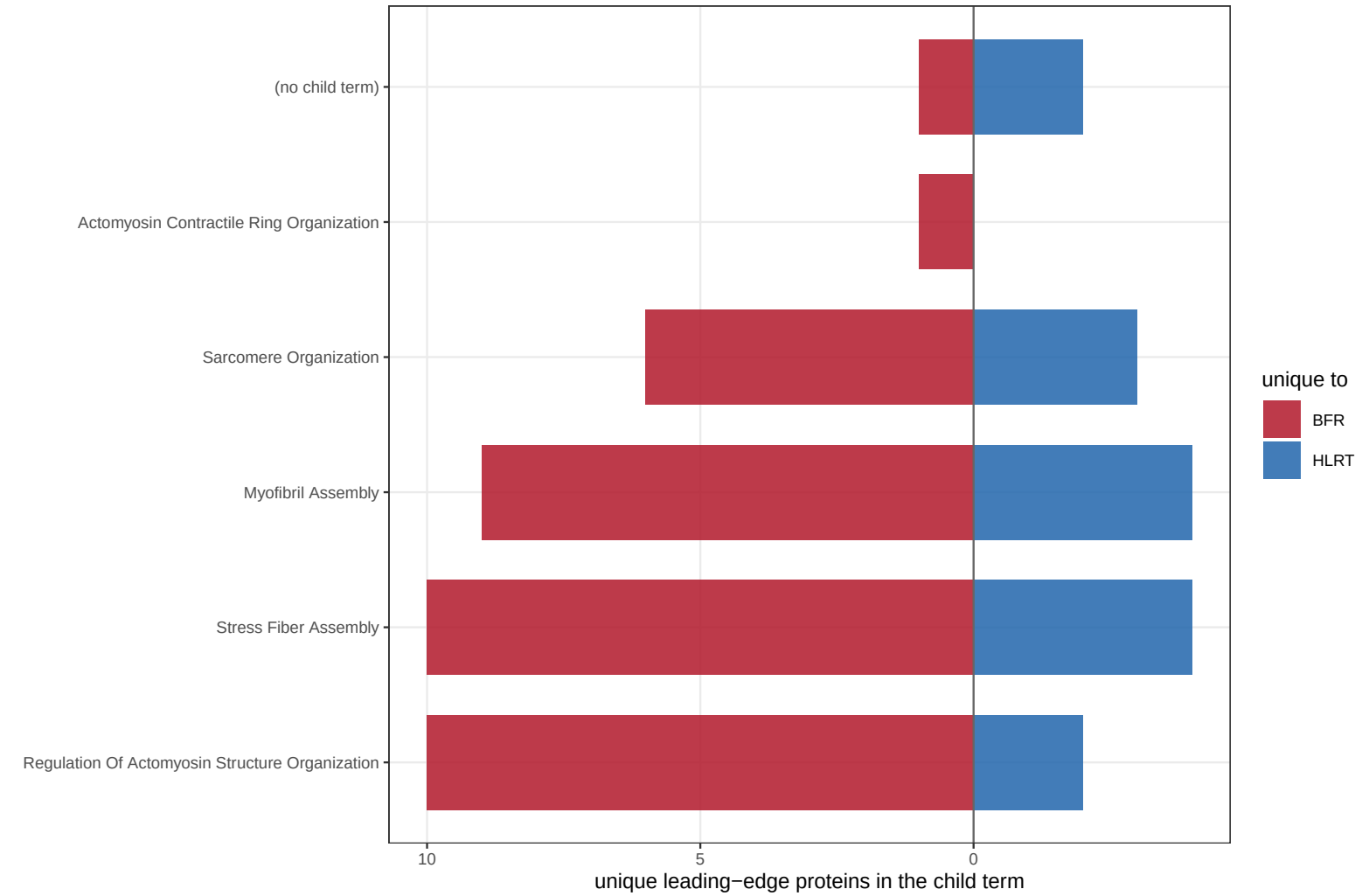
S18. Fatty Acid Metabolic Process

only BFR 16, only HLRT 8, shared 36; divergence 1.38, permutation p 0.394 (BH 0.92)



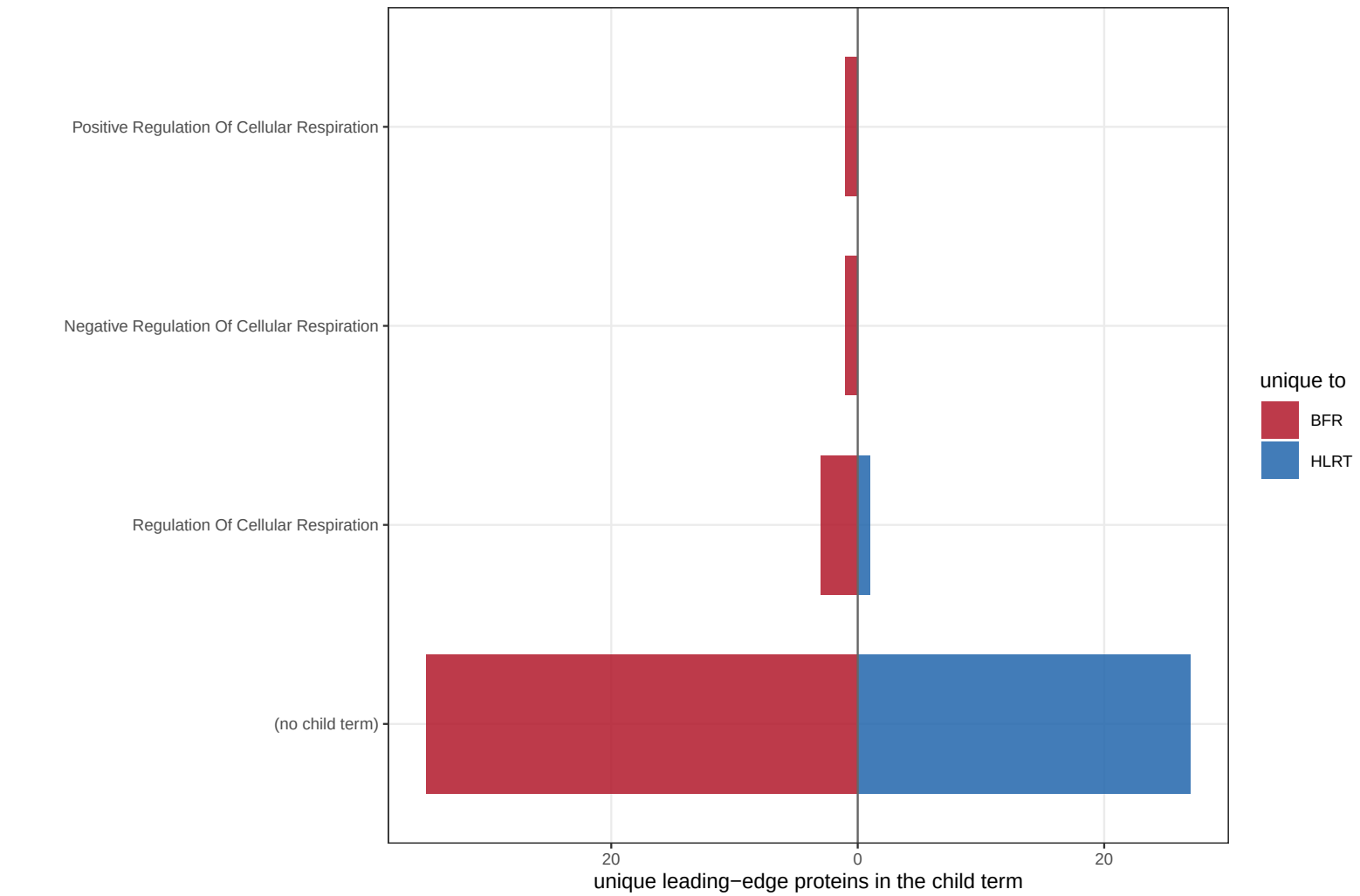
S18. Actomyosin Structure Organization

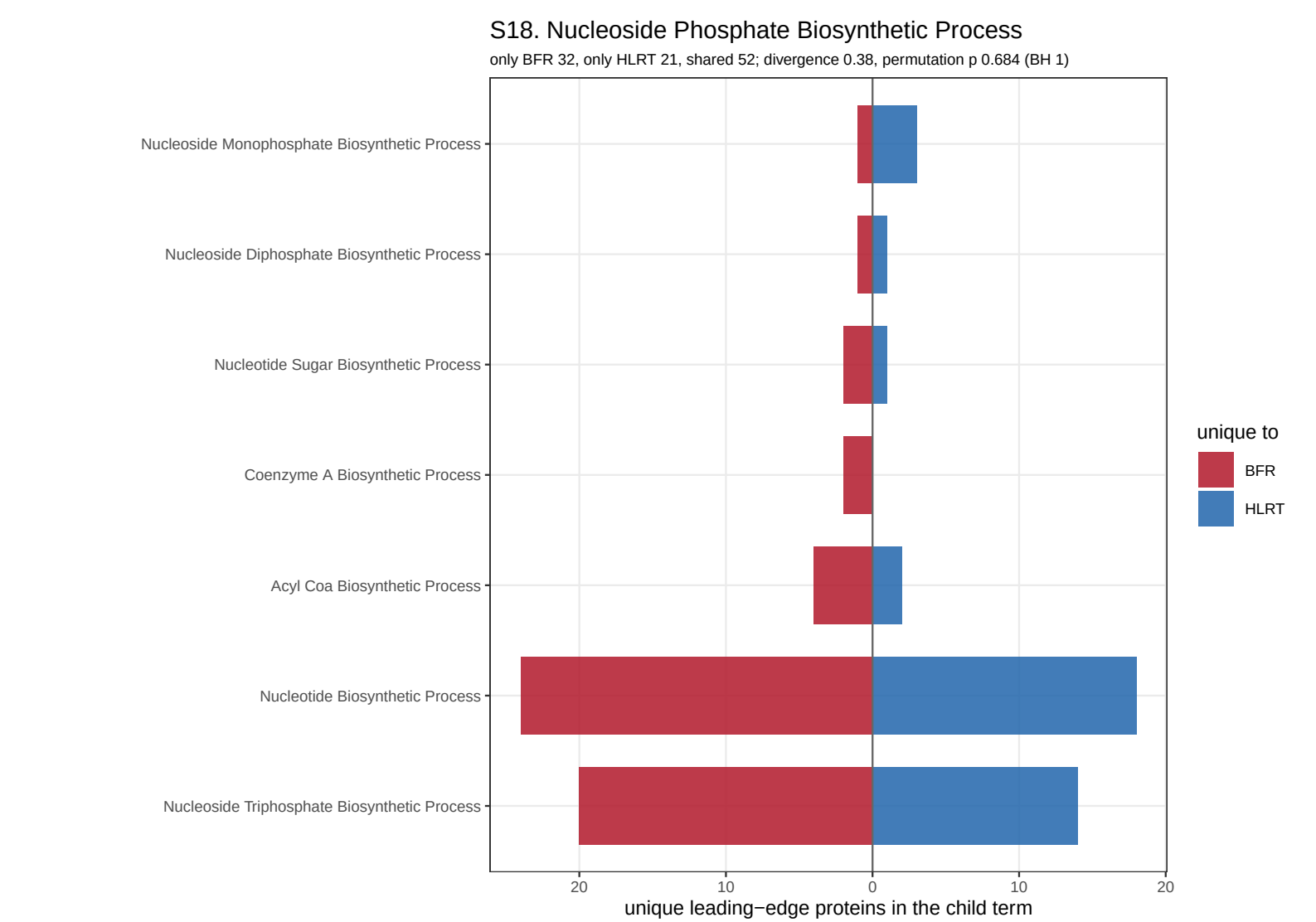
only BFR 20, only HLRT 10, shared 32; divergence 0.65, permutation p 0.599 (BH 1)

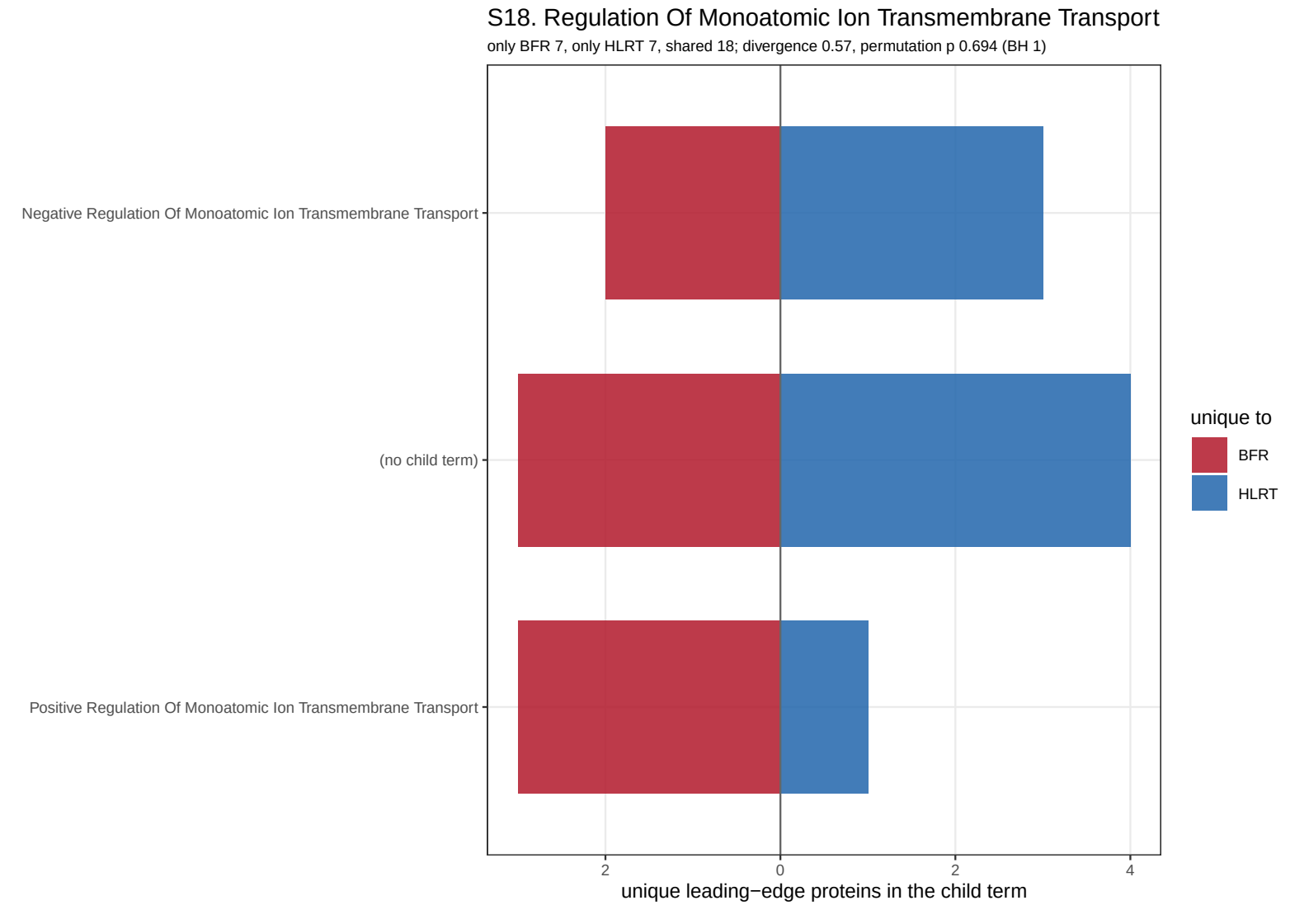


S18. Cellular Respiration

only BFR 38, only HLRT 28, shared 78; divergence 0.14, permutation p 0.642 (BH 1)

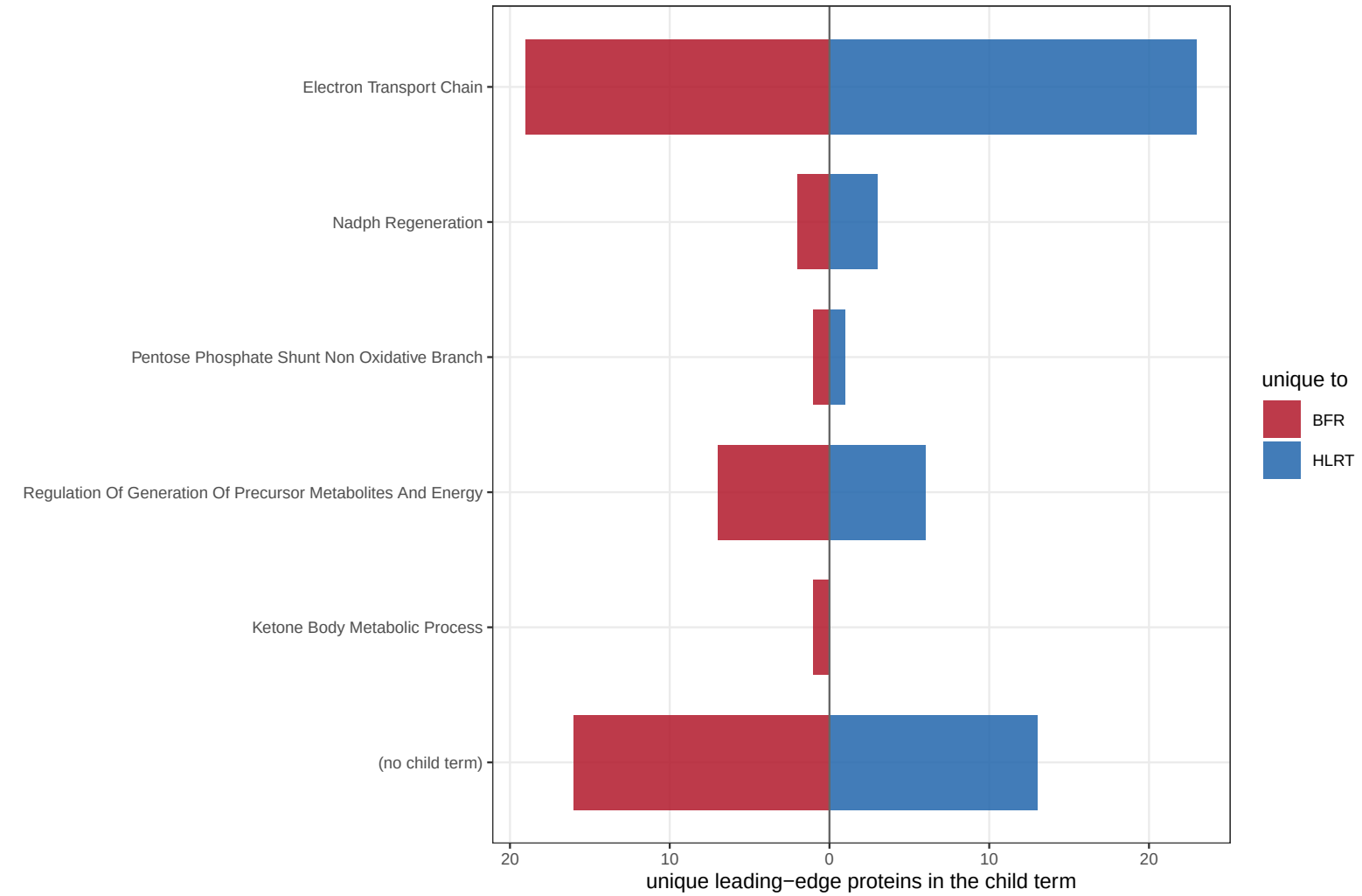



S18. Nucleoside Phosphate Biosynthetic Process
only BFR 32, only HLRT 21, shared 52; divergence 0.38, permutation p 0.684 (BH 1)
Nucleoside Monophosphate Biosynthetic Process
Nucleoside Diphosphate Biosynthetic Process
Nucleotide Sugar Biosynthetic Process
Coenzyme A Biosynthetic Process
Acyl Coa Biosynthetic Process
Nucleotide Biosynthetic Process
Nucleoside Triphosphate Biosynthetic Process
20
10
0
10
20
unique leading−edge proteins in the child term
unique to
BFR
HLRT


S18. Regulation Of Monoatomic Ion Transmembrane Transport
only BFR 7, only HLRT 7, shared 18; divergence 0.57, permutation p 0.694 (BH 1)
Negative Regulation Of Monoatomic Ion Transmembrane Transport
(no child term)
Positive Regulation Of Monoatomic Ion Transmembrane Transport
2
0
2
4
unique leading−edge proteins in the child term
unique to
BFR
HLRT

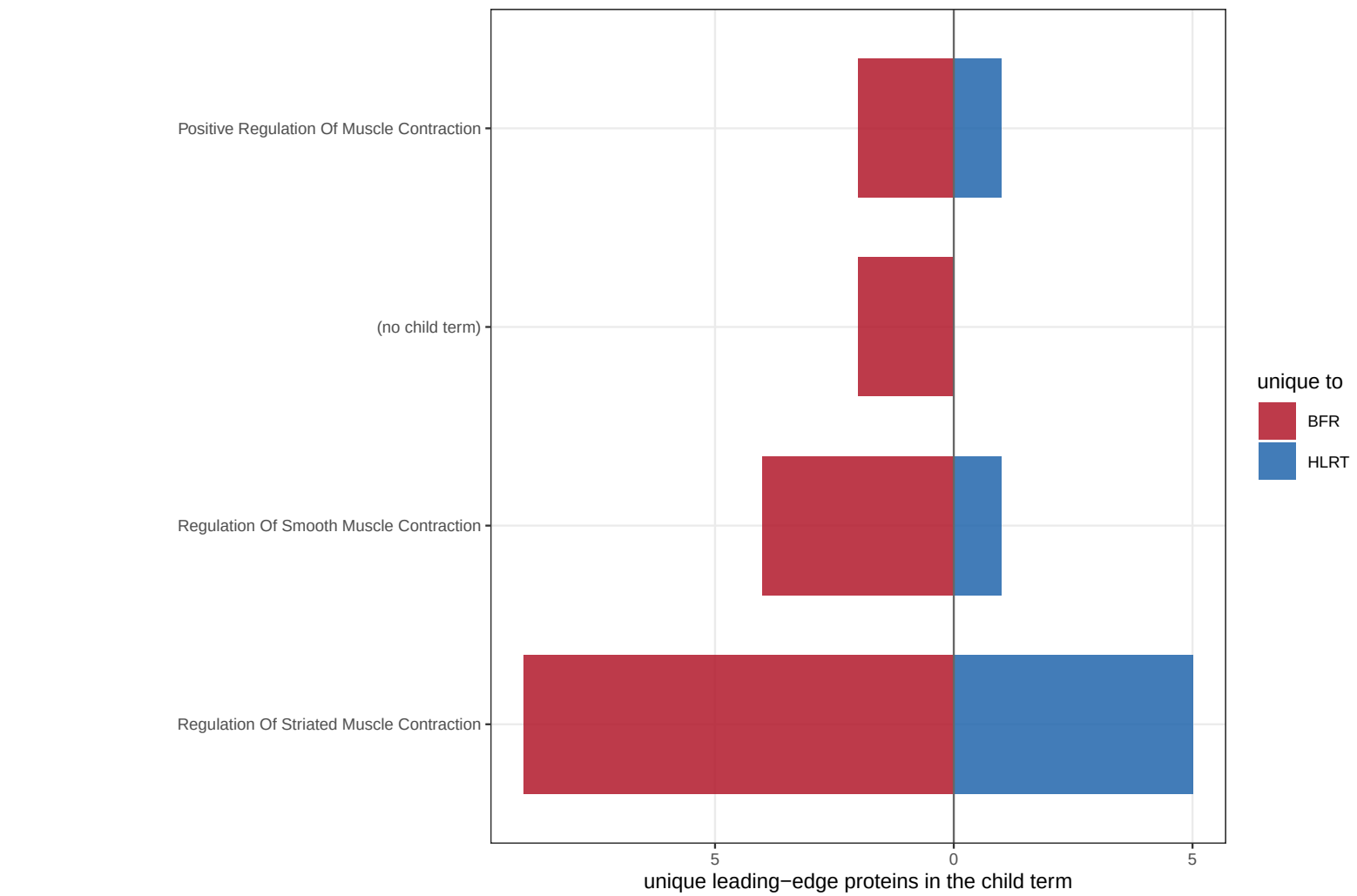
S18. Generation Of Precursor Metabolites And Energy

only BFR 43, only HLRT 45, shared 80; divergence 0.23, permutation p 0.737 (BH 1)



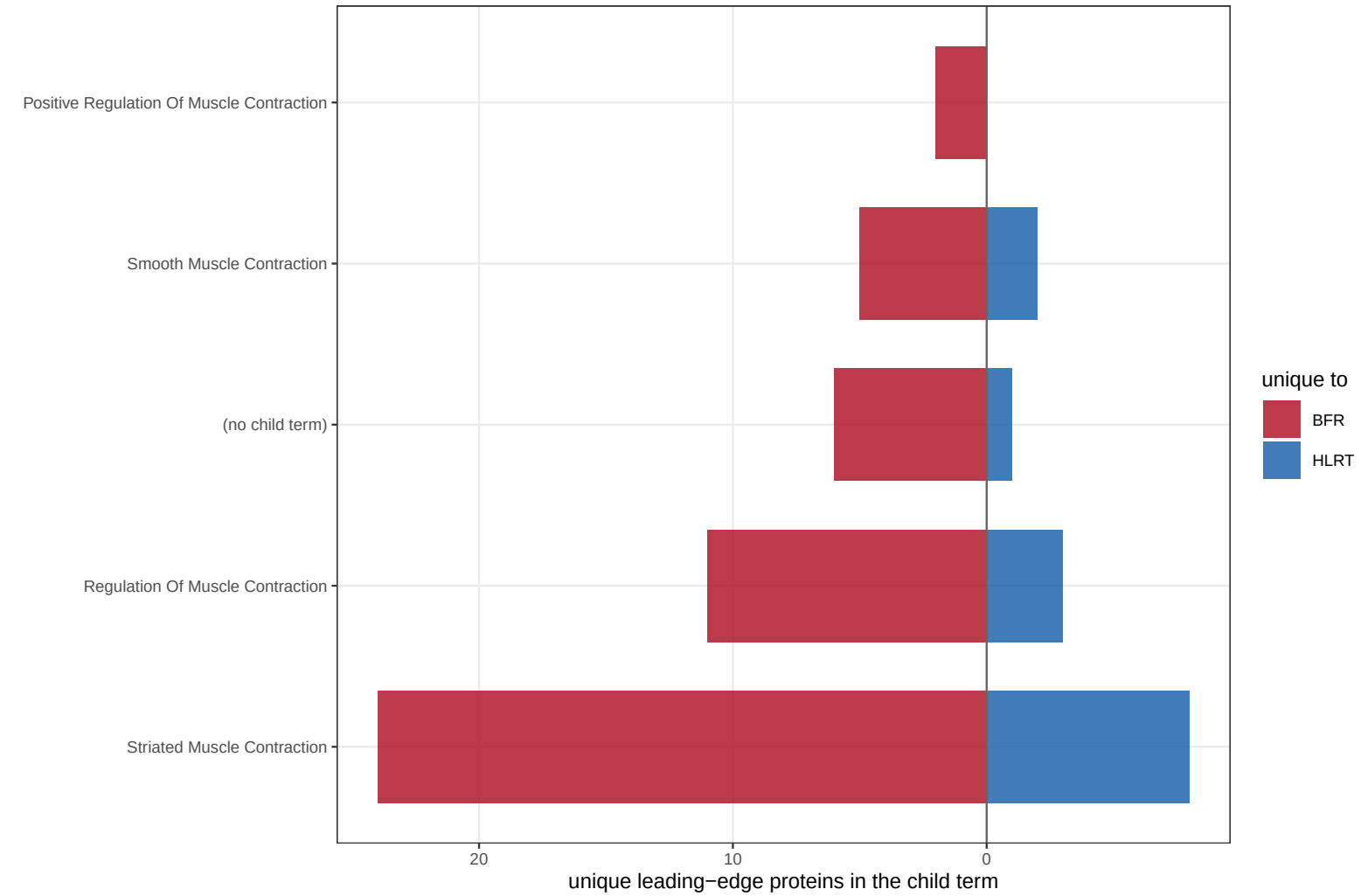
S18. Regulation Of Muscle Contraction

only BFR 14, only HLRT 6, shared 25; divergence 0.48, permutation p 0.818 (BH 1)



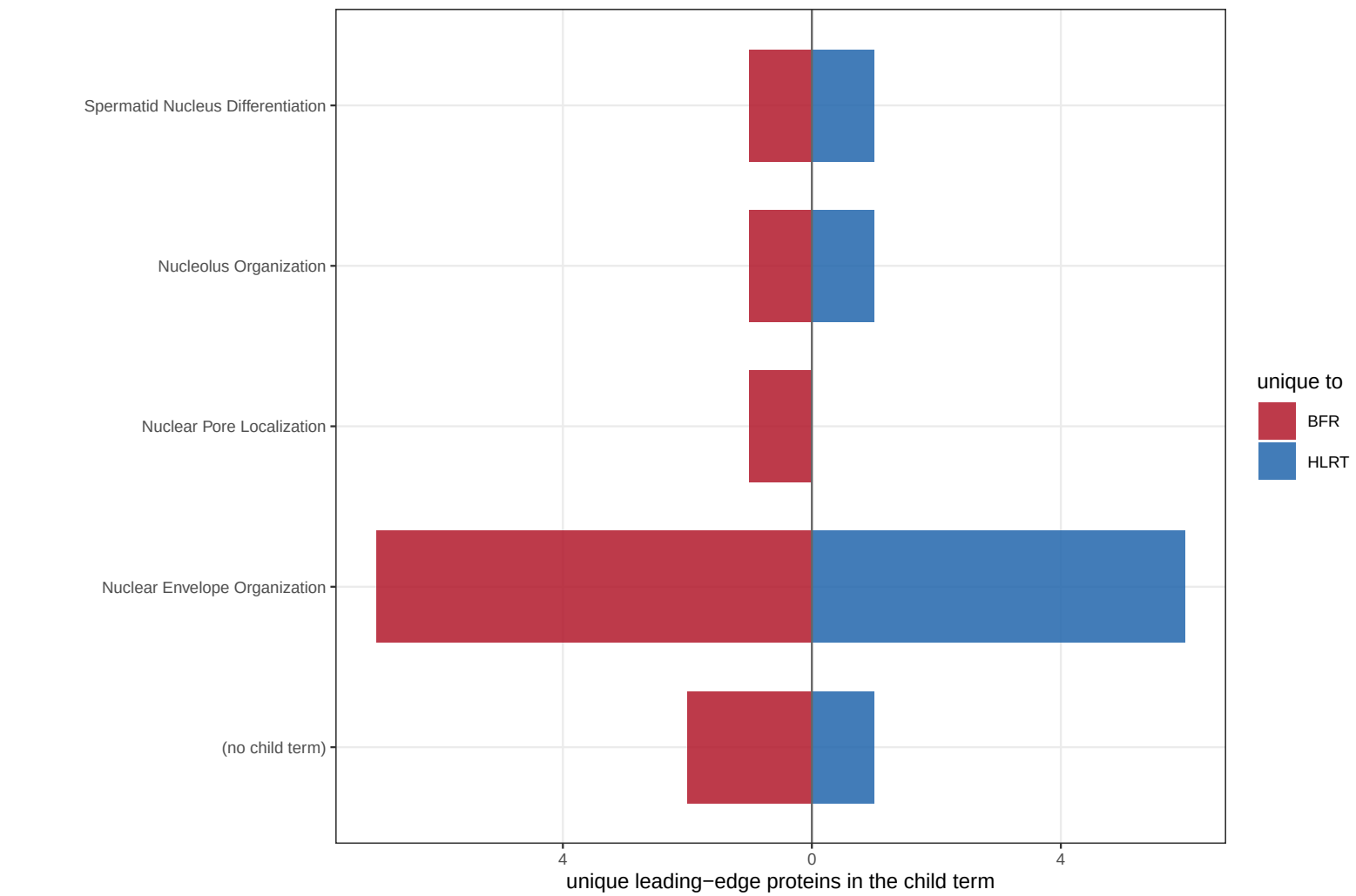
S18. Muscle Contraction

only BFR 35, only HLRT 11, shared 62; divergence 0.26, permutation p 0.974 (BH 1)



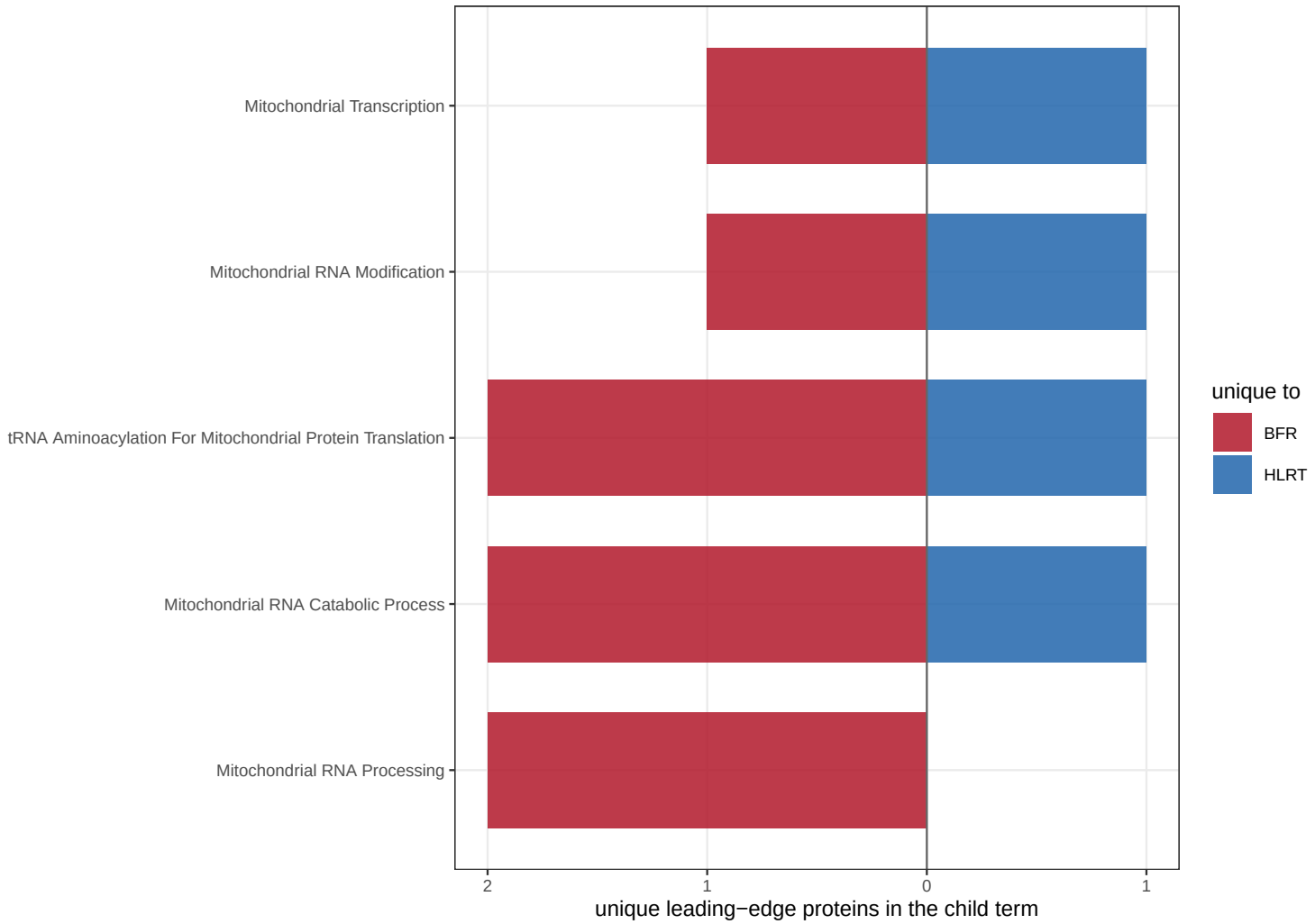
S18. Nucleus Organization

only BFR 11, only HLRT 9, shared 12; divergence 0.23, permutation p 1 (BH 1)



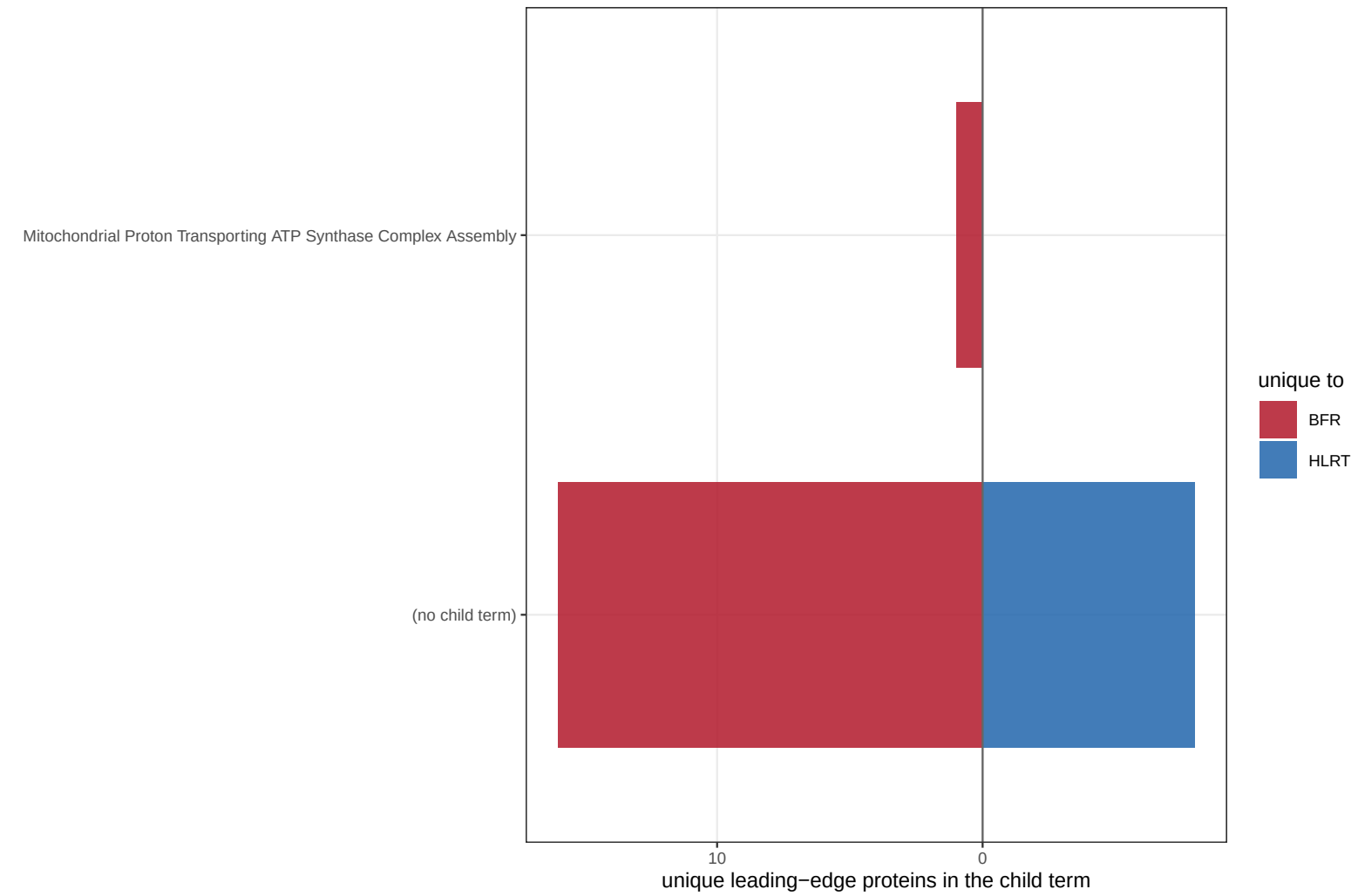
S18. Mitochondrial RNA Metabolic Process

only BFR 7, only HLRT 3, shared 10; divergence 0.76, permutation p 1 (BH 1)



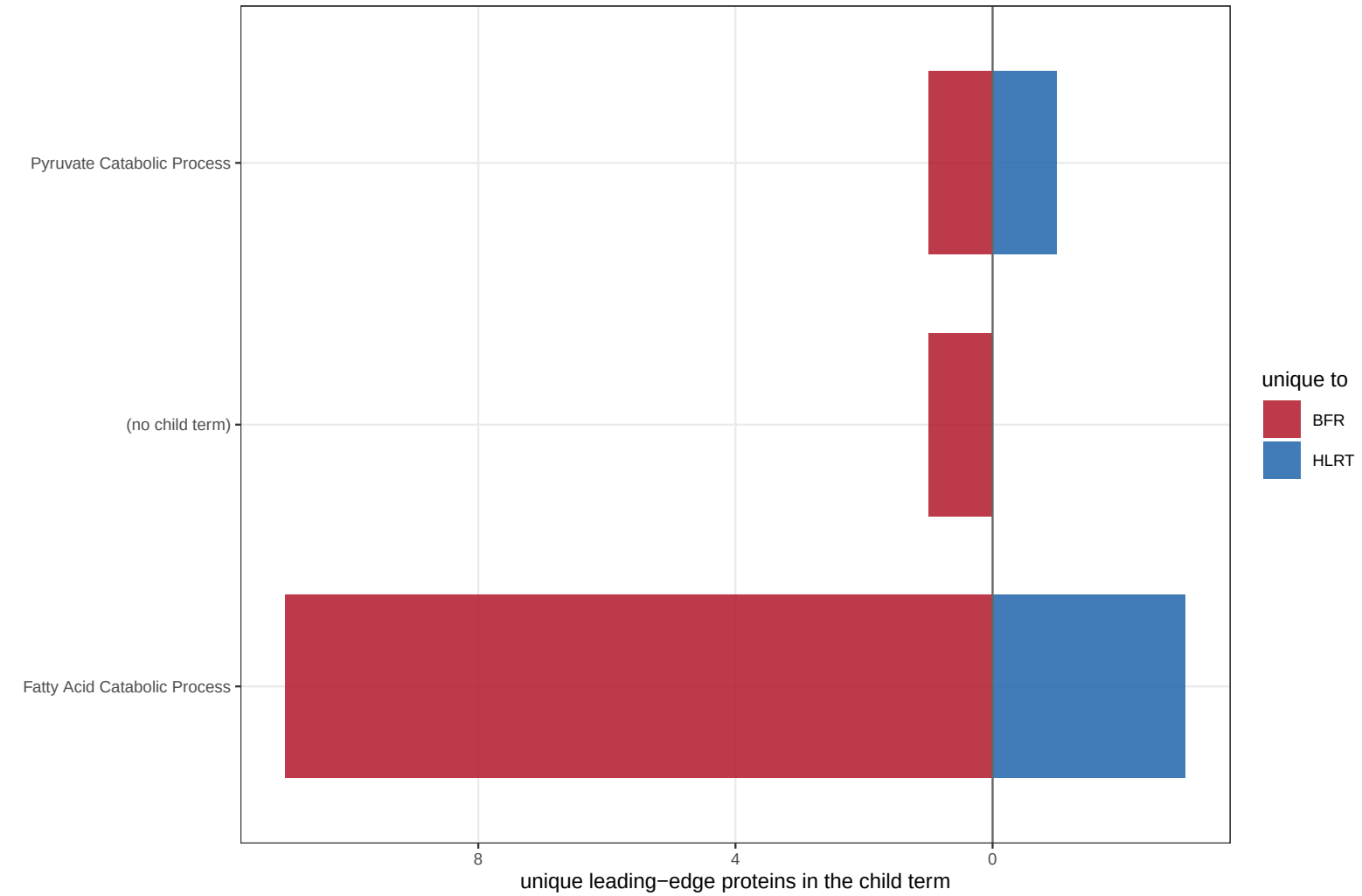
S18. Mitochondrial Respiratory Chain Complex Assembly

only BFR 17, only HLRT 8, shared 29; divergence 0.12, permutation p 1 (BH 1)



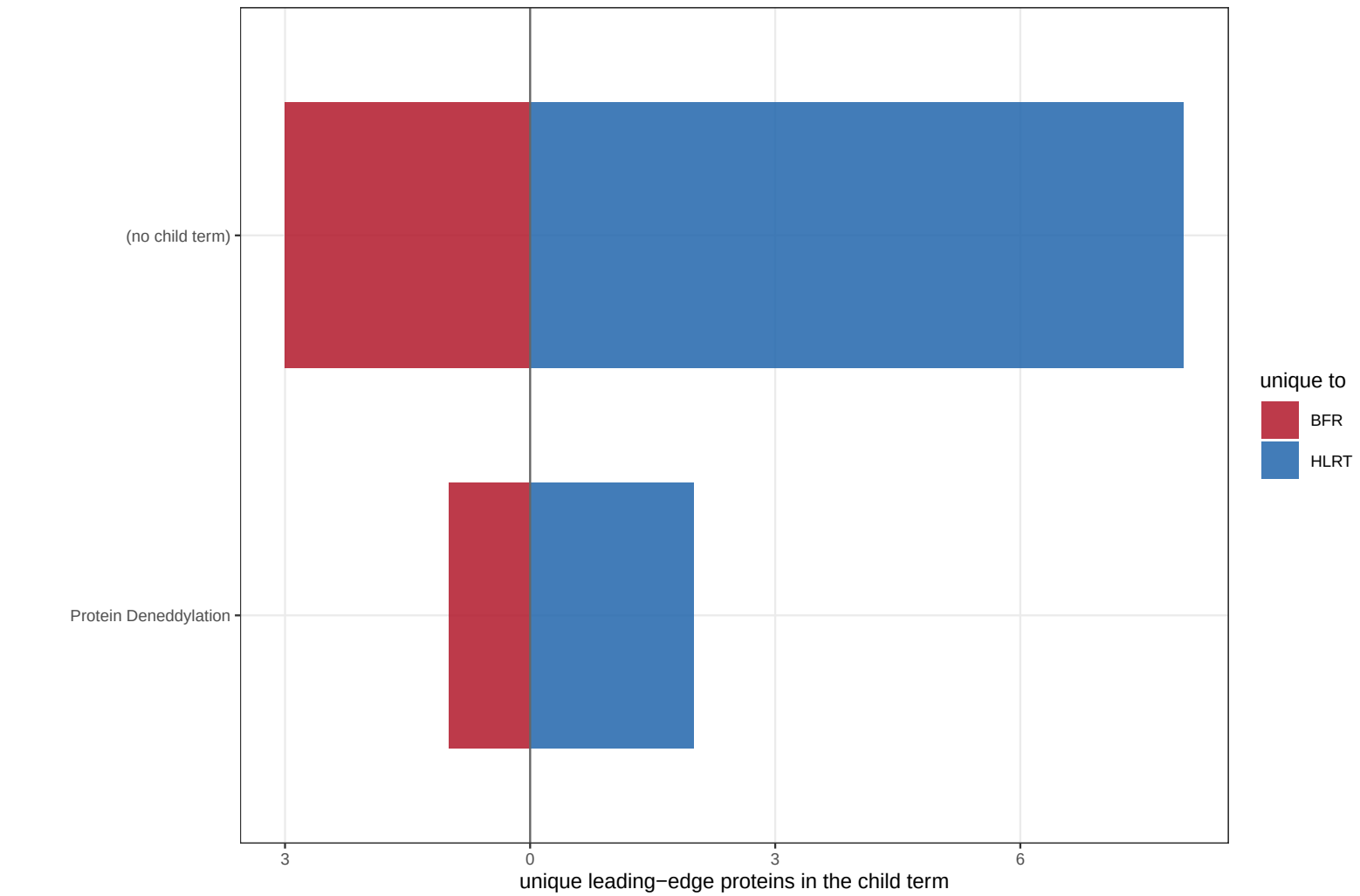
S18. Monocarboxylic Acid Catabolic Process

only BFR 13, only HLRT 4, shared 20; divergence 0.35, permutation p 1 (BH 1)



S18. Protein Modification By Small Protein Removal

only BFR 4, only HLRT 10, shared 17; divergence 0.10, permutation p 1 (BH 1)



S18. Muscle Cell Development

only BFR 23, only HLRT 4, shared 33; divergence 0.09, permutation p 1 (BH 1)

